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degli Studi
di Padova

LECTURE NOTES
OF

MODELS OF THEORETICAL PHYSICS

Collection of the lecture notes of Prof. Azaele.

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About these notes

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Some parts of these notes are based on and adapted from the material written by Dario Liotta, available at

`https://github.com/darioliotta`

My material has been revised, reorganized, expanded, and integrated with my own lecture notes and other course resources. Therefore, any modifications and errors in the present version are my own.

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Physics of Data MSc
University of Padua

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Chapter 1

Introduction

1.1 Introduction

1.1.1 Goal of the course

The aim of this course is to learn how to construct mathematical models, using a set of general techniques and tools. Rather than focusing on specific equations or physical theories, the course centers on **models** – such as ODEs, PDEs, and SDEs – and on **analytical and computational tools** that can be applied across a broad range of research fields.

The main objective is to become familiar with key mathematical models and to develop the ability to design new ones for diverse problems arising in different domains. The focus is on understanding their essential features and on making robust predictions. In essence, we will work with exact mathematical frameworks, but we will not rigorously develop all the underlying theory.

1.1.2 Importance of diffusion models

The techniques introduced in this course are highly versatile. They can be used to describe the shape of shells, species dynamics within a microbiome, or spatial patterns in ecosystems. They are also applicable to studying heterogeneity in disordered solid materials.

Among these, **diffusion models** play a central role. We will examine them in several ways because of their fundamental importance. Brownian motion was first observed by Robert Brown in 1828, but it was Einstein in 1905 who provided the theoretical framework by modeling it through the diffusion equation. This theory lies on **stochastic processes** and forms the backbone of all diffusion models.

Brownian motion – also known as the **Wiener process** in the mathematical framework – serves as the building block for more general stochastic processes, which can be formulated through *Stochastic Differential Equations* both in classical and quantum mechanics.

1.1.3 Example: stochastic resonance

An instructive example of stochastic effects is stochastic resonance (SR). Consider a classical oscillator driven by a periodic force with frequency ω and subject to damping:

$$m\frac{d^2x}{dt^2} = F_0 \sin(\omega t) - kx - c\frac{dx}{dt}$$

This linear ODE has the solution:

$$x(t) = \frac{F_0}{m\sqrt{(2\omega\omega_0\zeta)^2 + (\omega_0^2 - \omega^2)^2}} \sin(\omega t + \varphi)$$

where ω_0 is the natural frequency. We know that when $\omega \sim \omega_0$, resonance occurs.

Stochastic resonance exploits this phenomenon: a weak signal, normally undetectable by a sensor, can be amplified by adding white noise, which contains a broad spectrum of frequencies (this is why it is called white). The components of the noise that match the signal's frequencies resonate with it, selectively boosting the original signal while leaving the rest of the noise unamplified. As a result, the signal-to-noise ratio increases, making the original signal more prominent.

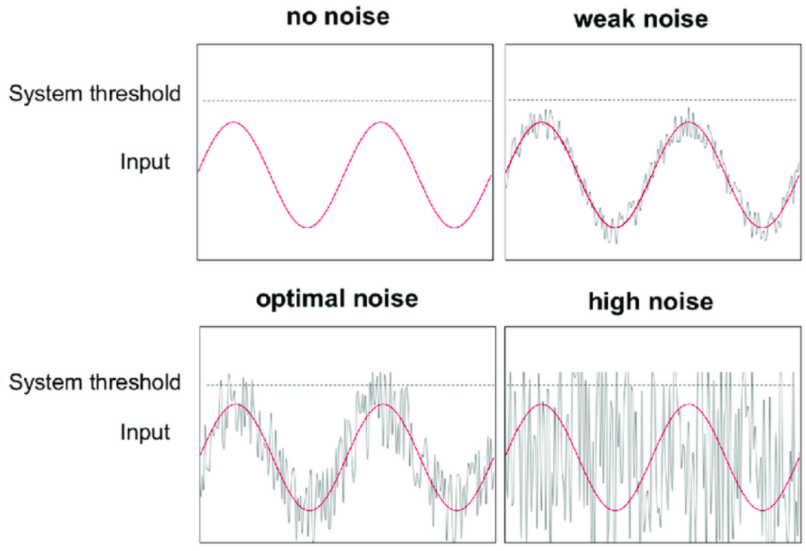


Figure 1.1

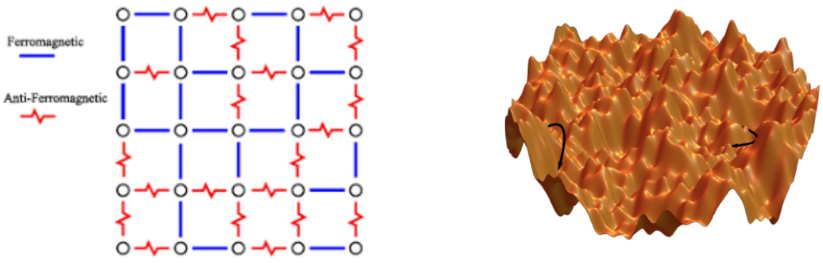
Importantly, adding noise does not merely produce a shaky version of a smooth signal. Noise can fundamentally alter system behavior, introducing new effects that do not exist without it.

1.1.4 Example: spin glasses

Another example involves spin glasses, which are **disordered systems**. In condensed matter physics, a spin glass models a special type of magnet, typically using the Ising model. Magnetic spins represent the orientation of atomic magnetic moments in space.

In ferromagnets, spins align uniformly, whereas in **spin glasses** they are arranged in a disordered, frozen pattern known as **quenched noise**. The consequence of this is an effect called *frustration*, in which spins cannot decide whether to align or anti-align, resulting in multiple local minima in the free energy landscape.

This frustrated structure is highly relevant to a variety of fields – such as optimization problems in computer science, neural network dynamics, social systems, materials with impurities, and learning algorithms – because it reflects the presence of many competing stable states.



(a) (b)

Figure 1.2

1.1.5 Why build models if the physics is known?

There are several reasons why modeling remains necessary even when the fundamental physics is known. Often, the exact equations are not known, as in the case of complex microbial interactions. In such scenarios, models provide a way to approximate the dynamics. Sometimes, the known equations are defined at a scale too coarse to be useful – for instance, applying QCD at the molecular level or Newtonian mechanics to interacting gases is meaningless. Finally, *emergent phenomena* can arise in complex systems, revealing new behaviors not evident from microscopic equations.

Depending on the problem, different modeling strategies are appropriate:

- **Microscopic approach**: all degrees of freedom are used.
- **Mesoscopic approach**: uses stochastic description such as Langevin and diffusion eqs.
- **Macroscopic approach**: relies on deterministic ODEs or PDEs, and mean-field approximations.

As physicists, the key challenge is to identify which features of a complex model are essential and which can be discarded, and to choose the appropriate technical tools accordingly. Typical guiding questions include: do we really need microscopic details? and which ingredients are essential for a macroscopic description?

Chapter 2

Revision stuff

Ref.: J. Zinn-Justin - Quantum Field Theory and Critical Phenomena, 1st Chapter.

2.1 Gaussian integrals

2.1.1 Simple form

Consider the Gaussian distribution (PDF):

$$p(x) = ce^{-\frac{ax^2}{2}}, \quad a > 0.$$

normalized on $(-\infty, +\infty)$:

$$\int_{-\infty}^{\infty} p(x)\, dx = 1 \quad \Rightarrow \quad c\int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}}\, dx = 1$$

Using the standard Gaussian integral,

$$\int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}}\, dx = \sqrt{\frac{2\pi}{a}} \tag{2.1}$$

the normalization constant is $c = \sqrt{a/(2\pi)}$.

2.1.2 General form

A more general form of the Gaussian integral is:

$$\int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}+bx}\, dx$$

To calculate this integral, we use a change of variables. First of all, the exponent attains its maximum at

$$\frac{d}{dx}\left(-\frac{ax^2}{2} + bx\right) = -ax + b = 0 \quad \Rightarrow \quad x = \frac{b}{a}$$

With the shift $y = x - \frac{b}{a}$,

$$-\frac{ax^2}{2} + bx = -\frac{a}{2}\left(y + \frac{b}{a}\right)^2 + b\left(y + \frac{b}{a}\right) = -\frac{ay^2}{2} + \frac{b^2}{2a}$$

Thus, we have:

$$\int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}+bx}\,dx = \int_{-\infty}^{\infty} e^{-\frac{ay^2}{2}+\frac{b^2}{2a}}\,dy = \sqrt{\frac{2\pi}{a}}e^{\frac{b^2}{2a}}, \quad a>0,\ b\in\mathbb{R} \tag{2.2}$$

where in the last equality we used the result of Equation 2.1. Now, let's calculate the integral 2.2 when $b = it$, with $t \in \mathbb{R}$. To do this, let's define:

$$\varphi(t) = \int_{-\infty}^{\infty} dx\, e^{ixt}\sqrt{\frac{a}{2\pi}}e^{-\frac{ax^2}{2}}$$

To compute this integral, we have to compute the derivative of $\varphi(t)$. Since it is in the form $\int f(x,t)dx$, this can happen under the following assumptions:

$$\frac{d}{dt}\int f(x,t)\,dx = \int \frac{\partial f(x,t)}{\partial t}\,dx \iff \begin{cases} f \text{ and } \partial_t f \text{ continuous} \\ f \text{ and } \partial_t f \text{ uniformly bounded} \end{cases}$$

The second condition is mathematically translated as:

$$\begin{cases} |f(x,t)| < A(x) \\ |\partial_t f(x,t)| < B(x) \end{cases} \qquad \text{where } \begin{cases} \int A(x)\,dx < \infty \\ \int B(x)\,dx < \infty \end{cases}$$

By assuming these conditions, we have:

$$\varphi'(t) = i\sqrt{\frac{a}{2\pi}}\int_{-\infty}^{\infty} xe^{-\frac{ax^2}{2}}e^{ixt}\,dx = -\frac{i}{\sqrt{2\pi a}}\int_{-\infty}^{\infty} dx\, e^{ixt}\frac{d}{dx}e^{-\frac{ax^2}{2}}$$

where we used $\frac{d}{dx}e^{-\frac{ax^2}{2}} = -axe^{-\frac{ax^2}{2}}$. Integrating by parts, we obtain:

$$\varphi'(t) = -\frac{t}{\sqrt{2\pi a}}\int_{-\infty}^{\infty} dx\, e^{ixt}e^{-\frac{ax^2}{2}} = -\frac{t}{a}\varphi(t)$$

Obtaining the linear differential equation:

$$\varphi'(t) = -\frac{t}{a}\varphi(t) \quad \Rightarrow \quad \varphi(t) = ce^{-\frac{t^2}{2a}}$$

Since $\varphi(0) = 1$, we have $c = 1$. Thus,

$$\int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}+bx}\,dx = \sqrt{\frac{2\pi}{a}}e^{\frac{b^2}{2a}}, \quad a>0,\ b\in\mathbb{C} \tag{2.3}$$

2.1.3 Hubbard-Stratonovich transformation

Equation 2.3 can be rewritten in the following manner:

$$e^{\frac{b^2}{2a}} = \sqrt{\frac{a}{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{ax^2}{2}+bx}\,dx \tag{2.4}$$

which is called **Hubbard-Stratonovich transformation**. This is useful to convert a quadratic expression in b into a linear one in b. Typically, it is used when b is a matrix, a field, or a set of vectors.

2.2 Characteristic functions

The characteristic function plays a central role in probability theory, especially when working with PDFs and proving major results such as the central limit theorem.

For a continuous random variable x with distribution $p(x)$, the **characteristic function** is defined as:

$$\varphi(k) = \int e^{ikx} p(x)\, dx \equiv \langle e^{ikx} \rangle_p \tag{2.5}$$

where the $\langle \cdot \rangle_p$ denotes the expectation value with respect to $p(x)$.

More generally, for any function $f(x)$,

$$\langle f \rangle_p \equiv \int f(x) p(x)\, dx$$

By differentiating $\varphi(k)$ with respect to k, we can extract successive moments of $p(x)$. For instance,

$$\frac{d\varphi}{dk} = \int ixe^{ikx} p(x)\, dx \quad \Rightarrow \quad -i\frac{d\varphi}{dk}\bigg|_{k=0} = \int xp(x)\, dx \equiv \langle x \rangle$$

In general,

$$(-i)^n \frac{d^n\varphi}{dk^n}\bigg|_{k=0} = \int x^n p(x)\, dx \equiv \langle x^n \rangle, \quad n = 1, 2, 3, \ldots \tag{2.6}$$

which correspond to the **n-th moments** of the distribution.

We can derive this starting from the Taylor expansion of e^{ikx}:

$$e^{ikx} = \sum_{n=0}^{\infty} \frac{(ikx)^n}{n!} \quad \Rightarrow \quad \varphi(k) = \langle e^{ikx} \rangle = \sum_{j=0}^{\infty} \frac{i^j k^j}{j!} \langle x^j \rangle$$

Differentiating n times with respect to k, all terms with power of k smaller than n will be zero. Thus:

$$\frac{d^n\varphi}{dk^n} = \sum_{j=n}^{\infty} \frac{i^j}{j!} j(j-1)\ldots(j-n+1) k^{j-n} \langle x^j \rangle = \sum_{j=n}^{\infty} \frac{i^j k^{j-n}}{(j-n)!} \langle x^j \rangle$$

Evaluating at $k = 0$, we have:

$$\frac{d^n\varphi}{dk^n}\bigg|_{k=0} = i^n \langle x^n \rangle \quad \Rightarrow \quad \langle x^n \rangle = (-i)^n \frac{d^n\varphi}{dk^n}\bigg|_{k=0}$$

2.2.1 Characteristic function of the Gaussian distribution

Consider the Gaussian distribution:

$$p(x) = \sqrt{\frac{a}{2\pi}} e^{-\frac{ax^2}{2}}$$

Introducing the variance $\sigma^2 = 1/a$, this can be rewritten as:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

The characteristic function can be derived directly from Equation 2.5:

$$\varphi(k) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} e^{ikx}\, dx \tag{2.7}$$

Using Hubbard-Stratonovich transformation in Equation 2.4, we find:

$$\varphi(k) = e^{-\frac{k^2\sigma^2}{2}}$$

Therefore, both $p(x)$ and $\varphi(k)$ share the same functional form. This self-similarity is a distinctive property of the Gaussian function, arising from its symmetry and the fact that the Fourier transform of a Gaussian is another Gaussian.

Exercise 2.2.1. *Show that when the mean of the Gaussian is μ, we have:*

$$\varphi(k) = e^{ik\mu - \frac{\sigma^2 k^2}{2}}$$

Exercise 2.2.2. *Calculate the characteristic function of the following distributions.*

- *Uniform distribution:*
$$p(x) = U([a, b])$$
- *Gamma distribution:*
$$p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad \alpha > 0,\ \beta > 0$$

It is useful to calculate the moments $\langle x^n \rangle$ for $n = 1, 2, 3, \ldots$.

- **case $n = 1$**
From Equation 2.6, we have:
$$\langle x \rangle = -i \frac{d}{dk} e^{-\frac{\sigma^2 k^2}{2}} \Big|_{k=0} = 0$$
- **case n odd**
This vanishes by symmetry, since the integrand is an odd function:
$$\langle x^n \rangle = 0, \quad n = 2k + 1,\ k \in \mathbb{N}^+$$
- **case n even**
In this case, we can either substitute explicit values or differentiate both sides of Equation 2.1 with respect to a. For instance, for the case $n = 4$ we have:
$$\int_{-\infty}^{\infty} x^4 e^{-\frac{ax^2}{2}}\, dx = \frac{3\sqrt{2\pi}}{a^{5/2}}$$
In general, we can write:
$$\int_{-\infty}^{\infty} x^n e^{-\frac{ax^2}{2}}\, dx = \frac{(n-1)(n-3)\cdot \ldots \cdot 5 \cdot 3 \cdot 1\sqrt{2\pi}}{a^{(n+1)/2}} = \frac{(n-1)!!\sqrt{2\pi}}{a^{(n+1)/2}}$$
where !! is the double factorial. In general, we can write:
$$\langle x^n \rangle = \frac{(n-1)!!}{a^{n/2}} = (n-1)!!\sigma^n \tag{2.8}$$

As an exercise, Derive the general expression fo $\langle x^n \rangle$ as a function of σ for n even.

2.3 Multidimensional Gaussian integrals

2.3.1 Simple form

We want to calculate:

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\vec{x}^T A\vec{x}}\, d^n x \tag{2.9}$$

with $\vec{x} = (x_1, \ldots, x_n)$ and A symmetric positive definite matrix. Since A is symmetric, it is also diagonalizable, meaning that there exists an orthogonal matrix O ($O^T O = OO^T = \mathbb{1}$) such that

$$\Lambda = OAO^T \quad \text{where } \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

where λ_i are the eigenvalues. Moreover, positive definite means that the eigenvalues are strictly positive: $\lambda_i > 0$. Thus, with the linear change of variables $\vec{y} = O\vec{x}$, it follows:

$$\vec{x}^T A\vec{x} = \vec{x}^T O^T \Lambda O\vec{x} = \vec{y}^T \Lambda\vec{y}$$

Thus, we can rewrite the integral 2.9 as:

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\vec{y}^T\Lambda\vec{y}} |\det J|\, d^n y$$

where $\det J$ is the determinant of the Jacobian:

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_n} \end{pmatrix}$$

However, it is possible to show that $|\det J| = 1$. In fact, since $\vec{y} = O\vec{x}$, we have $\vec{x} = O^T\vec{y}$. Thus,

$$|\det J| = \left|\left|\frac{\partial\vec{x}}{\partial\vec{y}}\right|\right| = |\det O^T| = 1$$

(for an orthogonal matrix $\det O = \pm 1$, show this as exercise).

Expanding the exponent:

$$\vec{y}^T\Lambda\vec{y} = \sum_{i,j} y_i \Lambda_{ij} y_j = \sum_{i,j} y_i \lambda_i \delta_{i,j} y_j = \sum_i \lambda_i y_i^2$$

and the integral factorizes

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\sum_i \lambda_i y_i^2}\, d^n y = \prod_{i=1}^n \int_{-\infty}^{\infty} e^{-\frac{1}{2}\lambda_i y_i^2}\, dy_i = \prod_{i=1}^n \sqrt{\frac{2\pi}{\lambda_i}} = \frac{(2\pi)^{n/2}}{\sqrt{\lambda_i \cdot \ldots \cdot \lambda_n}}$$

Since $\prod_i \lambda_i = \det\Lambda$ and $\det A = \det(O^T\Lambda O) = \det(\Lambda)\det(O)^2 = \det\Lambda$, we can write:

$$Z(A) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\vec{x}^T A\vec{x}}\, d^n x = \sqrt{\frac{(2\pi)^n}{\det A}} \tag{2.10}$$

which is the **multivariate normalized Gaussian**. This formula is also a practical tool for evaluating determinants.

Example 2.3.1

Show that:

$$\int dx_1 \int dx_2 \, e^{-\frac{3}{2}(x_1^2+x_2^2)+x_1x_2} = \frac{\pi}{\sqrt{2}}$$

We can write the exponent in the form:

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} -\frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{3}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Here $\left(\begin{smallmatrix} 3 & -1 \\ -1 & 3 \end{smallmatrix}\right) = A$. By applying Equation 2.10 with $n = 2$, we have:

$$Z(A) = \sqrt{\frac{(2\pi)^2}{8}} = \frac{\pi}{\sqrt{2}}$$

Example 2.3.2

Let

$$p(x, y) = \frac{\sqrt{\det A}}{2\pi} e^{-\frac{1}{2}(a_{11}x^2+2a_{12}xy+a_{22}y^2)}$$

with $A = \left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right)$, where $a_{11} > 0$, $a_{22} > 0$ and $a_{12} = a_{21}$. Show that the marginal $q(x) = \int p(x, y)\, dy$ remains a Gaussian and find the variance of x.

Note that:

$$\begin{aligned} q(x) &= \frac{\sqrt{\det A}}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(a_{11}x^2+2a_{12}xy+a_{22}y^2)}\, dy = \\ &= \frac{\sqrt{\det A}}{2\pi} e^{-\frac{1}{2}a_{11}x^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(a_{22}y^2-2a_{12}xy)}\, dy \end{aligned}$$

By using Equation 2.4 with $a = a_{22}$ and $b = -a_{12}x$:

$$q(x) = \frac{\sqrt{\det A}}{2\pi} e^{-\frac{1}{2}a_{11}x^2} \cdot \sqrt{\frac{2\pi}{a_{22}}} e^{\frac{a_{12}^2x^2}{2a_{22}}} = \sqrt{\frac{\det A}{2\pi a_{22}}} e^{-\frac{1}{2}\left(a_{11}-\frac{a_{12}^2}{a_{22}}\right)x^2} \tag{2.11}$$

It is important to verify that this is normalized in order to this being a Gaussian. We have that:

$$a_{11} - \frac{a_{12}^2}{a_{22}} = \frac{a_{11}a_{22} - a_{12}^2}{a_{22}} = \frac{\det A}{a_{22}}$$

By using Equation 2.1, we have:

$$\int_{-\infty}^{\infty} q(x)\, dx = \sqrt{\frac{\det A}{2\pi a_{22}}} \cdot \sqrt{\frac{2\pi a_{22}}{\det A}} = 1$$

From Equation 2.11, it is evident that $\langle x \rangle = 0$. While to compute $\langle x^2 \rangle$, we use Equation 2.8, from which follows:

$$\mathrm{Var}\,(x) = \langle x^2 \rangle = \frac{(2-1)!!}{a} = \frac{a_{22}}{\det A}$$

The key point of this example is that, because A is not diagonal, the Gaussian variables are correlated. In general, marginalizing over some variables changes the functional form of the distribution. However, for multivariate Gaussian distributions, the marginal distribution is still Gaussian.

2.3.2 General form

We now generalize the result of Equation 2.10 by adding a linear term:

$$Z(A, \vec{b}) = \int d^n x \, e^{-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b}}$$

where $\vec{b} \in \mathbb{C}^n$. To compute this, we use the same strategy of shifting to the stationary point. Thus, we take:

$$\vec{\nabla}_x \left(-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b} \right) = -A\vec{x} + \vec{b} = 0 \quad \Rightarrow \quad \vec{x} = A^{-1}\vec{b}$$

where we assume $\det A \neq 0$. We introduce the following change of variables:

$$\vec{y} = \vec{x} - A^{-1}\vec{b}$$

Thus,

$$-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b} = -\frac{1}{2}\vec{y}^T A \vec{y} + \frac{\vec{b}^T A^{-1} \vec{b}}{2}$$

where we used the commutativity $\vec{y}^T \vec{b} = \vec{b}^T \vec{y}$. Hence, we get:

$$Z(A, \vec{b}) = \int_{\mathbb{R}^n} d\vec{y} \, e^{-\frac{1}{2}\vec{y}^T A \vec{y} + \frac{\vec{b}^T A^{-1} \vec{b}}{2}} = e^{\frac{\vec{b}^T A^{-1} \vec{b}}{2}} \int_{\mathbb{R}^n} d\vec{y} \, e^{-\frac{1}{2}\vec{y}^T A \vec{y}} = e^{\frac{\vec{b}^T A^{-1} \vec{b}}{2}} Z(A, 0)$$

where $Z(A, 0)$ is known from Equation 2.10. Finally, we have:

$$Z(A, \vec{b}) = \int d^n x \, e^{-\frac{1}{2}\vec{x}^T A \vec{x} + \vec{x}^T \cdot \vec{b}} = \frac{(2\pi)^{n/2}}{\sqrt{\det A}} e^{\frac{\vec{b}^T A^{-1} \vec{b}}{2}} \tag{2.12}$$

2.3.3 Characteristic function

Equation 2.12 allows us to find the characteristic function of the multivariate Gaussian distribution. From Equation 2.10 the normalized Gaussian is:

$$p(\vec{x}) = \sqrt{\frac{\det A}{(2\pi)^n}} e^{-\frac{1}{2}\vec{x}^T A \vec{x}}$$

Thus, the characteristic function (Equation 2.5) is:

$$\varphi(\vec{k}) = \int_{\mathbb{R}^n} d^n x \, e^{i\vec{k} \cdot \vec{x}} p(\vec{x}) = e^{-\frac{\vec{k}^T A^{-1} \vec{k}}{2}} \tag{2.13}$$

2.4 Correlation function

We now examine the meaning of the matrix A^{-1}.

As shown in Equation 2.13, the characteristic function of the multivariate Gaussian distribution allows us to compute its moments by differentiation. Since we are dealing with several variables, we can select any subset of them and consider moments of the form $\langle x_i x_j \dots x_\ell \rangle$, where each variable may appear more than once. By differentiating the characteristic function with respect to the relevant components of $\vec{k}$ and setting $\vec{k} = 0$, we obtain the ***s*-point correlation function**:

$$\langle x_i x_j \dots x_\ell \rangle = (-i)^s \frac{\partial}{\partial k_i} \frac{\partial}{\partial k_j} \dots \frac{\partial}{\partial k_\ell} \varphi(\vec{k}) \Big|_{\vec{k}=0} = \int d^s x \, x_i x_j \dots x_\ell p(\vec{x}) \tag{2.14}$$

where s is the total number of variables, counting repetitions.

Let's compute the 2-point correlation function of the Gaussian distribution. Using Equation 2.14 together with Equation 2.13, we find:

$$\langle x_i x_j \rangle = (-i)^2 \frac{\partial}{\partial k_i} \frac{\partial}{\partial k_j} e^{-\frac{\vec{k}^T A^{-1} \vec{k}}{2}} \Big|_{\vec{k}=0} = (A^{-1})_{ij} \tag{2.15}$$

This result is very important because it gives a clear probabilistic interpretation of the matrix A: its inverse is precisely the matrix of 2-point correlation functions between two Gaussian random variables. In particular, if A is diagonal, the variables are uncorrelated, since all off-diagonal correlation terms vanish.

Furthermore, the symmetry of the correlation function implies that A must be symmetric as well, because the order in which we consider the variables cannot affect the value of the correlation. This observation justifies the initial assumption of symmetry for A.

Example 2.4.1

From 2.3.1, one can verify that:

$$\langle x_1^2 \rangle = \langle x_2^2 \rangle = \frac{3}{8} \qquad \langle x_1 x_2 \rangle = \langle x_2 x_1 \rangle = \frac{1}{8}$$

To check this explicitly, let's compute A^{-1}:

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \quad \Rightarrow \quad A^{-1} = \frac{1}{8} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

By applying Equation 2.15, we recover exactly the expected moments.

Thus, the Gaussian integral can be calculated exactly but also interpreted exactly in the sense that it gives us the distribution of two Gaussian random variables with mean zero and correlation given by the matrix A^{-1}

In summary, Gaussian integrals can not only be evaluated exactly, but they also can be interpreted exactly: they describe the joint distribution of Gaussian random variables with zero mean and a correlation structure encoded in the matrix A^{-1}.

Remark. Notice that, because of symmetry, the s-point correlation function for s-variable (s is odd) is zero (the Gaussian remains unchanged for $\vec{x} \to -\vec{x}$)

2.5 Wick's probability theorem

When we want to calculate higher-order moments such as $\langle x_i x_j \dots x_\ell \rangle$, do we really need to take all the derivatives of the characteristic function? The answer is no. If the variables are Gaussian, then we can use the following

Theorem 2.5.1 (Wick's theorem)**.** *For any set of zero-mean Gaussian random variables, every correlation involving an even number of variables can be expressed as a sum of products of 2-point correlation functions, which are given by the entries of* A^{-1}*.*

For instance, a 4-point correlation function can be decomposed as follows:

$$\langle x_a x_b x_c x_d \rangle = \underbrace{\langle x_a x_b \rangle}_{(A^{-1})_{ab}} \underbrace{\langle x_c x_d \rangle}_{(A^{-1})_{cd}} + \langle x_a x_c \rangle \langle x_b x_d \rangle + \langle x_a x_d \rangle \langle x_b x_c \rangle$$

In general, for an even number s of variables we have:

$$\langle x_i x_j \dots x_n x_m \rangle = \sum_p (A^{-1})_{i_p j_p} \dots (A^{-1})_{n_p m_p} \tag{2.16}$$

where the sum runs over all possible pairings of the s indexes, i.e., over all ways of grouping the set $\{i, j, \dots, n, m\}$ into pairs. Pairings are counted even if two indices are equal. Equivalently, it can be written as:

$$\langle x_1 x_2 \dots x_n \rangle = \sum_{p \in \mathcal{P}_n^2} \prod_{\{i,j\} \in \mathcal{P}} \langle x_i x_j \rangle$$

where $\mathcal{P}$ are the pairings.

Example 2.5.1

From 2.3.1 compute:

$$\langle x_1^2 x_2^2 \rangle \qquad \langle x_1^4 \rangle \qquad \langle x_1 x_2^2 \rangle$$

Using Theorem 2.5.1:

$$\begin{aligned} \langle x_1^2 x_2^2 \rangle &= \langle x_1^2 \rangle \langle x_2^2 \rangle + \langle x_1 x_2 \rangle \langle x_1 x_2 \rangle + \langle x_1 x_2 \rangle \langle x_1 x_2 \rangle = \\ &= (A^{-1})_{11} (A^{-1})_{22} + (A^{-1})_{12} (A^{-1})_{12} + (A^{-1})_{12} (A^{-1})_{12} = \\ &= \frac{3}{8} \cdot \frac{3}{8} + \frac{1}{8} \cdot \frac{1}{8} + \frac{1}{8} \cdot \frac{1}{8} = \frac{11}{64} \end{aligned}$$

Similarly,

$$\langle x_1^4 \rangle = 3 \cdot \frac{3}{8} \cdot \frac{3}{8} = \frac{27}{64}$$

Finally,

$$\langle x_1 x_2^2 \rangle = 0$$

by symmetry, since $s = 3$ is odd.

In summary, we began by considering Gaussian integrals. Given a Gaussian integral, the first step is to identify the corresponding matrix A. To proceed with the computation,

we must then verify that A is positive definite. Once this is established, we can invert A and carry out the calculation exactly with Theorem 2.5.1.

In statistical mechanics, characteristic functions are known as *partition functions*. Once the partition function of a system is known, the system is, in principle, fully determined. The same idea applies in probability theory: knowing the characteristic function of a stochastic process essentially gives us complete information about its probabilistic structure.

Chapter 3

Law of Large Numbers and Central Limit Theorem

We introduce now the concept of i.i.d. samples and the sum of i.i.d. variables. This will help us to prove two fundamental theorems:

- Law of large numbers (weak and stronge sense)
- Central limit theorem.

3.1 Independent and identically distributed variables

Consider a joint probability density function $p(x_1, x_2)$. If it factorizes as:

$$p(x_1, x_2) = p_1(x_1)p_2(x_2)$$

(where in general $p_1 \neq p_2$), then the two variables are said to be independent. If, in addition, $p_1 = p_2$, we say that the two random variables are **independent and identically distributed** (i.i.d.).

Given two i.i.d. random variables, can we compute the distribution of their sum? In other words, if $x_1 \sim q(x)$ ($\sim$ means is drawn) and $x_2 \sim q(x)$ – that is x_1 and x_2 are uncorrelated – what is the distribution of $x = x_1 + x_2$? A very common mistake is to take the sum of the distributions of the two random variables.

3.1.1 Distribution of sum of i.i.d. variables

Given x_1, x_2 i.i.d., we want to compute the distribution of the sum $x_1 + x_2$, denoted as $p(x)$. In this case, we must consider all possible combinations of x_1 and x_2 that yield to a given value x. To do this, we use a Dirac delta to enforce the constraint $x = x_1 + x_2$ while integrating over all possible x_1, x_2:

$$\begin{aligned} p(x) \equiv \langle \delta(x - x_1 - x_2) \rangle &= \int \delta(x - (x_1 + x_2)) p(x_1, x_2)\, dx_1\, dx_2 = \\ &= \int \delta(x - x_1 - x_2) q(x_1) q(x_2)\, dx_1 dx_2 = \\ &= \int_0^x q(x - y) q(y)\, dy \end{aligned} \tag{3.1}$$

In general, the domain is $A \cap (x - A)$ where $y \in A \cap (x - A)$, but we let $-\infty$ and $+\infty$ and let the zero of $q(y)$ do the trick. Thus, the distribution of the sum is exactly the **convolution** of $q(x_1)$ and $q(x_2)$.

Exercise 3.1.1. *Compute the distribution of $x = x_0 + x_1$ where x_0 and x_1 are i.i.d. drawn from:*

1. *a uniform distribution $\mathcal{U}([0, 1])$*
2. *a Gaussian distribution $\mathcal{N}([\mu, \sigma])$*
3. *a exponential distribution $\mathcal{E}(\lambda) = \lambda e^{-\lambda x}$*

Exercise 3.1.2. *Calculate the distribution of the product $x = x_1 x_2$, where x_1, x_2 are positive i.i.d.*

Exercise 3.1.3. *What is the distribution of the sum if they are independent but not identically distributed?*

3.1.2 Characteristic function of sum of i.i.d. variables

We can compute the characteristic function of $p(x)$:

$$\varphi(k) \equiv \left\langle e^{ikx} \right\rangle = \int e^{ikx} p(x)\, dx = \int dx\, e^{ikx} \delta(x - x_1 - x_2) q(x_1) q(x_2)\, dx_1\, dx_2 = \\ = \int e^{ik(x_1 + x_2)} q(x_1) q(x_2)\, dx_1 dx_2$$

where we used the Dirac delta. Since the variables are independent, the double integral factorizes:

$$\varphi(k) = \int q(x_1) e^{ikx_1}\, dx_1 \int q(x_2) e^{ikx_2}\, dx_2 = [\varphi_j(k)]^2 \tag{3.2}$$

where $\varphi_j(k) = \int q(x_j) e^{ikx_j}\, dx_j$ is the characteristic function of each variable.

Thus, the characteristic function of the sum is simply the product of the characteristic functions of the individual variables if these are i.i.d. This means that, instead of calculating the distribution itself, one can compute the characteristic function of the underlying variables and then obtain the distribution by inverse Fourier transform if needed.

Exercise 3.1.4. *What is the characteristic function of the distribution of the sum of n i.i.d. variables.*

3.2 Law of large numbers

Ref.: Grimmett and Stirzaker, Probability and Random Processes.

The core idea behind the law of large numbers is simple yet profound. Intuitively, consider a sequence of random variables obtained, for instance, by flipping a coin. Each flip yields either a head or a tail, and the outcome of each individual toss does not converge – every trial gives a different result.

However, if we take the average of the results, that is, we sum all observed values and divide by their number, this average converges to 1/2.

In probability theory and stochastic processes, one is often interested in reducing or controlling random fluctuations, i.e., noise. A single coin flip has no predictive power, but averaging over many flips, we obtain a deterministic and stable result: the average converges to a fixed number.

The fact that the average (or the sum) of an arbitrary number of random variables can converge to a deterministic constant, rather than remain a random quantity, is far from trivial. With appropriate averaging, randomness can be tamed.

There exist several types of convergence in stochastic processes. Depending on which notion of convergence is used, one can prove different forms of the law of large numbers. Here, we will focus only on weak and strong convergence.

Convergence in the weak sense (or in distribution) means that the distribution of the random variables converges, not the variables themselves. Equivalently, it means that the characteristic function of the random variable converges to a limit. This is a relatively weak form of convergence, since integrating over the distribution loses some information about individual realizations.

The strong law, instead, states that not only the distribution or the characteristic function, but also the random variable itself converges to a number. This means that for every realization of the process, the sample average will converge almost surely to the same constant value.

3.2.1 The weak law of large numbers

Theorem 3.2.1 (Weak law of large numbers)**.** *Let $x_1, x_2, \dots, x_n$ be i.i.d. random variables with PDF $q(x)$ and characteristic function $\varphi_1(k)$. Assume that the mean of each x_i is $\mu \equiv \int dx\, q(x)x < \infty$.*

Consider the sample mean

$$x = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

Then, as $n \to \infty$, x converges in probability to μ, that is for every $\varepsilon > 0$

$$\lim_{n\to\infty} P(|x-\mu| > \varepsilon) = 0$$

(Grimmett, p 193.)

Proof. Let $\varphi(k)$ be the characteristic function sample mean x:

$$\begin{aligned}\varphi(k) = \left\langle e^{ikx}\right\rangle = \left\langle e^{ik\frac{1}{n}\sum_i x_i}\right\rangle = \int e^{\frac{ik}{n}\sum_i x_i} q(x_1)\dots q(x_n)\, dx_1 \dots dx_n = \\ = \left(\int e^{\frac{ik}{n}x_1} q(x_1)\, dx_1\right)\dots\left(\int e^{\frac{ik}{n}x_n} q(x_n)\, dx_n\right) = \\ = \left[\varphi_1\left(\frac{k}{n}\right)\right]^n\end{aligned} \tag{3.3}$$

where we factorized the integrals since the random variables are independent and we used the characteristic function of the single variable, which is the same for each one. Moreover, since $n \to \infty$, we expand $\varphi_1(k/n)$ by using Taylor:

$$\varphi_1\left(\frac{k}{n}\right) = \int e^{i\frac{k}{n}x} q(x)\, dx = 1 + \frac{ik}{n}\langle x\rangle + O\left(\frac{1}{n^2}\right)$$

Therefore, from Equation 3.3 we have:

$$\left(1 + \frac{ik}{n}\langle x\rangle + \dots\right)^n \quad \xrightarrow{n\to\infty} \quad e^{i\mu k}$$

Since

$$e^{ik\mu} = \int_{p(x)} \delta(x-\mu)e^{ikx}\,dx$$

the characteristic function converges to that of a Dirac delta distribution centered at μ. Hence, the sample mean x converges in distribution to μ. □

It is indeed possible to prove not only convergence via the characteristic function, but that the empirical average itself converges to μ. This is the strong law of large numbers. To state it properly, we recall two notions of convergence.

Definition 3.2.2. A sequence of random variables x_n **converges in probability** to a random variable x if, for every $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|x_n - x| > \epsilon) = 0$$

Intuitively, the probability that x_n deviates from x more than ϵ goes to zero as n grows. This does not imply almost sure convergence in general.

Definition 3.2.3. A sequence x_n of random variables **converges almost sure** to a random variable x if

$$\mathbb{P}\left(\lim_{n\to\infty} x_n = x\right) = 1$$

Almost-sure convergence implies convergence in probability, but not conversely.

Theorem 3.2.4 (Strong law of large numbers)**.** *Let $x_1 \dots x_n$ be i.i.d. random variables with finite mean μ. Then, the empirical average*

$$\frac{1}{n}\sum_i x_i$$

converges to μ almost surely as $n \to \infty$ (Grimmett, page 329).

This result is stronger than the weak law: it asserts that for almost every sample path, the observed averages settle down to μ.

When the x_i have finite variance σ^2, two key factors hold:

1. The fluctuations of the sum around $n\mu$ are of order $\sqrt{n}\sigma$ (in probability):

$$\sum_i x_i - \mu n \simeq O_{\mathbb{P}}(\sqrt{n})$$

2. The normalized sum:

$$Y = \frac{\sum_i x_i - \mu n}{\sqrt{n}}$$

which is centered and normalized, approaches a Gaussian distribution as $n \to \infty$ *irrespective* of the PDF of x_i.

These two claims imply that probability distributions possess a kind of universal attractor to which they converge. This attractor – the Gaussian distribution – does not depend on the fine details of the original distribution, provided that the random variables have a finite mean and variance.

These claims are at the core of the following:

Theorem 3.2.5 (Central limit)**.** *Let $x_1, \ldots, x_n$ be a sequence of i.i.d. random variables with finite mean μ and finite (non-zero) variance σ^2. Then, the PDF of*

$$Y_n = \frac{\sum_{i=1}^{n} x_i - \mu n}{\sqrt{n}\sigma} \tag{3.4}$$

converges, as $n \to \infty$, in distribution to a standard normal random variable:

$$Y_n \xrightarrow{n\to\infty} \mathcal{N}(0,1)$$

where $N(0,1)$ denotes the Gaussian distribution with mean 0 and variance 1.

Obs. The expected value of Y_n is zero:

$$\langle Y_n \rangle = \frac{\langle \sum_i x_i \rangle - \mu n}{\sqrt{n}\sigma} = 0$$

Obs. By definition, Y_n is centered at 0, and its variance does not depend on n:

$$\mathrm{Var}(Y_n) = 1$$

Proof. Assume that each random variable x_i has a PDF $q(x)$ with characteristic function $\varphi_1(k)$. Let $\varphi(k)$ be the characteristic function of Y_n. Thus, we get

$$\begin{aligned} \varphi(k) = \left\langle e^{ikY_n} \right\rangle &= \int e^{ik\frac{\sum_i x_i - \mu n}{\sqrt{n}\sigma}} q(x_1)\ldots q(x_n)\, dx_1 \ldots dx_n = \\ &= e^{-\frac{ik\mu\sqrt{n}}{\sigma}} \left(\int e^{ikx/\sqrt{n}\sigma} q(x)\, dx \right)^n = e^{-ik\mu\sqrt{n}/\sigma} \varphi_1^n \left(\frac{k}{\sqrt{n}\sigma} \right) \end{aligned} \tag{3.5}$$

Since the variables are i.i.d., the integrals factorize.

As in Theorem 3.2.1, for large n, we can expand φ_1 around zero using a Taylor series:

$$\varphi_1 \left(\frac{k}{\sqrt{n}\sigma} \right) = 1 + \frac{ik}{\sqrt{n}\sigma} \langle x \rangle - \frac{k^2}{2n\sigma^2} \langle x^2 \rangle + O(n^{-3/2}) = e^{\frac{ik}{\sqrt{n}\sigma}\mu - \frac{k^2}{2n}}$$

Show this as exercise.

Substituting in Equation 3.5, we get:

$$\varphi(k) = e^{-\frac{ik\mu}{\sigma}\sqrt{n}} e^{\frac{ik\mu}{\sigma}\sqrt{n} - \frac{k^2}{2}} = e^{-\frac{k^2}{2}}$$

This is the characteristic function of the standard normal distribution:

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \equiv \mathcal{N}(0,1)$$

Therefore, $Y_n \xrightarrow{d} \mathcal{N}(0,1)$ □

Exercise 3.2.1. *Calculate:*

$$\mathrm{Var}(Y_n) = \ldots = 1$$

Exercise 3.2.2. *Let x_1, x_2 be two i.i.d. Gaussian random variables such that $\langle x_i \rangle = 0$, $\langle x_i^2 \rangle = 1$, $i = 1, 2$, and $\langle x_1 x_2 \rangle = 0$ (uncorrelated). Calculate $\langle y_i \rangle$, $\langle y_i^2 \rangle$, and $\langle y_1 y_2 \rangle$, $i = 1, 2$ where*

$$\begin{cases} y_1 = \rho + \sqrt{1-\rho^2} x_1 \\ y_2 = \rho\sqrt{1-\rho}(\gamma x_1 + \sqrt{1-\gamma^2} x_2) \end{cases}$$

where $|\rho| \leq 1$, $|\gamma| \leq 1$.

Exercise 3.2.3. *1. Show that $\sum_i x_i \sim N(n\mu, n\sigma^2)$*

2. Show that $\frac{1}{n}\sum_i x_i \sim N(\mu, \frac{\sigma^2}{n})$

The Theorem 3.2.5 not only ensures that the average converges to the expected value, but also tells us how fast it does so: the fluctuations shrink as $1/\sqrt{n}$. This universality explains why Gaussian behavior is so pervasive in physics – from thermal noise to measurement errors – whenever a large number of independent contributions are involved.

Chapter 4

Laplace Method

The **Laplace method** is a powerful approximation technique that also has a natural extension to the complex plane, known as the method of steepest descent.

Here, we present one of its simplest forms, which already captures its essential ideas. The method is widely used in many branches of physics, such as statistical mechanics of disordered systems and the study of matter's structure.

4.1 Basic idea

In many practical situations, one wishes to evaluate complicated real integrals of the form

$$I(\lambda) = \int_{x_1}^{x_2} dx\, g(x) e^{\lambda f(x)}, \quad \lambda \in \mathbb{R} \tag{4.1}$$

where $g(x)$ and $f(x)$ are continuous and infinitely differentiable functions.

Although such integrals are typically intractable for arbitrary values of λ, they can be well approximated – as $\lambda \to \infty$ – under suitable regularity conditions.

The key idea of the Laplace method is that, as $\lambda \to \infty$, the dominant contribution to $I(\lambda)$ comes from the neighborhood of the point $x_0 \in [x_1, x_2]$ where $f(x)$ attains its maximum.

Figure 4.1 shows how $e^{\lambda f(x)}$ becomes sharply peaked around x_0 as λ increases. The larger λ is, the more the integral is dominated by the vicinity of x_0.

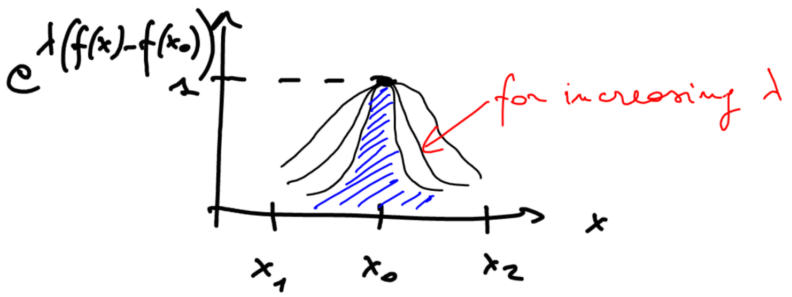


Figure 4.1

4.2 Method

There are essentially three cases:

Case 1 – Interior maximum ($x_1 < x_0 < x_2$). Since x_0 is an interior maximum, we have $f'(x_0) = 0$ and $f''(x_0) < 0$. We also assume that $g(x_0)$ is finite and non-zero.

As the dominant contribution arises from the neighborhood of x_0, we expand:

$$f(x) = f(x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + o(|x - x_0|^2), \quad \text{as } x \to x_0$$

Then, we can approximate $I(\lambda)$ as:

$$I(\lambda) \simeq g(x_0)e^{\lambda f(x_0)} \int_{x_1}^{x_2} e^{-\lambda \frac{|f''(x_0)|}{2}(x-x_0)^2}\, dx, \qquad \text{at leading order.}$$

Introducing the change of variable $s = (x - x_0)\sqrt{\frac{|f''(x_0)|}{2}\lambda}$, we obtain

$$I(\lambda) = g(x_0)e^{\lambda f(x_0)}\sqrt{\frac{2}{\lambda|f''(x_0)|}} \int_{(x_1-x_0)\sqrt{\frac{|f''(x_0)|}{2}\lambda}}^{(x_2-x_0)\sqrt{\frac{|f''(x_0)|}{2}\lambda}} e^{-s^2}\, ds \xrightarrow{\lambda\to\infty} \int_{-\infty}^{\infty} e^{-s^2}\, ds = \sqrt{\pi}$$

Thus, the leading order approximation is:

$$I(\lambda) = g(x_0)e^{\lambda f(x_0)}\sqrt{\frac{2\pi}{\lambda|f''(x_0)|}}, \qquad \text{as } \lambda \to \infty \text{ (leading order in } \lambda) \tag{4.2}$$

Show that this is the leading order as exercise.

Case 2 – Flat endpoint maximum ($x_0 = x_1$ or $x_0 = x_2$ with $f'(x_0) = 0$). Here, the dominant contribution comes from one side of the integration domain.

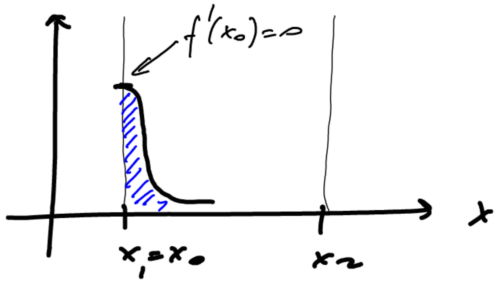


Figure 4.2

The leading-order term is then

$$I(\lambda) = g(x_0)e^{\lambda f(x_0)}\sqrt{\frac{\pi}{2\lambda|f''(x_0)|}} \tag{4.3}$$

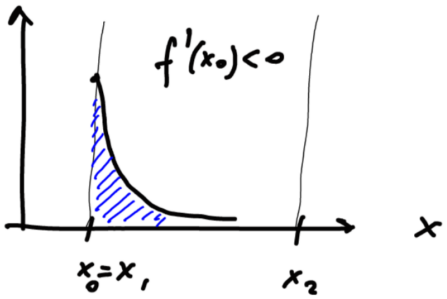


Figure 4.3

Case 3 – Non-flat endpoint ($x_0 = x_1$ or $x_0 = x_2$, $f'(x_0) \neq 0$). Without loss of generality, let $x_0 = x_1$, and $f'(x_0) < 0$ (Figure 4.3).

We expand

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \dots$$

and approximate

$$I(\lambda) \simeq \int_{x_1=x_0}^{x_2} g(x)e^{\lambda[f(x_0)+(x-x_0)f'(x_0)]}\,dx = g(x_0)e^{\lambda f(x_0)} \int_{x_0}^{x_2} e^{\lambda(x-x_0)f'(x_0)}\,dx =$$
$$\overset{s=x-x_0}{=} g(x_0)e^{\lambda f(x_0)} \int_0^{x_2-x_0} e^{\lambda f'(x_0)s}\,ds = \frac{g(x_0)e^{\lambda f(x_0)}}{\lambda f'(x_0)} \left[e^{\lambda s f'(x_0)}\right]_0^{x_2-x_1} =$$
$$= \frac{g(x_0)e^{\lambda f(x_0)}}{\lambda f'(x_0)} \left(e^{\lambda(x_2-x_1)f'(x_0)} - 1\right)$$

As $\lambda \to \infty$ and $f'(x_0) < 0$, we get

$$I(\lambda) \simeq g(x_0)\frac{e^{\lambda f(x_1)}}{\lambda|f'(x_1)|}, \qquad \text{as } \lambda \to \infty \tag{4.4}$$

Exercise 4.2.1. *Show that the next-to-leading-order term of $I(\lambda)$ in Equation 4.2 is:*

$$I(\lambda) = e^{\lambda f(x_0)}\sqrt{\frac{2\pi}{\lambda|f''(x_0)|}}\left(g(x_0) + \frac{c}{\lambda}\right) \quad \text{as } \lambda \to \infty$$

where c is a constant depending on the derivatives of f up to the fourth order at $x = x_0$, as well as on $g(x_0)$ and $g'(x_0)$. Basically, we have to find the constants: that is, prove that the first correction is $1/\lambda$ as shown before and c tells us how to calculate the next leading order.

Example 4.2.1

The modified Bessel function of the second kind is defined by

$$K_\nu(x) = \int_0^\infty e^{-x\cosh t}\cosh \nu t\,dt, \qquad x > 0$$

Here, the role played by λ is x.

To estimate $K_\nu(x)$ as $x \to \infty$ with fixed ν, we note that the maximum of $e^{-x\cosh t}$ occurs at $t = 0$ ($\cosh'(0) = 0$).

We can therefore apply Equation 4.3 (flat endpoint at the left), using the expansion of $\cosh t$ around zero. Thus,

$$K_\nu(x) \simeq e^{-x}\sqrt{\frac{\pi}{2x}}, \qquad \text{as } x \to \infty$$

to leading order.
Notice that the leading order does not depend on ν.

Exercise 4.2.2. *Show that*

$$K_\nu(x) = \sqrt{\frac{\pi}{2x}}e^{-x}\left(1 + \frac{c(\nu)}{x} + \dots\right)$$

and calculate $c(\nu) = \frac{2\nu^2-1}{8}$.

Example 4.2.2

Compute the leading order approximation of:

$$I(\lambda) = \int_0^1 x^m e^{\lambda(3x^2+2x^3)}\,dx, \qquad \text{as } \lambda \to \infty$$

The maximum of $f(x) = 3x^3 + 3x^3$ is at $x = 1$, where $f(1) = 5$, $f'(1) = 12$, and $g(1) = 1$. Since the maximum is at a non-flat endpoint, we are in Case 3. Applying Equation 4.4:

$$I(\lambda) \simeq \frac{e^{5\lambda}}{12\lambda}$$

At leading order, the result is independent of m.
Calculate the leading-order approximation for the integrals $\int_{-1}^1 \dots = ?$ and $\int_{-2}^0 \dots = ?$.

4.3 Applications

4.3.1 Stirling formula

Stirling formula provides an asymptotic approximation of the factorial. One of the most classical applications of the Laplace method is indeed the derivation of Stirling's formula. The advantage of using Laplace's methods is that it allows us to understand where the formula comes from and how to compute its systematic corrections.

We want to estimate how fast $N!$ diverges as $N \to \infty$. To do this, we apply Laplace's method to the Gamma function, defined by:

$$\Gamma(\lambda) = \int_0^\infty x^{\lambda-1}e^{-x}\,dx, \qquad \lambda > 0 \tag{4.5}$$

Consider:

$$\Gamma(\lambda+1) = \int_0^\infty x^{\lambda}e^{-x}\,dx$$

If we write it as $\int_0^\infty e^{-x} e^{\lambda \ln x}\, dx$, we cannot apply Laplace's method because $\ln x$ is an increasing function and its maximum occurs at $x = +\infty$. It is therefore more convenient to define

$$f(x) = -x + \lambda \ln x, \qquad g(x) = 1.$$

The maximum of $f(x)$ occurs at $x_0 = \lambda$, which is finite. This suggests introducing a change of variable $x = \lambda t$, so that the maximum is fixed at $t_0 = 1$. We obtain:

$$\Gamma(\lambda + 1) = \int_0^\infty x^\lambda e^{-x}\, dx \overset{x=\lambda t}{=} \lambda^{\lambda+1} \int_0^\infty t^\lambda e^{-\lambda t}\, dt$$

Thus, we can now identify

$$t^\lambda e^{-\lambda t} = e^{\lambda(\ln t - t)}, \quad h(x) = \ln t - t, \quad h'(1) = 0, \quad h''(1) = -1$$

In this way, we can apply the Laplace approximation Equation 4.2 with $g(t) = 1$ and $t_0 = 1$:

$$\Gamma(\lambda + 1) = \lambda! \simeq \lambda^{\lambda+1} e^{-\lambda} \sqrt{\frac{2\pi}{\lambda}} \qquad \text{as } \lambda \to \infty \tag{4.6}$$

This is precisely Stirling' formula.

Exercise 4.3.1. *Show that* $\Gamma(\lambda + 1) = \lambda\Gamma(\lambda)$. *Hence,* $\lambda! = \Gamma(\lambda + 1)$, *which extends the factorial to complex values of* λ.

Exercise 4.3.2. *Show that at leading order:*

$$\int_0^\infty e^{-\lambda t} e^{-1/t}\, dt \simeq \frac{\sqrt{\pi} e^{-2\sqrt{\lambda}}}{\lambda^{3/4}}, \qquad \textit{as } \lambda \to \infty$$

Example 4.3.1

Consider the family of integrals

$$I_m(x) = \int_0^\infty t^m e^{-\frac{t^2}{2} - \frac{x}{t}}\, dt, \quad x > 0$$

We want to compute $I_m(x)$ for large x and fixed m.

The function in the exponential,

$$\phi(t) = \frac{t^2}{2} + \frac{x}{t}$$

has a movable minimum determined by $\phi'(t) = t - \frac{x}{t^2} = 0$, which gives

$$t = \bar{t} = x^{1/3}$$

We introduce the rescaled variable $t = x^{1/3}\tau$, so that the minimum becomes fixed at $\tau = 1$. Then,

$$I_m(x) = x^{(m+1)/3} \int_0^\infty \tau^m e^{-x^{2/3}(\frac{\tau^2}{2} + \frac{1}{\tau})}\, d\tau$$

Since the minimum is interior ($\tau_0 = 1$) (we look at the minimum because there is a minus sign in the exponent which we want to maximize), we apply again Equation 4.2:

$$I_m(x) = x^{\frac{m+1}{3}} e^{-\frac{3}{2}x^{2/3}} \sqrt{\frac{2\pi}{x^{2/3} \cdot 3}} =$$
$$= x^{m/3} e^{-\frac{3}{2}x^{2/3}} \sqrt{\frac{2\pi}{3}} \qquad \text{as } x \to \infty \text{ at leading order, fixed } m$$

Exercise 4.3.3. *Show, using Laplace's method, the following asymptotic equalities as* $\lambda \to \infty$*:*

$$\int_{-2}^{0} e^t e^{\lambda(3t^2+2t^3)}\, dt \simeq e^{\lambda-1}\sqrt{\frac{\pi}{3\lambda}}$$
$$\int_{0}^{1} e^t e^{\lambda(3t^2+2t^3)}\, dt \simeq \frac{e^{5\lambda+1}}{12\lambda}$$
$$\int_{0}^{1} \sqrt{1+t} e^{\lambda(2t-t^2)} dt \simeq e^{\lambda}\sqrt{\frac{\pi}{2\lambda}}$$
$$\int_{-1}^{2} e^{\lambda(t^3-1)}(1+t^2) dt \simeq \frac{5e^{6\lambda}}{11\lambda}$$

4.3.2 L^p-norm in real analysis

Another useful application of Laplace's method arises in real analysis. The **L^p-norm** of a function $g(t)$ defined on $[a, b]$ is

$$||g||_p \coloneqq \left(\int_a^b |g(t)|^p\, dt\right)^{1/p}, \qquad p > 0$$

We want to study the behavior of $||g||_p$ as $p \to \infty$.
Assume that $g \in C^4(a, b)$ and that g has a unique interior maximum at t_0, with $g(t_0) \neq 0$.
We define the integral

$$I(p) \equiv \int_a^b |g(t)|^p\, dt, \qquad ||g|| = I^{1/p}$$

which can be written as

$$I = \int_a^b e^{p \ln |g(t)|}\, dtt$$

In this setting, $f(t) = \ln|g(t)|$ plays the role of the exponent in the Laplace method. Although $f(t)$ diverges to $-\infty$ wherever $g(t) = 0$, such regions contribute negligibly to the integral for large p, because they are exponentially suppressed.

Applying Equation 4.2, we find:

$$I(p) = e^{p \ln |g(t_0)|} \sqrt{\frac{2\pi g(t_0)}{p|g''(t_0)|}} \left(1 + o\left(\frac{1}{p}\right)\right) =$$
$$= |g(t_0)|^p \sqrt{\frac{2\pi g(t_0)}{p|g''(t_0)|}} \left(1 + o\left(\frac{1}{p}\right)\right) \qquad \text{as } p \to \infty$$

Obs. Notice that if $a > 0$ is a constant, we can write

$$a^{1/p}p^{-1/2p} = e^{\frac{\ln a}{p}}e^{-\frac{\ln p}{2p}} =$$
$$= \left(1 + \frac{\ln a}{p} + \dots\right)\left(1 - \frac{\ln p}{2p} + \dots\right) \simeq 1 - \frac{\ln p}{2p} + \dots \qquad \text{as } p \to \infty$$

Given this observation, we can rewrite:

$$I^{1/p} = g(t_0)\left(\frac{2\pi|g|}{p|g''|}\right)^{\frac{1}{2p}} + \text{h.o.t.} = |g(t_0)|\left(1 - \frac{\ln p}{2p} + o\left(\frac{1}{p}\right)\right)$$

Hence,

$$||g||_p = \max_{t\in[a,b]}|g(t)|\left[1 - \frac{\ln p}{2p} + \dots\right] \qquad \text{as } p \to \infty$$

This justifies the usual definition:

$$||g||_\infty = \max_{t\in[a,b]}|g|$$

Example 4.3.2

Consider the integral

$$\int_0^{\frac{3\pi}{2}} e^{-\lambda \sin t} f(t)\, dt \qquad \text{as } \lambda \to \infty$$

where f is continuous and differentiable on $\left[0, \frac{3\pi}{2}\right]$.

We separate the contributions from the two endpoint minima, $t = 0$ and $t = \frac{3\pi}{2}$:

$$I(\lambda) = \underbrace{\int_0^{\pi/2} e^{-\lambda \sin t} f(t)\, dt}_{I_1} + \underbrace{e^{\lambda}\int_{\pi/2}^{3\pi/2} e^{-\lambda \sin t - \lambda} f(t)\, dt}_{I_2}$$

For I_1, the minimum at $t = 0$ corresponds to a non-flat endpoint, so that we apply Equation 4.4, while for I_2, the minimum at $t = \frac{3\pi}{2}$ is flat and we apply Equation 4.3. Thus, we have:

$$I_1 = f(0)\frac{e^{\lambda\cdot 0}}{\lambda\cos(0)} \simeq \frac{f(0)}{\lambda} \qquad \text{as } \lambda \to \infty$$
$$I_2 = f\left(\frac{3\pi}{2}\right)e^{\lambda\cdot 0}\sqrt{\frac{\pi}{2\lambda\sin(\frac{3\pi}{2})}} \simeq f\left(\frac{3\pi}{2}\right)\sqrt{\frac{\pi}{2\lambda}} \qquad \text{as } \lambda \to \infty$$

Therefore, the leading contribution comes from I_2:

$$I(\lambda) \simeq f\left(\frac{3\pi}{2}\right)e^{\lambda}\sqrt{\frac{\pi}{2\lambda}} + \text{h.o.t.} \qquad \text{as } \lambda \to 0$$

The contribution from $t = 0$ is subleading and becomes relevant only in higher-order terms.

Exercise 4.3.4. *Calculate the leading contribution to*

$$\int_0^\pi e^{-\lambda \sin t} f(t)\, dt \qquad \text{as } \lambda \to \infty$$

This example illustrates that not all minima contribute equally at leading order: the dominant contribution depends on their position and relative weight in the exponent.

Chapter 5

Review of Dynamical Systems

5.1 Vector field, phase space, trajectories, phase portraits

In many scientific fields we often need to study non-linear autonomous systems of ODEs:

$$\begin{cases} \dot{\vec{x}}(t) = \vec{f}(\vec{x}(t)) \\ \vec{x}(0) = \vec{x}_0 \end{cases} \tag{5.1}$$

where $\vec{x}(t) = (x_1(t), \ldots, x_N(t)) \in U \subseteq \mathbb{R}^N$, and U is an open, connected subset of $\mathbb{R}^N$

In general, since the function $\vec{f}$ is generic, usually continuous and differentiable, the system is non-linear. Here, *autonomous* means that $\vec{f}$ does not depend explicitly on time t; time enters only through $\vec{x}(t)$. These are first-order ODEs.

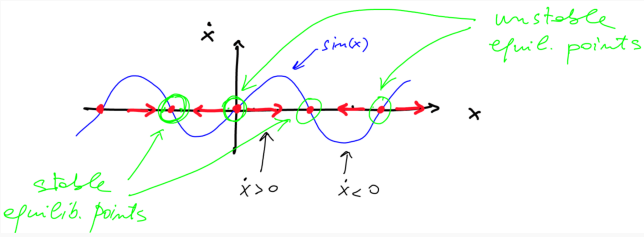
Each point $\vec{x}$ in the domain U is associated with a vector $\vec{f}(\vec{x})$; the collection of all these vectors forms a **vector field**. The domain U is also called the **phase space** The solutions $\vec{x}(t)$ of the system 5.1 are **trajectories** – smooth, time-parametrized curves in phase space that describe the evolution of the system.

Example 5.1.1

Let's consider ($N = 1$):

$$\dot{x}(t) = \sin[x(t)]$$

In this case, the vector field is one-dimensional and lies along the x-axis. When $x > 0$, $\dot{x} > 0$ and arrows point to the right; when $x < 0$, $\dot{x} < 0$ and they point left. Points where the arrows converge correspond to **stable equilibria** (**attractors**), while points where they diverge correspond to **unstable equilibria**. The set of initial conditions that lead to a given stable equilibrium is called its **basin of attraction**. With one dof, stability can often be determined graphically.

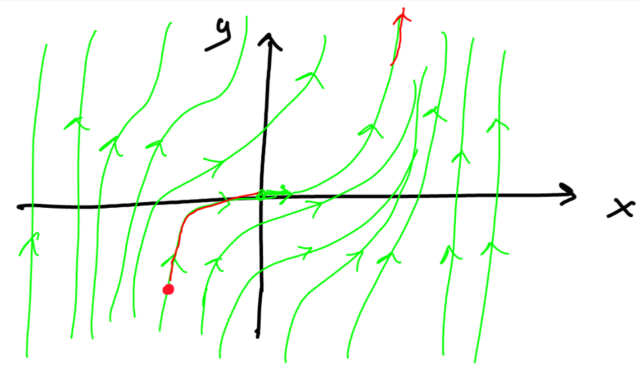


Example 5.1.2

For $N = 2$:

$$\begin{cases} \dot{x} = 1 \\ \dot{y} = x^2 + y^2 \end{cases}$$

Here the vector field here is two-dimensional: $\vec{x} = (x, y) \mapsto \vec{f}(\vec{x}) = (1, x^2 + y^2)$. The arrows show the local direction of motion – in other words, where the system tends to move next. Knowing the vector field allows us to predict or compute the trajectories, at least numerically.



In general, the vector field $\vec{f}$ is tangent to a trajectory in every point. In fact, let $\vec{x}(t)$ be a trajectory, then the equation of a tangent line to the trajectory at a point $\vec{x}_a \equiv \vec{x}(t_a)$ is

$$\vec{Y}(u) = \vec{x}_a + (u - t_a)\dot{\vec{x}}(t_a) = \vec{x}_a + (u - t_a)\vec{f}(\vec{x}_a)$$

Thus, $\vec{f}(\vec{x}_a)$ gives the direction of the tangent.

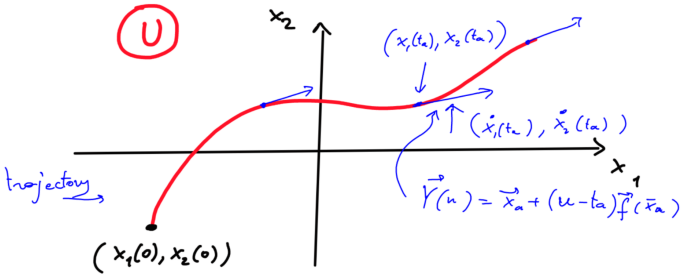


Figure 5.1

Here, the trajectory is the red curve, the starting point is $(x_1(0), x_2(0))$ and the vector field is represented by the blue arrows.

A **phase portrait** is the graphical representation of several trajectories with the corresponding direction of the vector field. In two dimensions,

$$\begin{cases} \dot{x}_1 = f_1(x_1, x_2) \\ \dot{x}_2 = f_2(x_1, x_2) \end{cases} \tag{5.2}$$

each trajectory is a curve in the (x_1, x_2) plane representing the evolution of the system from its initial condition. Thus, by flowing along the vector field, a point traces out the trajectory $\vec{x}(t)$ (solution of Equation 5.2).

In general, we cannot solve Equation 5.1. Thus, we would like to analyze qualitative behavior. For instance, one can study **nullclines** – curves where one of the derivatives vanishes ($\dot{x}_i = 0$ for a given i). For $N = 2$, setting $f_1(x_1, x_2) = 0$ or $f_2(x_1, x_2) = 0$ gives two families of curves whose intersections correspond to possible fixed points.

5.2 Two interesting examples

Example 5.2.1

Find the solution of $\dot{y} = y^2$ with $y(0) = 1$ and compute $y(2)$.

Integrating

$$\int \frac{dy}{y^2} = t - c \quad \Rightarrow \quad y = \frac{1}{c - t}$$

Using $y(0) = 1$ gives $c = 1$, so $y(t) = \frac{1}{1-t}$.
At $t = 2$, this formula gives $y(2) = -1$, which is clearly impossible because $\dot{y} = y^2 \geq 0$ implies that $y(t)$ must be increasing. In fact, the solution only exists in the interval $(0, 1)$: it diverges to infinity at $t = 1$.
The lesson here is that having an analytic expression does not guarantee it's valid for all times – the solution may only exist on a finite interval.

Example 5.2.2

Solve $\dot{y} = \sqrt{y}$ with $y(0) = 0$ and compute $y(2)$.

Integrating

$$\int \frac{dy}{\sqrt{y}} = 2\sqrt{y} = t - c \quad \Rightarrow \quad y = \frac{(t - c)^2}{4}$$

Applying $y(0) = 0$ gives $y(t) = \frac{t^2}{4}$, so $y(2) = 1$. But notice that $y(t) = 0$ also satisfies both the equation and the initial condition. Even more, any function of the form

$$y(t) = \begin{cases} 0 & 0 \leq t \leq T \\ \frac{(t-T)^2}{4} & t > T \end{cases}$$

for any $T > 0$ is a valid solution. So there are infinitely many solutions – the initial value problem is not unique.

In the first example, the solution ceases to exist after a finite time; in the second, it's not unique. These pathologies motivate the following

Theorem 5.2.1 (Picard's theorem)**.** *If $\vec{f}$ and all its partial derivatives $\frac{\partial f_i}{\partial x_j}$ are continuous in U, then for every $\vec{x}_0 \in U$ the initial value problem defined in Equation 5.1 has a unique solution in some interval $t \in [-\delta, \delta]$, with $\delta > 0$.*

As a direct consequence, trajectories in such systems never intersect.

5.3 Fixed points

A **fixed point** $\vec{x}^*$ satisfies

$$\vec{f}(\vec{x}^*) = \vec{0} \tag{5.3}$$

At a fixed point, time evolution stops. However, the existence of fixed points does not imply that trajectories necessarily approach them; if the point is unstable, the system will move away instead.

5.3.1 Local stability of fixed points

Once the fixed points are known, we can examine their **local stability**. Intuitively, a fixed point is locally stable if small perturbations around it decay over time. Typically, one studies local stability to avoid studying global stability of a system, which may be more complicated.

Consider the system of ODEs:

$$\begin{cases} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{cases}$$

and let (x^*, y^*) be a fixed point, that is $f(x^*, y^*) = g(x^*, y^*) = 0$. Let's introduce a small perturbation $u(t) = x(t) - x^*$, $v(t) = y(t) - y^*$. Since $|u|$ and $|v|$ are *small* compared to the dimensions of the system, we can expand the system to the first order:

$$\begin{aligned} \dot{u} &= u\frac{\partial f}{\partial x}\bigg|_* + v\frac{\partial f}{\partial y}\bigg|_* + O(u^2, v^2, uv) \\ \dot{v} &= u\frac{\partial g}{\partial x}\bigg|_* + v\frac{\partial g}{\partial y}\bigg|_* + O(u^2, v^2, uv) \end{aligned}$$

where the $*$ denotes computation at fixed point (x^*, y^*). This means that at leading order, we get the linearized system:

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = A \begin{pmatrix} u \\ v \end{pmatrix} \qquad A \equiv \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x^*, y^*)} \tag{5.4}$$

where A is the Jacobian matrix at fixed points.

In general form with N dof, where $\vec{u} \equiv (u_1, \ldots, u_n)$ with $u_i = x_i - x_i^*$, Equation 5.1 can be written as

$$\dot{\vec{u}} = A\vec{u} \qquad A_{ij} \equiv \frac{\partial f_i}{\partial x_j}\bigg|_{\vec{x}^*} \tag{5.5}$$

To understand if it's safe to neglect the non-linear terms, we introduce the following

Definition 5.3.1. A fixed point of an N-order system is **hyperbolic** if all eigenvalues of the linearized system are such that $\mathfrak{Re}(\lambda_i) \neq 0$ for any $i = 1, \ldots, N$.

Definition 5.3.2. Topological equivalence means there's a homeomorphism – a continuous map with a continuous inverse – that carries the phase portrait of the non-linear system to that of its linearization without reversing time.

Intuitively, near a hyperbolic fixed point, one can reshape the non-linear flow – stretching or compressing it locally without tearing or gluing – so that it looks like the linear flow. This reshaping does not change whether trajectories approach or leave the fixed point. Hence, local linearization is trustworthy for stability analysis around a hyperbolic fixed point. In fact, for hyperbolic points, it can be proved the following:

Theorem 5.3.3 (Hartmann-Grobman Theorem)**.** *Near a hyperbolic fixed point, the nonlinear system is* ***topologically equivalent*** *to its linearization – that is, their phase portraits are qualitatively the same.*

Hence, if all eigenvalues have negative real parts, the fixed point is asymptotically stable, and it's always safe to linearize the system **locally**.

If $\mathfrak{Re}(\lambda_i) < 0 \ \forall i = 1, \ldots, N$, the fixed point x^* is **asymptotically stable**.

Let's focus on Equation 5.4 and 5.5, and solve the linearized system with $\vec{u}(t) = e^{\lambda t}\vec{v}$, where $\vec{v}$ is the eigenvector, then

$$\lambda\vec{v} = A\vec{v}$$

For a 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the eigenvalues satisfy $\det(\lambda\mathbb{1} - A) = 0$, which can be written as

$$\det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = (a-\lambda)(d-\lambda) - bc = \lambda^2 - T\lambda + D = 0$$

where $T = \text{trace}(A) = a + d$ and $D = \det(A)$. Then. the eigenvalues are

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4D}}{2}$$

and the general solution is

$$\vec{u}(t) = c_1 e^{\lambda_1 t}\vec{v}_1 + c_2 e^{\lambda_2 t}\vec{v}_2$$

If $\mathfrak{Re}(\lambda_i) < 0$, then $\lim_{t\to\infty} \vec{u}(t) = 0$.

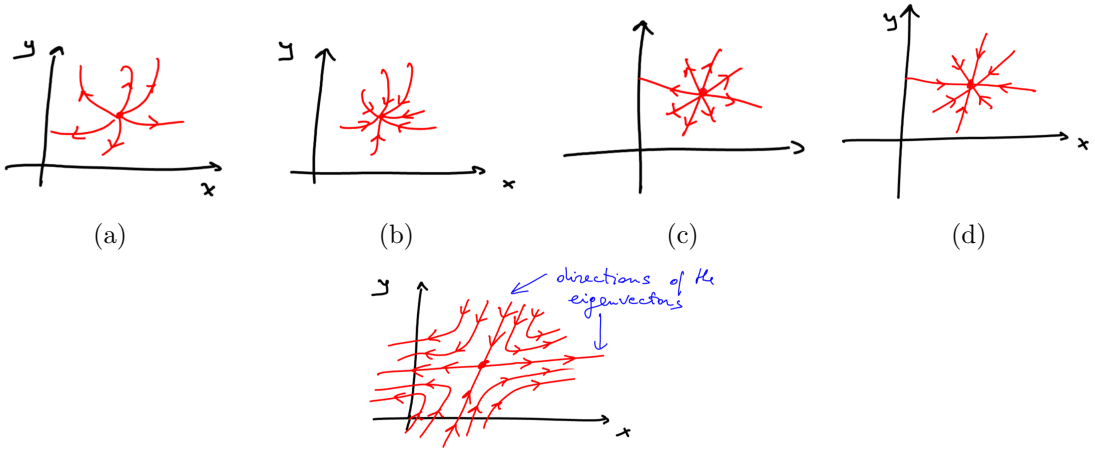
Summarizing, for a non-linear system, first find the fixed points, then check if they're hyperbolic. If they are, analyze the linearized system: the eigenvalues determine whether each fixed point is stable, unstable, or of another type.

5.3.2 A gallery of phase portraits (2×2 case)

Different combinations of eigenvalues correspond to different qualitative behaviors:

Real eigenvalues:

- $0 < \lambda_1 < \lambda_2$: **unstable node** – trajectories move away from the fixed point (Figure 5.2a).
- $\lambda_2 < \lambda_1 < 0$: **stable node** – trajectories converge (Figure 5.2b).
- $\lambda_2 < 0 < l\lambda_1$: **saddle point** – unstable (Figure 5.2e).
- $0 < \lambda_1 = \lambda_2$ ($A = \lambda_1\mathbb{1}$): **unstable star** (Figure 5.2c).
- $\lambda_2 = \lambda_1 < 0$ ($A = \lambda_1\mathbb{1}$): **stable star** (Figure 5.2d).
- If $A \neq \lambda_1\mathbb{1}$, we have **degenerate (un)stable nodes**, with distorted geometry.



(a) (b) (c) (d) (e)

Figure 5.2: Phase portraits for real eigenvalues.



(a) (b)

Figure 5.3: Phase portraits for complex eigenvalues.

Complex eigenvalues: Linear analysis fails when some eigenvalues have zero real part: If $\lambda_{1,2} = \gamma \pm i\omega$

- $\gamma < 0$: **stable focus** (spiral) (Figure 5.3a)
- $\gamma > 0$: **unstable focus** (spiral) (Figure 5.3b).

All the above fixed points are hyperbolic. For non-hyperbolic fixed points we have:

- $\lambda_1 = 0$, $\lambda_2 > 0$ or $\lambda_1 < 0$: **line of fixed points** – possibly attractive or not (Figure 5.4a).
- $\gamma = 0$, $\lambda_2 = \pm i\omega$: **center** – trajectories orbit around the point without convergence or divergence (Figure 5.4b).



(a) (b)

Figure 5.4: Phase portraits for non-hyperbolic fixed points.

Exercise 5.3.1. *Find the fixed points and classify them.*

$$\begin{cases} \dot{x} = -x + x^3 \\ \dot{y} = -2y \end{cases}$$

Exercise 5.3.2. *Show that the linearization of the system*

$$\begin{cases} \dot{x} = -y + ax(x^2 + y^2) \\ \dot{y} = x + ay(x^2 + y^2) \end{cases}$$

incorrectly predicts that the origin is a center for all values of a. Indeed, the origin is not a center, but is a stable spiral if $a < 0$, and unstable spiral if $a > 0$.

[Hint: to analyze the non-linear system, use polar coordinates $x = r\cos\theta$, $y = r\sin\theta$ and find the equations for r and θ. Then, interpret the non-linear system.]

Exercise 5.3.3. *The equation of motion of a particle is*

$$\ddot{x} = x - x^3$$

Find the fixed points and classify them. Show that the function

$$E = \frac{\dot{x}^2}{2} - \frac{x^2}{2} + \frac{x^4}{4}$$

which is the energy, is conserved by the dynamics, and that the trajectories are closed curves defined by the contours of E. Draw the phase portrait.

Exercise 5.3.4. *Study the system of ODEs:*

$$\begin{cases} \dot{x} = x(3 - x - 2y) \\ \dot{y} = y(2 - x - y) \end{cases}$$

where $x, y \geq 0$.

Find the fixed points and classify them (many). Show that the phase portrait is given by the Figure 5.5

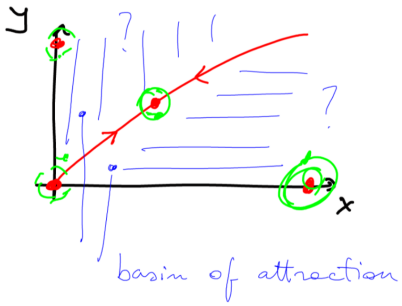


Figure 5.5

*where the blue points are the so-called **basin of attraction**, which is the set of all initial conditions which end up in one fixed point as $t \to \infty$.*

Chapter 6

Diffusion

To introduce stochastic processes, we start with the concept of diffusion. We will derive the diffusion equation in different contexts using two methods:

- one at the **mesoscopic scale**, describing observable phenomena such as Brownian motion;
- one at **microscopic scale**, leading to the random walk model.

Diffusion is a fundamental process, both physically and mathematically, as it's the cornerstone of stochastic dynamics.

6.1 Einstein derivation

Consider a long, thin tube filled with water. At time $t = 0$, we inject a unit amount of ink at $x = 0$, where *ink* represents particles that are not very different in size from those in the surrounding fluid. The random motion of these particles is known as **Brownian motion**, a typical of stochastic behavior.

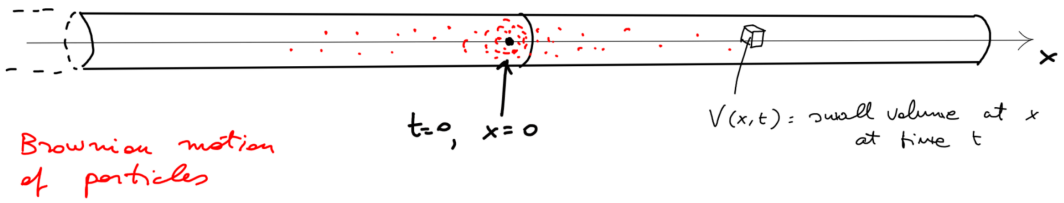


Figure 6.1

Let's consider a small volume $V(x,t)$ containing a certain number of particles. We define the function

$$w(x,t) = \lim_{\substack{V\to 0 \\ N\to\infty}} \frac{[\#\text{ of particles in } V(x,t)]}{V(x,t)}$$

as the **particle density** at position $x \in \mathbb{R}$ and time $t \geq 0$.

The meaning of $w(x,t)$ is that the integral over a region A gives the probability of finding a particle there:

$$\int_A w(x,t)\,dx \equiv \text{probability to find particle in the region } A \subseteq \mathbb{R} \text{ at time } t$$

Thus, $w(x,t)$ is smooth, integrable, and normalized

$$\int_{-\infty}^{\infty} w(x,t)\,dx = 1$$

which means there is no loss or creation of particles. At time $t = 0$, $w(x,0) = \delta(x)$, indicating that all particles start at the origin.

Assume that the probability density for a Brownian particle to move from position x to $x + y$ during a short time interval Δt is given by $f(y|x,\Delta t) = f(y,\Delta t)$. This is a conditional probability, and the equality shows that f does not depend on x – a homogeneity assumption.

Since f is a probability density, it must satisfy:

$$\int_{-\infty}^{\infty} f(y,\Delta t)dy = 1$$

We already know the probability of finding a particle at x at time t ($w(x,t)$) and the probability of making a jump from any position to y ($f(y,\Delta t)$). We now want to find the distribution of particles at position x and time $t + \Delta t$ – in other words, the time evolution of $w(x,t)$.

To do this, consider that particles located at $x-y$ at time t may move to x after a jump of size y in the interval Δt – thus, y is arbitrary and x is fixed.



Figure 6.2

The density of particles at $x - y$ is $w(x - y, t)$, and each of these makes a jump y with a probability $f(y,\Delta t)\,dy$. Therefore, the particle density at x at time $t + \Delta t$ is given by

$$w(x, t + \Delta t) = \int_{-\infty}^{\infty} w(x - y, t) f(y, \Delta t)\,dy \tag{6.1}$$

This is the basic diffusion equation for a generic jump distribution f. Here, we assume that all particles jump independently. For small Δt, we can simplify the function f by further assuming:

1. $f(y,\Delta t)$ decays quickly with $|y| \to$ long jumps are much less probable than short ones.
2. $f(-y,\Delta t) = f(y,\Delta t) \to$ jumps are equally likely in both directions (symmetry assumption).

With these two assumptions, we can expand $w(x - y, t)$ in a Taylor series:

$$\int w(x-y,t)f(y,\Delta t)\,dy \simeq \int (w(x,t) - (\partial_x w)y + \frac{1}{2}(\partial_x^2 w)y^2 + \ldots)f(y,\Delta t)\,dy =$$

$$= \int w(x,t)f(y,\Delta t)\,dy - \int \underbrace{(\partial_x w)yf(y,\Delta t)}_{\int yf(y,\Delta t)\,dy=0}\,dy + \frac{1}{2}\int \left((\partial_x^2 w)y^2 + \text{h.o.t.}\right) f(y,\Delta t)\,dy$$

where the first term simply gives $w(x,t)$ (because $\int f(y,\Delta t)\,dy = 1$). The second term disappears because $yf(y,\Delta t)\,dy = 0$ – by symmetry, it means there is no drift: particles diffuse but on average stay there because jumps are equally likely. The third term represents the effect of diffusion. Thus, we have

$$\int w(x-y,t)f(y,\Delta t)\,dy \simeq w(x,t) + \frac{1}{2}\partial_x^2 w \underbrace{\int_{-\infty}^{\infty} y^2 f(y,\Delta t)\,dy}_{\text{assume finite}} + \text{h.o.t.}$$

We now assume that the second moment of f is finite and proportional to Δt, i.e.,

$$\int y^2 f(y,\Delta t)\,dy \overset{\Delta t\to 0}{=} D\Delta t + \text{h.o.t.}$$

where D is called the diffusion coefficient. This is a linear approximation and reflects the intensity of fluctuations. In general, it may not be finite. Then,

$$w(x,t+\Delta t) = w(x,t) + \frac{1}{2}D\Delta t\partial_x^2 w + \text{h.o.t.}$$

By rearranging it

$$\frac{w(x,t+\Delta t) - w(x,t)}{\Delta t} = \frac{D}{2}\partial_x^2 w + o(\Delta t)$$

and taking the limit $\Delta t \to 0$, we get

$$\frac{\partial w(x,t)}{\partial t} = \frac{D}{2}\frac{\partial^2 w(x,t)}{\partial x^2} \tag{6.2}$$

which is the **diffusion (or heat) equation** for Brownian motion in one dimension. This is a PDE that link the spatial and temporal evolution of the particle distribution.

In d-dimensions, it becomes:

$$\frac{\partial w(F,t)}{\partial t} = \frac{D}{2}\nabla^2 w(F,t), \qquad \text{where } \nabla^2 \equiv \sum_{i=1}^{d}\frac{\partial^2}{\partial x_i^2}$$

This equation must be solved with an initial condition, usually $w(x,0) = \delta(x-x_0)$, and possibly with boundary conditions depending on the system.

6.2 Solution of the diffusion equation with scaling assumption

We now solve Equation 6.2 by exploiting its symmetries. This is not the most general solution, but a valid one.

In general, finding the solution of a PDE is not straightforward. To simplify the problem, it is often useful to exploit the symmetries of the system. Rather than solving the equation blindly, one should first look for possible symmetries. Incorporating these symmetries into the solution can greatly simplify the problem, often reducing a PDE to an ODE, which is much easier to handle.

Here, we will use the dilation (or scaling) symmetry under which Equation 6.2 preserves the dimensionless ratio x^2/t.

Obs. If $w(x,t)$ solves Equation 6.2, then so does $w(\lambda x, \lambda^2 t)$ for any $\lambda \neq 0$ (try this as exercise by changing $x \to \lambda x$ and $t \to \lambda^2 t$). Physically, this reflects a change of units (dilation symmetry) and implies that x^2/t is invariant.

Motivated by this, we assume a similarity form where w depends on x and t only through the dimensionless combination

$$z = \frac{x}{\sqrt{Dt}}$$

which is a specific combination of the two variables x and t. Hence, we have

$$w(x,t) = u\left(\frac{x}{\sqrt{Dt}}\right) = u(z)$$

We now verify this ansatz. By computing the derivatives, we get

$$\partial_t w = \frac{\mathrm{d}u}{\mathrm{d}z}\frac{\partial z}{\partial t} = \frac{\mathrm{d}u}{\mathrm{d}z} \cdot \left(-\frac{x}{\sqrt{D}} \cdot \frac{1}{2} \cdot \frac{1}{t^{3/2}}\right)$$

$$\partial_x^2 w = \partial_x \left(\frac{\mathrm{d}u}{\mathrm{d}z}\frac{\partial z}{\partial x}\right) = \frac{\mathrm{d}^2 u}{\mathrm{d}z^2}\left(\frac{\partial z}{\partial x}\right)^2 + \frac{\mathrm{d}u}{\mathrm{d}z}\frac{\partial^2 z}{\partial x^2} = \frac{\mathrm{d}^2 u}{\mathrm{d}z^2}\frac{1}{Dt}$$

Plugging into Equation 6.2, we get

$$\partial_t w = -\frac{x}{2\sqrt{D}t^{3/2}}\frac{\mathrm{d}u}{\mathrm{d}z} = \frac{D}{2}\frac{1}{Dt}\frac{\mathrm{d}^2 u}{\mathrm{d}z^2} = \frac{1}{2t}\frac{\mathrm{d}^2 u}{\mathrm{d}z^2}$$

By multiplying both sides by t, we get:

$$-\frac{x}{\sqrt{Dt}}\frac{\mathrm{d}u}{\mathrm{d}z} = \frac{\mathrm{d}^2 u}{\mathrm{d}z^2}$$

Since $z = \frac{x}{\sqrt{Dt}}$, we have

$$\frac{\mathrm{d}^2 u}{\mathrm{d}z^2} + z\frac{\mathrm{d}u}{\mathrm{d}z} = 0 \tag{6.3}$$

which is no longer a PDE, but a ODE. This means that our assumptions on dilation symmetry were indeed correct.

To find a solution for $u(z)$, let $\rho = \frac{\mathrm{d}u}{\mathrm{d}z}$. Then

$$\frac{\mathrm{d}\rho}{\mathrm{d}z} + z\rho = 0 \qquad \Rightarrow \qquad \rho(z) = ae^{-z^2/2}$$

Thus, integrating, the final solution is:

$$u(z) = a\int_0^z e^{-s^2/2}\,ds + b$$

where a and b are fixed by initial/boundary conditions.

Therefore, the final solution of Equation 6.2 is

$$u\left(\frac{x}{\sqrt{Dt}}\right) = a\int_0^{\frac{x}{\sqrt{Dt}}} e^{-s^2/2}\,ds + b$$

For $x > 0$ and $t \to 0^+$, $z \to +\infty$ and

$$u \to a\int_0^{\infty} e^{-s^2/2}\,ds + b = \sqrt{\frac{\pi}{2}}a + b$$

which is the first initial condition. At $x = 0$, $t > 0$, $z = 0$ and $u \to b$, which is the value of the distribution at $x = 0$ for any time.

In general, arbitrary a, b do not satisfy the delta initial condition $w(x,0) = \delta(x - x_0)$. Later we will introduce the **fundamental solution** that matches arbitrary initial data.

6.3 A more microscopic derivation: the random walk

We now derive the diffusion equation from a microscopic discrete perspective, starting from the random walk. This approach highlights the mathematical nature of diffusion and can be applied to many systems unrelated to the original physical model.

Consider a particle moving on a one-dimensional lattice with spatial step $\Delta\ell$ and time step Δt. Here, Δt is fixed and finite, and x is no longer a continuous variable – it takes discrete values $x_i = i\Delta\ell$.

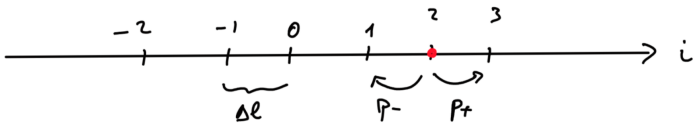


Figure 6.3

We introduce the following notation:

- $w_i(k)$: the probability (previously it was the density) of finding the particle at position $i = 0, \pm1, \pm2, \ldots$ at time step $k = 0, 1, 2, \ldots$. Hence, $x_i = i\Delta\ell$, $t_k = k\Delta t$.
- p_+: probability to jump one site to the right at the next time step; p_- is the probability to jump one site to the left. Since one of these events must occur at each step, $p_+ + p_- = 1$.

We now write an equation for $w_i(k)$. Consider Figure 6.4. At time $k+1$, the particle can reach site i only from its nearest neighbors $i-1$ and $i+1$, since longer jumps are excluded and staying still is not allowed.

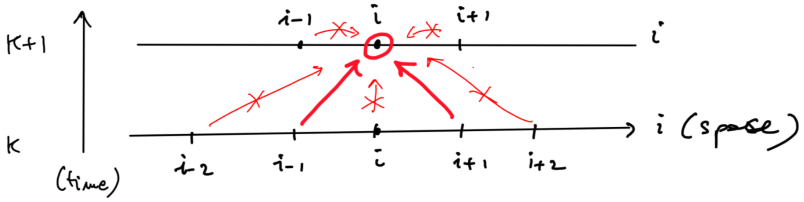


Figure 6.4

Therefore, the probability to be at site i at time $k+1$ is

$$w_i(k+1) = w_{i-1}(k)p_+ + w_{i+1}(k)p_- \tag{6.4}$$

This is called the **master equation** of the random walk.
It is natural to ask:

1. Can this equation be solved explicitly?
2. Does it reduce to the diffusion equation in an appropriate continuous limit?

The answer to both questions is yes.

The random walk is very easy to simulate. Starting from an initial position, the particle jumps right or left with probability p_+ or p_-. To estimate $w_i(k)$, we perform R independent

realizations of the process up to time step $k = K$ with the same initial condition. Then, for larger R,

$$w_i(k) = \frac{[\#\text{ of runs in which the particle is at position } i \text{ at time } k]}{R}$$

Since each realization is stochastic, the individual paths differ, but the ensemble average approximates the probability distribution. For example, for $R = 3$ and an initial condition $w_i(0) = \delta_{i,1}$, a typical simulation (Figure 6.5) might yield $w_2(1) = 2/3$, $w_0(1) = 1/3$ at time $k = 1$.

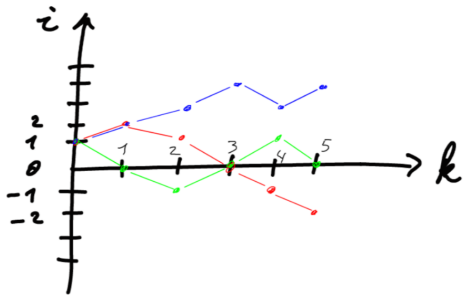


Figure 6.5

An important property of Equation 6.4 is *normalization*. Because the particle cannot disappear or be created, total probability is conserved, and summing over all states, it must be

$$\sum_{i=-\infty}^{+\infty} w_i(k) = 1$$

for all time k.

To verify this, sum both sides of Equation 6.4 over i:

$$\sum_{i=-\infty}^{+\infty} w_i(k+1) = p_+ \sum_{i=-\infty}^{+\infty} w_{i-1}(k) + p_- \sum_{i=-\infty}^{+\infty} w_{i+1}(k)$$

Since the indices are dummy variables in infinite sums, we can re-name them and we get:

$$\sum_{i=-\infty}^{+\infty} w_i(k+1) = (p_+ + p_-) \sum_{i=-\infty}^{+\infty} w_i(k) = 1$$

Hence, if the distribution is normalized initially, it remains normalized at all times. For master equations , this conservation property always holds in the absence of external fluxes.

6.4 Solution of master equation with generating function

We now solve Equation 6.4 using the generating function method. This method can be seen as a kind of projection technique: Equation 6.4 could in principle be written in matrix form, but for large systems this matrix becomes intractable. Instead, we project the equation onto a simpler basis, choosing z^i as the projection basis.

We multiply Equation 6.4 by z^i, with $z \neq 0$, and sum over i, where in general $z \in \mathbb{C}$:

$$\sum_{i=-\infty}^{+\infty} z^i w_i(k+1) = p_+ \sum_{i=-\infty}^{+\infty} z^i w_{i-1}(k) + p_- \sum_{i=-\infty}^{+\infty} z^i w_{i+1}(k)$$

Define the generating function:

$$\tilde{w}(z,k) = \sum_{i=-\infty}^{+\infty} z^i w_i(k) \tag{6.5}$$

Then Equation 6.4 becomes an equation for $\tilde{w}$:

$$\begin{aligned} \tilde{w}(z,k+1) &= p_+ z \sum_{i=-\infty}^{+\infty} z^{i-1} w_{i-1}(k) + p_- \frac{1}{z} \sum_{i=-\infty}^{+\infty} z^{i+1} w_{i+1}(k) = \\ &= p_+ z \tilde{w}(z,k) + p_- \frac{1}{z} \tilde{w}(z,k) = \left(z p_+ + \frac{p_-}{z} \right) \tilde{w}(z,k) \end{aligned} \tag{6.6}$$

Iterating this relation gives

$$\tilde{w}(z,k) = \left(z p_+ + \frac{p_-}{z} \right)^k \tilde{w}(z,0)$$

where $\tilde{w}(z,0)$ is the generating function of the initial condition.

Obs. Notice that at $z = 1$

$$\tilde{w}(1,k) = (p_+ + p_-)^k \tilde{w}(1,0) = \tilde{w}(1,0)$$

which confirms that the distribution remains normalized during the evolution.

If the particle starts at $i = 0$ at time $k = 0$, then

$$\tilde{w}(z,0) = \sum_{i=-\infty}^{+\infty} z^i \delta_{i,0} = 1$$

Hence, for this initial condition, the solution of Equation 6.6 is

$$\tilde{w}(z,k) = \left(z p_+ + \frac{p_-}{z} \right)^k \tag{6.7}$$

The generating function allows an easy calculation of moments such as the mean and variance. A key feature of the random walk is that both the mean and the variance scale linearly with time.

Using Equation 6.5, the average position at time step k is

$$\frac{\mathrm{d}\tilde{w}}{\mathrm{d}z} = \sum_{i=-\infty}^{+\infty} i z^{i-1} w_i(k) \quad \xrightarrow{z=1} \quad \left. \frac{\mathrm{d}\tilde{w}}{\mathrm{d}z} \right|_{z=1} = \sum_{i=-\infty}^{+\infty} i w_i(k) = \langle i(k) \rangle \equiv \mu(k)$$

From Equation 6.7,

$$\frac{\mathrm{d}\tilde{w}}{\mathrm{d}z} = k \left(z p_+ + \frac{p_-}{z} \right)^{k-1} \left(p_+ - \frac{p_-}{z^2} \right) \quad \xrightarrow{z=1} \quad \mu(k) = k(p_+ - p_-)$$

where we evaluated at $z = 1$. Thus, the mean displacement grows linearly with time.

For the variance we have by definition:

$$\sigma^2(k) = \left\langle i^2(k) \right\rangle - \langle i(k) \rangle^2 \tag{6.8}$$

Since

$$\frac{\mathrm{d}^2\tilde{w}}{\mathrm{d}z^2} = \sum_{i=-\infty}^{+\infty} i(i-1)z^{i-2}w_i(k) \quad \xrightarrow{z=1} \quad \langle i(i-1) \rangle = \left\langle i^2 \right\rangle - \langle i \rangle = \left\langle i^2(k) \right\rangle - \mu(k)$$

the second derivative at $z = 1$ is not the variance itself. Using the definition Equation 6.8, we get:

$$\begin{aligned}
&\sigma^2(k) = \left.\frac{\mathrm{d}^2\tilde{w}}{\mathrm{d}z^2}\right|_{z=1} + \mu(k) - [\mu(k)]^2 = \\
&= \left[k(k-1)\left(zp_+ + \frac{p_-}{z}\right)^{k-2}\left(p_+ - \frac{p_-}{z}\right)^2 + k\left(zp_+ + \frac{p_-}{z}\right)^{k+1}\frac{zp_-}{z}\right]_{z=1} + \mu(k) - \mu^2(k) = \\
&= k(k-1)(p_+ - p_-)^2 + 2kp_- + k(p_+ - p_-) - k^2(p_+ - p_-)^2 = \\
&= -k(p_+ - p_-)^2 + 2kp_- + k(p_+ - p_-) = \\
&= 2kp_-(1 + p_+ - p_-) = 4kp_+p_-
\end{aligned}$$

Summarizing, the variance is

$$\sigma^2(k) = 4kp_+p_- \tag{6.9}$$

Thus, both the mean and the variance of a random walk depend linearly on time. In general, if a process exhibits a mean and a variance that scale linearly with time, one should consider whether the underlying dynamics can be described as a random walk.

Finally, let's compute the distribution $w_i(k)$. Since k is an integer, Equation 6.7 can be expanded as a polynomial

$$\left(zp_+ + \frac{p_-}{z}\right)^k = \sum_{\ell=0}^{k} \binom{k}{\ell} (zp_+)^\ell \left(\frac{p_-}{z}\right)^{k-\ell}$$

Setting $\ell = \frac{k+i}{2}$ gives

$$\begin{aligned}
\tilde{w}(z,k) &= \sum_{i=-k}^{k} \binom{k}{\frac{k+i}{2}} (zp_+)^{(k+i)/2} \left(\frac{p_-}{z}\right)^{(k-i)/2} = \\
&= \sum_{\ell=-k}^{k} \binom{k}{\frac{k+i}{2}} (p_+)^{(k+i)/2}(p_-)^{(k-i)/2} z^i = \sum_{i=-\infty}^{+\infty} w_i(k) z^i
\end{aligned}$$

Comparing coefficients of z^i, we obtain the explicit solution:

$$w_i(k) = \begin{cases} \binom{k}{\frac{k+i}{2}} (p_+)^{(k+i)/2}(p_-)^{(k-i)/2}, & -k \le i \le k \\ 0, & \text{otherwise} \end{cases} \tag{6.10}$$

This satisfies $w_i(0) = \delta_{i,0}$ for $k = 0$ and normalization for $k = 1, 2, \ldots$.

Exercise 6.4.1. *Find $w_i(k)$ for an initial condition $w_i(0) = \delta_{i,i_0}$, and then for a general initial condition*

$$w_i(0) = g_i \ge 0, \qquad \sum_i g_i = 1$$

This generalizes what we have obtained up to now to any initial condition with generic point.

Exercise 6.4.2. *Compute the mean $\mu(k)$ and variance $\sigma^2(k)$ directly from Equation 6.10.*

6.5 Continuum limit of the random walk

We now want to connect the random walk to the diffusion equation derived earlier. In Einstein's derivation we obtained a first-order PDE involving the Laplacian. We now ask: under what conditions do the discrete random walk and the diffusion process describe the same physics?

We will show that in an appropriate limit – when both the time step and the spatial step go to zero in a controlled way – the random walk converges to the diffusion equation. This means that a discrete stochastic process can be mapped to a continuous one in a suitable regime.

Let's introduce two variables:

- $x_i = i\Delta\ell$ as the distance between two adjacent lattice sites
- $t_k = k\Delta t$ as the discrete time.

We explicitly keep these spacings because we will later show that, when $\Delta\ell$ and Δt scale in a particular way, we can take the continuum limit, linking the two descriptions.

$$x_{i\pm1} = (i \pm 1)\Delta\ell = x_i \pm \Delta\ell \equiv x \pm \Delta\ell$$
$$t_{k+1} = (k+1)\Delta t = k\Delta t + \Delta t \equiv t + \Delta t$$

After taking the limit, we will drop the indices, as the variables become continuous.

From Equation 6.4, we have

$$\begin{aligned} w(x, t+\Delta t) &= p_+ w(x-\Delta\ell, t) + p_- w(x+\Delta\ell, t) \overset{\Delta\ell\to 0}{=} \\ &\simeq p_+ \left(w(x,t) - (\partial_x w)\Delta\ell + \frac{1}{2}\partial_x^2 w(\Delta\ell)^2 + \text{h.o.t.} \right) + \\ &\quad + p_- \left(w(x,t) + (\partial_x w)\Delta\ell + \frac{1}{2}\partial_x^2 w(\Delta\ell)^2 + \text{h.o.t.} \right) \end{aligned}$$

where we expanded w as a Taylor series. Collecting terms and using $p_+ + p_- = 1$, we obtain:

$$w(x, t+\Delta t) = w(x,t) - (p_+ - p_-)\Delta\ell\partial_x w + \frac{1}{2}(\Delta\ell)^2\partial_x^2 w + \text{h.o.t.}$$

At the same time, as $\Delta t \to 0$:

$$w(x, t+\Delta t) = w(x,t) + \partial_t w \Delta t =$$

Equating the two expressions and rearranging, we find:

$$\frac{w(x,t+\Delta t) - w(x,t)}{\Delta t} = -(p_+ - p_-)\frac{\Delta\ell}{\Delta t}\partial_x w + \frac{1}{2}\frac{(\Delta\ell)^2}{\Delta t}\partial_x^2 w + \text{h.o.t.} \tag{6.11}$$

To obtain a meaningful continuum limit, the coefficients of the spatial derivatives must remain finite as $\Delta\ell, \Delta t \to 0$. Therefore, the two quantities below must approach constants:

$$\lim_{\substack{\Delta\ell\to 0 \\ \Delta t\to 0}} (p_+ - p_-)\frac{\Delta\ell}{\Delta t} = v$$
$$\lim_{\substack{\Delta\ell\to 0 \\ \Delta t\to 0}} \frac{(\Delta\ell)^2}{\Delta t} = D > 0$$

Here v and D are finite constants representing the **drift velocity** and the **diffusion coefficient**, respectively.

These limits represent a rescaling: as we zoom in on space, we must also rescale time accordingly. If we zoom in space but not in time (or vice versa), the limit becomes trivial. In this scaling regime, discreteness in space is lost while time is approximated as continuous (we are looking the system under a different perspective).

In the continuum limit, Equation 6.11 becomes

$$\partial_t w(x,t) = -v\partial_x w(x,t) + \frac{D}{2}\partial_x^2 w(x,t) \tag{6.12}$$

This is the **diffusion equation with constant drift** v. This generalizes the standard diffusion equation by including a deterministic drift term. If $v > 0$, particles drift to the right ($+\infty$); if $v < 0$, to the left ($-\infty$). When $v = 0$, the equation reduces to the symmetric case, where the mean displacement is zero and the process is purely diffusive.

6.6 Fundamental solution of the diffusion equation

We will show that the solution of Equation 6.12 (PDE) is a Gaussian distribution. For $v = 0$, the Gaussian is centered at the origin and broadens over time. For $v \neq 0$, the Gaussian peak drifts with velocity v, moving right or left depending on the sign of v.

We first compute the fundamental solution of Equation 6.12 for $v = 0$ (to be generalized to general v as exercise). This solution is called *fundamental* because it allows us to solve any initial value problem by convolution. We use the Fourier transform and assume that the solution is bounded – otherwise uniqueness is not guaranteed – since the equation is defined on $\mathbb{R}$.

The initial value problem reads:

$$\begin{cases} \partial_t w(x,t) = \frac{D}{2}\partial_x^2 w(x,t) & x \in \mathbb{R} \\ w(x,0) = \delta(x-y) & t > 0 \end{cases} \tag{6.13}$$

where y is the initial position.

Taking the Fourier transform of Equation 6.13

$$\tilde{w}(\omega,t) = \int_{-\infty}^{\infty} w(x,t)e^{i\omega x}\,dx$$

we obtain

$$\begin{cases} \partial_t \tilde{w}(\omega,t) = -\frac{D}{2}\omega^2 \tilde{w}(\omega,t) \\ \tilde{w}(\omega,0) = e^{i\omega y} \end{cases}$$

which is no longer a PDE, but an ODE in time with solution

$$\tilde{w}(\omega,t) = e^{i\omega y - \frac{D\omega^2 t}{2}}$$

By using the inverse Fourier transform,

$$w(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega x}\tilde{w}(\omega,t)$$

we obtain

$$w(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega\, e^{-i\omega x}e^{i\omega y - \frac{D\omega^2 t}{2}} = \frac{1}{\sqrt{2\pi Dt}}e^{-\frac{(x-y)^2}{2Dt}}$$

which is a Gaussian integral.

Hence, the fundamental function is:

$$w(x,t) = \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x-y)^2}{2Dt}} \tag{6.14}$$

Let's see some properties. As $t \to 0^+$, $w(x,t) \to 0$ for any $x \neq y$, while $\int_{-\infty}^{\infty} w(x,t)\,dx = 1$ for any t. Therefore,

$$\lim_{t\to 0^+} w(x,t) = \delta(x-y)$$

and $w(x,t)$ indeed solves the system Equation 6.13.

In physics, $w(x,t)$ is called the **propagator** of Equation 6.13, often denoted $w(x,t|y,0)$. It represents the probability of finding a particle at position x at time t, given that it started at y at time 0. This is also the propagator of the **Wiener process** (Brownian motion).

More generally,

$$w(x,t|y,t_0) = \frac{1}{\sqrt{2\pi D(t-t_0)}} e^{-\frac{(x-y)^2}{2D(t-t_0)}}, \qquad t > t_0$$

Because $w(x,t|y,t_0) = w(x-y,t-t_0|0,0)$, the propagator depends only on the differences $x-y$ and $t-t_0$, which means that its increments are independent.

Exercise 6.6.1. *Show that the solution of Equation 6.12 with drift v is*

$$w_v(x,t) = \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x-y-vt)^2}{2Dt}}$$

and verify that $\langle x(t)\rangle = y + vt$, $v \in \mathbb{R}$, that is, the mean position drifts linearly with v.

Using the fundamental solution 6.14, we can solve the general initial value problem:

$$\begin{cases} \partial_t W(x,t) = \frac{D}{2}\partial_x^2 W(x,t) & x \in \mathbb{R} \\ W(x,0) = f(x) & t > 0 \end{cases} \tag{6.15}$$

where $f(x)$ is an arbitrary initial condition.

The solution of Equation 6.15 is the convolution

$$W(x,t) = \int_{-\infty}^{\infty} w(x,t|y,0) f(y)\,dy \tag{6.16}$$

which shows explicitly why w i called a propagator: it propagates the initial distribution $f(y)$ forward in time. Show that this solves Equation 6.15 as exercise.

From Equation 6.14, all statistical moments $\langle x^n(t)\rangle$ can be computed. For instance,

$$\langle x\rangle_t = \int_{-\infty}^{\infty} x w(x,t|y,0)\,dx = y \tag{6.17}$$

with $\langle x\rangle_0 = y$ when $v = 0$.

Obs. Do not confuse integration over x and y because they give different information. Integrating over x gives averages of observables for a fixed initial condition; integrating over y instead describes how initial conditions propagate to position x at later times.

The variance is:

$$\mathrm{Var}(x_t) = \int_{-\infty}^{\infty} (x - \langle x \rangle_t)^2 w(x,t|y,0)\, dx = Dt \tag{6.18}$$

Equation 6.17 and 6.18 describe the typical behavior of the Wiener process – linear growth of the variance in time – and reproduce the scaling of the discrete random walk in the continuum limit.

Given any observable $f(x)$ (not the initial condition), its average at time t is defined by the integral over the entire state space:

$$\langle f(x) \rangle_t \equiv \int_{\mathbb{R}} f(x) w(x,t|y,0)\, dx \tag{6.19}$$

This can be justified starting from the law of large numbers and can be used for a general propagator (or PDF).

Differentiating with respect to time and using Equation 6.15, we get the temporal evolution of $\langle f(x) \rangle_t$:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \langle f(x) \rangle_t &= \int f(x) \frac{\partial w(x,t|y,0)}{\partial t}\, dx \overset{6.15}{=} \frac{D}{2} \int f(x) \frac{\partial^2 w}{\partial x^2}\, dx = \\ &= \frac{D}{2} \int w \partial_x^2 f(x)\, dx \end{aligned}$$

where the last step follows from integrating by parts twice (the boundary terms vanish because w decays exponentially). Hence, we can find evolution equations for expectations $\langle f \rangle_t$ without explicitly knowing the solution w, but only the PDE that satisfies.

For instance, if $\langle f(x) \rangle = x$, the mean is:

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle x \rangle_t = 0 \quad \Rightarrow \quad \langle x \rangle_t = y$$

which is much easier than using the full propagator. For $f(x) = x^2$, we get the variance:

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle x^2 \rangle_t = D, \qquad \mathrm{Var}(x_t) = Dt$$

Exercise 6.6.2. *Compute the time evolution of the general moment* $\langle x^n \rangle_t$.

6.7 Diffusion on a semi-infinite line: absorbing boundary condition.

Up to now, we have solved the diffusion equation without boundaries, allowing particles to move freely from $-\infty$ to $+\infty$. In that case, there were no walls, sources, or sinks – particles could diffuse indefinitely.

We now study a more complex situation: diffusion in the presence of an absorbing wall. Although this specific case can still be solved analytically, in general, one must rely on numerical methods.

This setup also reveals a key property of Brownian motion: the survival probability – the probability that a particle has not yet been absorbed – decays in a universal way, with exponents that do not depend on the microscopic parameters of the process.

We introduce an absorbing wall at $x = 0$. Whenever a particle reaches this wall, it is immediately removed (absorbed) from the system. Hence $x = 0$ acts as an absorbing boundary, and all particles are initially located in the positive half-line $\mathbb{R}^+$.

The absorbing boundary condition requires that the probability to find a particle at $x = 0$ be zero for all $t > 0$.

The initial-boundary value problem reads:

$$\begin{cases} \partial_t w_a(x,t) = \frac{D}{2}\partial_x^2 w_a(x,t), & x \in \mathbb{R}^+,\ t > 0 \\ w_a(0,t) = 0, & \forall t > 0 \\ w_a(x,0) = \delta(x-y), & y > 0 \end{cases} \tag{6.20}$$

The PDE is identical to the free diffusion equation within the domain, but the boundary condition at $x = 0$ enforces the absorbing wall.

By symmetry arguments, the solution can be obtained via the method of images. The function

$$\frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x-y)^2}{2Dt}}$$

solves Equation 6.20 without the boundary condition at $x = 0$. Similarly,

$$\frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x+y)^2}{2Dt}}$$

also solves the same PDE, but corresponds to an initial condition $\delta(x + y)$; it is the mirror image of the first solution with respect to the origin $x = 0$.

Because Equation 6.20 is linear, any linear combination of these two functions is also a solution;

$$\frac{a}{\sqrt{2\pi Dt}} e^{-\frac{(x-y)^2}{2Dt}} + \frac{b}{\sqrt{2\pi Dt}} e^{-\frac{(x+y)^2}{2Dt}}$$

corresponding to the initial condition $a\delta(x - y) + b\delta(x + y)$.

By choosing $a = 1$ and $b = -1$, the solution satisfies the absorbing condition $w_a(0, t) = 0$ for all $t > 0$. Thus,

$$w_a(x,t|y,0) = \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x-y)^2}{2Dt}} - \frac{1}{\sqrt{2\pi Dt}} e^{-\frac{(x+y)^2}{2Dt}} \tag{6.21}$$

This is the fundamental solution (or **absorbing boundary propagator**) for Equation 6.20 with an absorbing boundary at $x = 0$. Indeed, it is a propagator because

$$w(0,t|y,0) = 0$$

for any $t > 0$.

If the initial position $y > 0$ is distributed according to $f(y)$ (with $\int_0^\infty f(y),\, dy = 1$), the second term $\delta(x + y)$ plays no role, since $x, y > 0$ and the wall cannot be crossed.

Hence, for a general initial distribution, any other solution can be written as

$$W_a(x,t) = \int_0^\infty w_a(x,t|y,0) f(y)\, dy$$

The integral decreases over time, since particles are progressively absorbed at the wall.

Obs. Initially,

$$\int_0^\infty w_a(x,0|y,0)\,dx = 1$$

meaning that all particles start in the positive half-line. However, for any $t > 0$,

$$\int_0^\infty w_a(x,t|y,0)\,dx < 1$$

which indicates that some particle have been absorbed at $x = 0$.

This quantity represents the **survival probability** of the particles, i.e., the probability that the particle is still present in $(0, +\infty)$ at time $t > 0$. As time increases, the survival probability decreases monotonically due to the absorption boundary. This is a non-conservational probability: total probability is not conserved in $(0, +\infty)$, because there is a continuous outgoing probability flux through the absorbing boundary at $x = 0$.

The non-conservation of probability for $x > 0$ results from the fact that particles are removed upon reaching the boundary $x = 0$. Consequently, there must be a flux of particles leaving the system at $x = 0$.

From Equation 6.20,

$$\partial_t w_a(x,t) = \frac{D}{2}\partial_x^2 w_a(x,t) = -\partial_x\left(-\frac{D}{2}\partial_x w_a(x,t)\right) \equiv -\partial_x j_a(x|y)$$

where we define the probability flux $j_a(x|y)$ at position x from the divergence theorem. Thus,

$$j_a(x,t|y,0) \equiv -\frac{D}{2}\partial_x w_a(x,t|y,0) \tag{6.22}$$

For a particle starting at $y > 0$ at $t = 0$, the flux at the boundary $x = 0$ and time $t > 0$ is

$$j_a(0|y) = -\frac{D}{2}\partial_x w_a(x,t)\Big|_{x=0} = -\frac{y}{t^{3/2}}\frac{1}{\sqrt{2\pi D}}e^{-\frac{y^2}{2Dt}} < 0 \tag{6.23}$$

which represents the outgoing probability flux at the absorbing wall.

The flux is directly related to the rate of decrease of total probability in the domain $(0, +\infty)$:

$$\partial_t \int_0^\infty w_a(x,t|y,0)\,dx \overset{6.20}{=} \frac{D}{2}\int_0^\infty (\partial_x^2 w_a)\,dx = \frac{D}{2}\partial_x w_a\Big|_0^\infty = -\frac{D}{2}\partial_x w_a\Big|_{x=0}$$

meaning that the loss of probability exactly equals the flux through the boundary, as expected. This means that the variation of the probability is exactly given by the flux, as it must be. Thus,

$$\partial_t \int_0^\infty w_a(x,t|y,0)\,dx = j_a(0|y) < 0 \tag{6.24}$$

Since we want to check this result, we define the survival probability

$$P(t) = \int_0^\infty w_a(x,t|y,0)\,dx$$

which is the probability that the particle has not yet been absorbed at time $t > 0$, given that its motion started at $y > 0$ at $t = 0$.

From this distribution, we can derive the **lifetime distribution** $q(t|y)$, which is a PDF giving the probability that a particle is absorbed between times t and $t + dt$. By definition,

$$q\,dt = -\underbrace{(P(t+\Delta t) - P(t))}_{<0} \quad \xrightarrow{\Delta t \to 0} \quad -\frac{\partial P(t)}{\partial t}\,dt$$

Using Equation 6.23, we have

$$q(t|y) = -\frac{\partial P(t)}{\partial t} \overset{6.23}{=} -j_a(0|y) \geq 0$$

Thus, the lifetime distribution for a Brownian particle starting at $y > 0$ is

$$q(t|y) = \frac{y}{t^{3/2}\sqrt{2\pi D}} e^{-\frac{y^2}{2Dt}} \tag{6.25}$$

Obs. For long times ($y \ll \sqrt{Dt}$),

$$q(t|y) \sim t^{-3/2}$$

The decay exponent $-3/2$ is universal – it does not depend on the diffusivity D or other microscopic detail of the process. This is one of the characteristic signatures of Brownian motion.

In practice, to verify whether a stochastic process is diffusive, one often plots $\log q$ vs $\log t$: a straight line with slope $-3/2$ indicates typical Brownian behavior.

From any arbitrary initial distribution $f(y)$:

$$\bar{q}(t) = \int_0^\infty q(t|y) f(y)\,dy$$

which gives the overall lifetime distribution averaged over all possible starting position y.

6.8 Diffusion on a semi-finite line: reflecting boundary condition

Unlike the previous case, particles can now bounce off the wall and return to the positive half-line $\mathbb{R}^+$. As soon as a particle hits the boundary at $x = 0$, it is reflected back to the interior of the domain. Therefore, there is no loss of probability, no leakage at $x = 0$, and the flux at the boundary is zero. Mathematically, this is implemented through a reflecting boundary condition, which requires that the probability flux vanish at $x = 0$.

From Equation 6.22, we have

$$j_r(0|y) = -\frac{D}{2}\partial_x w_r(x,t|y,0)\Big|_{x=0} = 0$$

Therefore, the new initial value boundary problem reads:

$$\begin{cases} \partial_t w_r(x,t) = \frac{D}{2}\partial_x^2 w_r(x,t), & x \in \mathbb{R}^+,\ t > 0 \\ \partial_x w_r(x,t)\Big|_{x=0} = 0 & \forall t \\ w_r(x,0) = \delta(x-y) \end{cases} \tag{6.26}$$

For the solution, we proceed as before and consider the linear combination:

$$\frac{a}{\sqrt{2\pi Dt}}e^{-\frac{(x-y)^2}{2Dt}} + \frac{b}{\sqrt{2\pi Dt}}e^{-\frac{(x+y)^2}{2Dt}}$$

which is general solution of Equation 6.26 for appropriate constants a, b.

We now impose the reflecting boundary condition:

$$\partial_x e^{-\frac{(x\pm y)^2}{2Dt}} = -\frac{x \pm y}{Dt}e^{-\frac{(x\pm y)^2}{2Dt}} \quad \xrightarrow{x=0} \quad \mp\frac{y}{Dt}e^{-\frac{y^2}{2Dt}}$$

The requirement $\partial_x w_r(x,t)|_{x=0} = 0$ implies $a = b = 1$. Hence, the reflecting solution is

$$w_r(x,t|y,0) = \frac{1}{\sqrt{2\pi Dt}}e^{-\frac{(x-y)^2}{2Dt}} + \frac{1}{\sqrt{2\pi Dt}}e^{-\frac{(x+y)^2}{2Dt}} \tag{6.27}$$

This is the propagator of the Brownian process with reflecting boundary at $x = 0$. Starting from an arbitrary initial distribution, we get

$$W_r(x,t) = \int_0^\infty w_r(x,t|y,0)f(y)\,dy$$

where $\int_0^\infty f(y)\,dy = 1$.

Notice that normalization is preserved $\int_0^\infty w_r(x,t|y,0)\,dx = 1$ since there is no absorption. However, notice also that $\lim_{t\to\infty} w_r(x,t|y,0) = 0$ because the probability density spreads and the particle drifts away to infinity.

Most of the derivations above extend to arbitrary dimensions n. Let $w(\vec{x},t)$ denote the probability density (PDF) and $\vec{x} \in \mathbb{R}^n$, and let $V \subseteq \mathbb{R}^n$ be a smooth region. The rate of change of the total number of particles (or substance) in V equals the negative net flux through the surface $\Sigma(V)$ (its boundary):

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V w(\vec{x},t)\,d^n x = -\int_{\Sigma(V)} \vec{J}\cdot\hat{n}\,dS = -\int_V \vec{\nabla}\cdot\vec{J}\,d^n x$$

where in the last equality follows from the divergence theorem, and $\vec{J}(\vec{x},t)$ is the flux density.

If particles cannot leave the region V, then $\vec{J} = \vec{0}$ on $\Sigma(V)$, implying that $\int_V w\,d^n x = \text{const}$. These are called reflecting boundaries, while $w(\vec{x},t) = 0$ on $\Sigma(V)$ corresponds to absorbing boundaries.

In general, for $\vec{J} \neq \vec{0}$ and arbitrary smooth V, we get the continuity equation

$$\partial_t w(\vec{x},t) = -\vec{\nabla}\cdot\vec{J} \tag{6.28}$$

If the system is closed ($\mathbb{R}^n$), then $\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^n} w(\vec{x},t)d^n x = 0$.

In the absence of external fields, we assume that $\vec{J} \propto -\vec{\nabla}w$, as particles diffuse from high to low concentration regions. Thus, we can set the constant as $D/2$, and we get

$$\vec{J}(\vec{x},t) = -\frac{D}{2}\vec{\nabla}w(\vec{x},t) \tag{6.29}$$

which is **Fick's law**.

In general, D may depend on $\vec{x}$ and w (when there are crowding effects), so from Equation 6.28

$$\partial_t w(\vec{x},t) = \vec{\nabla}\cdot\left(\frac{D}{2}\vec{\nabla}w\right)$$

If D is constant

$$\partial_t w(\vec{x},t) = \frac{D}{2}\nabla^2 w$$

which is the diffusion equation in n dimensions.

Chapter 7

Introduction to Stochastic Processes

Ref.: Introduction to Stochastic Calculus with Application, F.C. Kleboner, Imperial P. – An Introduction to SDEs, Lawrence C.Evans, AMS. – Stochastic Processes and Applications, G.A. Pavliotis, Springer. – Stochastic Methods, C. Gardiner, Springer – Stochastic Differential Equations, B. Oksendal, Springer. – Stochastic Processes in Physics and Chemistry, N.G. van Kampen.

7.1 Introduction

We now introduce stochastic processes, which are naturally connected to everything we have done so far. Up to this point, we derived the diffusion equation from a physical perspective, without placing stochastic processes at the center of the discussion.

From now on, the focus shifts to stochastic processes as modeling tools. These appear not only in physics – where randomness plays a crucial role – but also in economics, chemistry, biology, neuroscience, and many other fields.

We begin with continuous stochastic processes, whose paths are continuous functions that are nowhere differentiable. This property is a direct consequence of the fact that these processes have no memory (they satisfy Markov assumption). We will then move to discrete processes and eventually connect the material to topics we have already encountered, such as path integrals and disordered systems.

The mathematical tools required for a rigorous treatment is quite heavy, and we will not develop them fully here. To understand every technical detail, one should consult the books listed at the beginning of the chapter. In these notes, we will performs calculations by assuming certain properties rather than proving them formally. The goals is to introduce the basic language needed to read mathematical and physical literature, especially the definitions that will be used throughout the course.

7.2 Basic notion of probability theory

Before introducing stochasticity, we need some fundamental elements of **probability theory**, which is the mathematical basis of all stochastic processes.

Consider tossing a coin three times (heads→0, tails→1). We want to analyze the outcome given the sum of the three tosses. We describe this using the formal language of

stochastic processes.

The set of realizations – here $2^3 = 8$ sequences – is the **sample space**

$$\Omega = \{\omega_i\}$$

The set of possible values taken by the sum is the **state space** (measurements):

$$E = \{y_i\}$$

Then, a **random variable** Y a function that maps each realization in Ω to an outcome in E

$$Y : \Omega \to E$$

In our example:

$$Y : \Omega = \{\underbrace{(0,0,0)}_{\omega_1}, \underbrace{(0,0,1)}_{\omega_2}, \ldots, \underbrace{(1,1,1)}_{\omega_8}\} \mapsto E = \{0,1,2,3\}$$

In physics, we usually think: *repeat an experiment and get different results.* Mathematically, however, a random variable is simply a function from a sample space to a state space, and this function must be **measurable**.

More generally, we want to assign probabilities to events such as, for example, the sum is less than 2, or the sum is a certain set. In general, that must satisfy a specific condition. To do this, we look at sets of the form

$$Y^{-1}(B) \subseteq \Omega, \qquad B \subseteq E$$

These are the realizations that produce outcomes in B. Assigning probabilities to all such sets requires Ω and E to be measurable spaces.

A function Y is measurable (and therefore a random variable) if the pre-image of any measurable set in the state space is a measurable subset of the sample space. More formally,

Definition 7.2.1. A set S is **measurable** if there exists an algebra of sets F such that:

1. $\emptyset, S \subseteq F$
2. $\forall A, B \in F,\ A \cup B \in F,\ A \cap B \in F,\ A \setminus B \in F.$

This level of structure is unnecessary for discrete variables (where every subset is measurable), but essential for continuous ones.

A probability measure is a function

$$\mathbb{P} : \Omega \to [0,1]$$

such that

$$\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$$

From this, we define the probability of an outcome $y_i \in E$ as calculated on subsamples of Ω as:

$$p_i = \sum_{\omega : Y(\omega) = y_i} \mathbb{P}(\omega) = P(Y = y_i)$$

The expectation of a random variable Y is:

$$\mathbb{E}(Y) = \sum_{\omega \in \Omega} Y(\omega)\mathbb{P}(\omega) = \sum_{i=1}^{N} y_i P(Y = y_i) = \sum_i y_i p_i$$

where $\mathbb{P}(\omega)$ gives the weight of each realization, multiplied by the value assigned by the random variable.

If the sample space Ω is uncountable, we must ensure that probabilities are defined only on measurable sets. In this case, a random variable $Y : \Omega \to \mathbb{R}$ is well defined if the pre-image of any interval such as $\{Y \leq y\}$ or $\{a < Y \leq b\}$ is measurable in Ω.

This generalizes the discrete definition and allows us to handle continuous stochastic processes properly.

7.3 Stochastic processes

When we introduce time, we obtain what is called a stochastic process.

Definition 7.3.1. A **stochastic process** is a collection of random variables $x(t, \omega)$

$$\{x(t, \omega),\ t \in T,\ \omega \in \Omega\}$$

where T is an ordered set, such as $\mathbb{N}$ or $\mathbb{R}^+$, and Ω is a sample space.

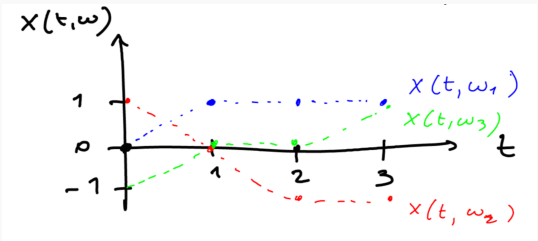
For fixed ω, the map $t \mapsto x(t, \omega)$ is a function of time and is called a **sample path** or **stochastic realization** of the process. In practice, the dependence on ω is often omitted and one simply writes $x(t)$. Thus, a stochastic process can be viewed as a two-index object: one index for time and one for the realization. Summarizing,

- Fixing t gives a random variable from the sample space to the state space.
- Fixing ω gives a time-dependent trajectory of the process.

Let's see some examples.

Example 7.3.1

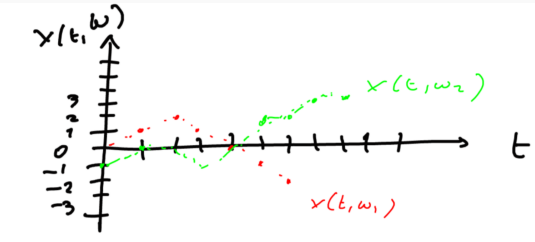
Let $E = \{0, -1, 1\}$ be the state space of a stochastic process X, and let $T = \{0, 1, 2, 3\}$ be discrete time. There are $3^4 = 81$ possible sample paths, then the cardinality is $|\Omega| = 81$. Three of them are shown here:



Because the state space is finite, all paths can be enumerated. This becomes impossible when the state space is infinite or uncountable.

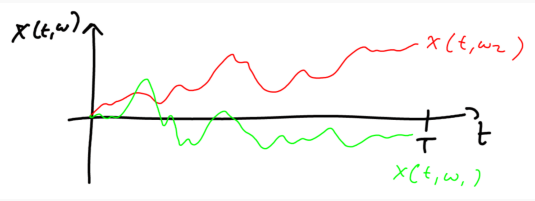
Example 7.3.2

In a discrete-time random walk, the state space is $E = \mathbb{Z}$ and $T = \mathbb{N}$. Each trajectory $x(t, \omega)$ is a function of time, but the set of all possible paths is uncountable.



Example 7.3.3

The position of a Brownian particle is described by a Brownian process in which $T = \mathbb{R}^+$ and $E = \mathbb{R}$. Both the sample space and state space ar continuous, and each realizations is a continuous (but nowhere differentiable) function of time: $x(t, \omega) \in C^0([0, 1])$. This is the most difficult one.



Remark. In physics, the term *random variable* is often used informally to mean something that gives different outcomes when repeated. In mathematics, a random variable is rigorously defined as a measurable function from a sample space to a state space. When we introduce time, we obtain a sequence (or family) of random variables indexed by t. Fixing ω gives a path of the system in time. Fixing t gives a random variable describing the distribution of states at that time.

In what follows, we will omit the explicit ω-dependence and simply write $x(t) \in E$.

7.3.1 Brownian or Wiener process

Now we can formalize Brownian motion, also known as the **Wiener process**. This is one of the most important stochastic processes and forms the basis of more general continuous-time stochastic models.

A standard one-dimensional Brownian process $B(t)$, is a real-valued stochastic process ($E = \mathbb{R}$, $T = \mathbb{R}^+$, that satisfies the following three criteria:

A) Initialization at the origin. All trajectories at time zero are in the origin: $B(0) = 0$

B) Gaussian increments. For any pair of times (s, t), with $0 \leq s < t < \infty$, the increment $B(t) - B(s)$ is normally distributed with mean 0 and variance $t - s$

$$B(t) - B(s) \sim \mathcal{N}(0, t - s) = \frac{1}{\sqrt{2\pi(t-s)}} e^{-\frac{(\Delta x)^2}{2(t-s)}}$$

where Δx is the displacement of the process from time s to t.

C) Independent increments. For any sequence of times $0 \leq t_0 < t_1 < t_2 < \ldots < t_n$ the increments $\Delta B_i \coloneqq B(t_i) - B(t_{i-1})$ are mutually independent.

7.3.2 Important properties

1. **Stationarity of increments** (the increments are stationary, not the process itself). Consider the increment at time s and $t = s + \Delta t$. From property (B), we have

   $$B(s + \Delta t) - B(s) \sim \mathcal{N}(0, \Delta t)$$

   which shows that the distribution of the increment $B(s+\Delta t) - B(s)$ does not depend on the specific time s. This means that not only increments are mutually independent, but they do not depend neither from the moment in which the increment is considered

   Therefore, the increment depends only on the time difference

   $$\Delta B(s + \Delta t) - B(s) \sim B(\Delta t)$$

   That is, the variation of the process has the same law as the process evaluated at the variation interval. This invariance under time translation is the stationarity of increments.

   Since $\mathrm{Var}(\Delta B) = \mathrm{Var}(B(\Delta t)) = \Delta t$, we expect that $|\Delta B| = |B(\Delta t)| \simeq \sqrt{\Delta t}$ as $t \to 0^+$.

2. **Connection with diffusion equation**. We have seen that a realization of the Wiener process satisfies the diffusion equation with $D = 1$:

   $$\begin{cases} \partial_t w(x,t) = \frac{1}{2}\partial_x^2 w(x,t), & x \in \mathbb{R},\ t > t_0 \\ w(x, t_0) = \delta(x - x_0) \end{cases}$$

   The solution is the propagator of Brownian motion:

   $$w(x, t|x_0, t_0) = \frac{1}{\sqrt{2\pi(t - t_0)}} e^{-\frac{(x-x_0)^2}{2(t-t_0)}} \tag{7.1}$$

   which is exactly the Gaussian form used in (B).

3. **Correlation function**. Let $0 \leq s \leq t$. From (B), we get $\mathbb{E}[B(s)] = 0$ and $\mathbb{E}[B^2(s)] = s$. Thus, we compute the two-point correlation

   $$\begin{aligned} \mathbb{E}[B(t)B(s)] &= \mathbb{E}[(B(s) + B(t) - B(s))B(s)] = \\ &= \mathbb{E}[B(s)^2 + (B(t) - B(s))B(s)] = \\ &= \mathbb{E}[B(s)^2] + \mathbb{E}[(B(t) - B(s))B(s)] = \\ &= s + \mathbb{E}[B(t) - B(s)] \underbrace{\mathbb{E}[B(s)]}_{0} = s \end{aligned}$$

   The first term comes from (B) and the second term vanishes because $B(t) - B(s)$ is independent of $B(s)$ and has zero mean. Therefore, $\mathbb{E}[B(t)B(s)] = s$. For arbitrary $s, t \geq 0$

   $$\mathbb{E}\left[B(t)B(s)\right] = \min(t, s) \tag{7.2}$$

   This is the correlation of Brownian motion.

   Interpretation:

- For fixed s, the correlation with $B(t)$ decreases as t moves away from s.
- For fixed t, the correlation with future values does not add new information: the future increment is independent of the past.

Exercise 7.3.1. *Let* $V(t) = e^{-t}B(e^{2t})$ *with* $B(t)$ *a Brownian process. Show that* $\mathbb{E}[V(t)] = 0$ *and*

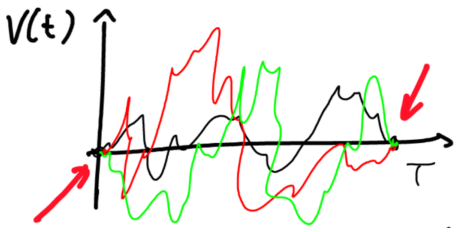
$$\mathbb{E}[V(t)V(s)] = e^{-|t-s|}$$

i.e. the process has exponentially correlations.

Exercise 7.3.2. *Define* $V(t) = TB(t) - tB(T)$ *for any* $t \in [0, T]$. *Show that* $\mathbb{E}[V(t)] = 0$ *and*

$$\mathbb{E}[V(t)V(s)] = T^2 \min(t, s) - Tts$$

for any $t, s \in [0, T]$. *This is called* ***Brownian bridge*** *and satisfies* $V(0) = V(T) = 0$



Exercise 7.3.3. *Show that for Brownian motion*

$$\mathbb{E}\left[e^{i\lambda B(t)}\right] = e^{-\frac{\lambda^2 t}{2}}, \qquad t \geq 0$$

Exercise 7.3.4. *Using the Law of large numbers and* $B(t) = \sum_{i=1}^{t} \Delta B_i$ *show that*

$$\lim_{t\to\infty} \frac{B(t)}{t} = 0$$

See (C) for property of ΔB_i.

Exercise 7.3.5. *Let* $z(t) = \frac{1}{\sqrt{c}}B(ct)$ *with* $c > 0$. *Show that* $z(t)$ *is distributed like* $B(t)$

$$\mathbb{E}\left[e^{i\lambda z(t)}\right] = \ldots = e^{-\frac{\lambda^2 t}{2}}$$

Exercise 7.3.6. *Let* $z(t) = tB(\frac{1}{t})$, $t > 0$, *with* $z(0) = 0$. *Show that* $z(t)$ *is also distributed like* $B(t)$.

$$\mathbb{E}\left[e^{i\lambda z(t)}\right] = \ldots = e^{-\frac{\lambda^2 t}{2}}$$

Notice that $\lim_{t\to\infty} \frac{z(t)}{t} = 0$ *(see 7.3.4).*

7.3.3 Path properties of the Brownian motion

In general, proving rigorous pathwise properties of Brownian motion is highly non-trivial. Here, however, we develop some heuristic but very informative arguments that capture the correct qualitative behavior.

To investigate the regularity of Brownian paths, consider the ratio

$$\frac{\Delta B}{(\Delta t)^\alpha}, \qquad \Delta t = t_2 - t_1 > 0, \quad \alpha > 0, \quad \Delta B = B(t_2) - B(t_1)$$

and study its behavior as $\Delta t \to 0$. Basically, instead of proving almost-sure statements, which is extremely hard, we study the probability that this ratio remains bounded:

$$P\left(\left|\frac{\Delta B}{(\Delta t)^\alpha}\right| \le k\right)$$

with $k \in \mathbb{R}^+$. Since $|\Delta B|$ is a random process, it is distributed normally. Thus, we obtain:

$$\begin{aligned} P\left(\left|\frac{\Delta B}{(\Delta t)^\alpha}\right| \le k\right) = P\left(|\Delta B| \le k(\Delta t)^\alpha\right) = \int_{-k(\Delta t)^\alpha}^{k(\Delta t)^\alpha} dx\, \frac{e^{-\frac{x^2}{2\Delta t}}}{\sqrt{2\pi\Delta t}} = \\ = \frac{2}{\sqrt{2\pi\Delta t}} \int_0^{k(\Delta t)^\alpha} dx\, e^{-\frac{x^2}{2\Delta t}} \overset{z=\frac{x}{\sqrt{\Delta t}}}{=} \sqrt{\frac{2}{\pi}} \underbrace{\int_0^{k(\Delta t)^{\alpha-\frac{1}{2}}} dz\, e^{-\frac{z^2}{2}}}_{\sqrt{2\pi}/2} \end{aligned} \tag{7.3}$$

Consequences:

1. **Hölder continuity when** $0 < \alpha < \frac{1}{2}$. If $0 < \alpha < \frac{1}{2}$, then as $\Delta t \to 0^+$, $k(\Delta t)^{\alpha-\frac{1}{2}} \to +\infty$, so the integral tends to:

   $$P\left(\left|\frac{\Delta B}{\Delta t^\alpha}\right| \le k\right) \to 1$$

   regardless of k. Thus, heuristically, as $|t-s| \to 0^+$, we should expect

   $$|B(t) - B(s)| \le \text{const}\, |t-s|^\alpha$$

   with probability close to 1.

   This states that Brownian motion is Hölder continuous of any order $\alpha < \frac{1}{2}$. These arguments are not a proof but correctly predict the known theorem: Brownian trajectories are continuous everywhere almost surely.

2. **Brownian motion is nowhere differentiable**. To examine differentiability, set $\alpha = 1$. Then as $\Delta t \to 0^+$

   $$P\left(\left|\frac{\Delta B}{\Delta t}\right| \le k\right) \to 0, \quad \forall k \in \mathbb{R}^+$$

   Thus, for every fixed $k > 0$, the ratio $\frac{\Delta B}{\Delta t}$ exceeds k with probability 1 as $\Delta t \to 0^+$. Heuristically,

   $$|\Delta B| \sim \sqrt{\Delta t} \quad \Rightarrow \quad \frac{|\Delta B|}{\Delta t} \sim \frac{1}{\sqrt{\Delta t}} \xrightarrow{\Delta t \to 0^+} \infty$$

   which means the derivative cannot exist anywhere.

Hence, Brownian paths are:

- continuous,
- extremely irregular,
- nowhere differentiable.

The last property is compatible with the Markov property: if the derivative existed, it would carry information about the future, contradicting the memoryless nature of the process.

7.3.4 Brownian motion with drift

We can add a deterministic drift and a general diffusion coefficient without difficulty. A Brownian motion with drift μ and variance parameter D is defined as:

$$x(t) = x_0 + \mu t + \sqrt{D}B(t) \tag{7.4}$$

Since $B(t)$ has mean 0 and variance t,

$$\mathbb{E}\left[x(t)\right] = x_0 + \mu t, \qquad \mathrm{Var}(x(t)) = Dt$$

(verify that).

Formally, we write the stochastic differential equation (SDE):

$$dx(t) = \mu dt + \sqrt{D}dB(t), \qquad \underbrace{\frac{\mathrm{d}x}{\mathrm{d}t} = \cancel{\mu + \sqrt{D}\frac{\mathrm{d}B}{\mathrm{d}t}}}_{\text{does not exist}} \tag{7.5}$$

In particular, we emphasized that the derivative of $x(t)$ does not exist. Only the increment $dB(t)$ is meaningful in the stochastic sense.

7.3.5 How to simulate a simple SDE

We now want to simulate a sample path of the SDE in Equation 7.5, starting from a given initial condition x_0. For a Brownian motion, increments satify

$$\Delta B \sim B(\Delta t) \sim \sqrt{\Delta t}B(1)$$

meaning that for any time step $\Delta t = t_i - t_{i-1}$ the increment is Gaussian with mean zero and variance Δt. Using this property, we discretize Equation 7.5 over a finite set of times $t_0 = 0 < t_1 < t_2 < \ldots < t_n$. The Euler-Maruyama approximation gives

$$\begin{cases} x(t_i) = x(t_{i-1}) + \mu\Delta t + \sqrt{D\Delta t}\varepsilon_i, & i = 1, 2, \ldots, n \\ x(0) = x_0 \end{cases} \tag{7.6}$$

where each $\varepsilon_i \sim \mathcal{N}(0, 1)$ is an independent standard normal variable. Thus, to generate one realization of the process:

1. Start from x_0
2. Draw n i.i.d. samples $\varepsilon_1, \ldots, \varepsilon_n \sim \mathcal{N}(0, 1)$.
3. Update the position iteratively using Equation 7.6

Each increment introduces an independent random shock, reflecting the fact that the SDE has no memory: the noise at time t_i is independent of all past values. The parameters μ and D determine, respectively, the deterministic drift and the strength of the fluctuations.

In particular, choosing $\mu = 0$, $D = 1$, and $x_0 = 0$ generates a standard Brownian motion, satisfying properties (A), (B), and (C).

7.4 Markov processes

We now discuss the memoryless property previously mentioned for Brownian motion. This naturally leads to the notion of Markov processes, a broad class of stochastic processes for which many analytical calculations become tractable. Non-Markovian processes can also be treated, but they typically require more sophisticated tools.

Consider a stochastic process observed at ordered times $t_0 < t_1 < \ldots < t_n$ and assume that a conditional pdf exists at the last observation time t_n:

$$p_{1|n-1}(x_n, t_n | x_{n-1}, t_{n-1}; \ldots; x_1, t_1; x_0, t_0)$$

where the probability is conditioned on $n-1$ previous variables, which are not random variables, but given numbers.

This quantity expresses the probability of observing the value x_n at time t_n, given the entire past history of the process.

To make predictions under this description, one must know the full trajectory realized up to time t_{n-1}, which is a large amount of information. However, many important processes – including Brownian motion – have the property that the influence of the past can be fully summarized by the most recent value. Formally, if for every $n \geq 1$

$$p_{1|n-1}(x_n, t_n | x_{n-1}, t_{n-1}; \ldots; x_1, t_1; x_0, t_0) = p_{1|1}(x_n, t_n | x_{n-1}, t_{n-1}) \tag{7.7}$$

then the process is said to be **Markovian**. In this case, predictions at time t_n depend only on the immediately preceding value at time t_{n-1}: the process is memoryless.

The function $p_{1|1}$ is called the **transition probability** or **propagator**.

A Markov process can be fully characterized by two functions:

1. the initial distribution $p_1(x_0, t_0)$
2. the propagator $p_{1|1}$

Indeed, the 3-point joint pdf of three observations at times $t_0 < t_1 < t_2$ factorizes as:

$$\begin{aligned} p_3(x_2t_2; x_1t_1; x_0t_0) &= \underbrace{p_{1|2}(x_2t_2|x_1t_1; x_0t_0)}_{=p_{1|1}(x_2t_2|x_1t_1)}\, p_2(x_1t_1; x_0t_0) = \\ &\overset{\text{Markov}}{=} p_{1|1}(x_2t_2|x_1t_1)\, p_{1|1}(x_1t_1|x_0t_0)\, p_1(x_0t_0) \end{aligned} \tag{7.8}$$

where we used the Markov property, which states that the conditional probability depends only on the immediately preceding step, together with the standard identity that expresses a three-point joint probability as the product of a conditional probability and a two-point joint probability. By iterating this reasoning, for any sequence of times, we obtain the general factorization:

$$p_n(x_nt_n; x_{n-1}t_{n-1}; \ldots; x_0t_0) = \left[\prod_{i=1}^{n} p_{1|1}(x_it_i|x_{i-1}t_{i-1})\right] p_1(x_0t_0) \tag{7.9}$$

which expresses the entire joint distribution in terms of the initial condition and the propagator. This multiplicative structure is exactly what makes Markov processes mathematically manageable.

Obs. The joint pdf $p_n(x_nt_n; \ldots; x_0t_0)$ is a probability density in all its arguments. Thus, integrating over all variables yields 1. This is different from a conditional density such as $p_{1|1}(x, t|x', t')$, which is normalized only with respect to its first argument.

7.4.1 Chapman-Kolmogorov equation

A natural question is whether any positive and normalizable function can serve as the propagator of a Markov process. The answer is no: not every such function is admissible. A Markov process is fully determined by two functions – the propagator $p_{1|1}$ and the initial distribution p_1 – but these two functions must satisfy specific identities. One of them is the Chapman-Kolmogorov equation.

Starting from the L.H.S of Equation 7.8, integrating with respect to x_1:

$$\int dx_1\, p_3(x_2t_2; x_1t_1; x_0t_0) = p_2(x_2t_2; x_0t_0)$$

Now consider the R.H.S.:

$$\int dx_1\, p_{1|1}(x_2t_2|x_1t_1)\, p_{1|1}(x_1t_1|x_0t_0)\, p_1(x_0t_0)$$

Using the identity

$$p_2(x_2t_2; x_0t_0) = p_{1|1}(x_2t_2|x_0t_0)\, p_1(x_0t_0)$$

we obtain

$$\int dx_1\, p_{1|1}(x_2t_2|x_1t_1)\, p_{1|1}(x_1t_1|x_0t_0)\, p_1(x_0t_0) = p_{1|1}(x_2t_2|x_0t_0)\, p_1(x_0t_0) \tag{7.10}$$

Because we integrate over x_1, the factor $p_1(x_0t_0)$ can be pulled out of the integral and canceled from both sides. We obtain, for $t_2 > t_1 > t_0$

$$p_{1|1}(x_2t_2|x_0t_0) = \int dx_1\, p_{1|1}(x_2t_2|x_1t_1)\, p_{1|1}(x_1t_1|x_0t_0) \tag{7.11}$$

which is the **Chapman-Kolmogorov identity**. Every valid Markov propagator must satisfy it.

If we further integrate Equation 7.10 also with respect to x_0, we obtain:

$$\int dx_0\, p_{1|1}(x_2t_2|x_0t_0)\, p_1(x_0t_0) = p_1(x_2t_2)$$

and analogously,

$$\int dx_0\, p_{1|1}(x_1t_1|x_0t_0)\, p_1(x_0t_0) = p_1(x_1t_1)$$

Thus, for $t_2 > t_1$

$$p_1(x_2t_2) = \int dx_1\, p_{1|1}(x_2t_2|x_1t_1)\, p_1(x_1t_1) \tag{7.12}$$

which is the second identity linking propagator and initial distribution. Therefore, p_1 and $p_{1|1}$ cannot be chosen independently.

In summary, a Markov process is defined by

- the propagator $p_{1|1}$
- the initial distribution p_1

but these must satisfy:

- the Chapman-Kolmogorov identity Equation 7.11
- the consistency relation Equation 7.12

It can be shown that any pair of non-negative functions satisfying these two identities defines a valid Markov process.

7.4.2 Stationary Markov process

If the initial distribution p_1 does not depend on time and coincides with the equilibrium distribution of the process, and if the propagator $p_{1|1}$ depends only on the time difference $|t_2 - t_1|$, then the process is called a **stationary Markov process**.

7.4.3 The Brownian motion is a Markov process

This property follows directly from the independence of increments used in the definition of Brownian motion. This can be proven in different ways. In our context, we can verify it by using the propagator in Equation 7.1

$$p_{1|1}(x_1t_1|x_0t_0) \equiv w(x_1, t_1|x_0, t_0) = \frac{1}{\sqrt{2\pi(t_1 - t_0)}} e^{-\frac{(x_1-x_0)^2}{2(t_1-t_0)}} \tag{7.13}$$

and show by direct integration that this propagator satisfies (exercise) the Chapman-Kolmogorov identity (Equation 7.11).

Now take the initial distribution

$$p_1(y, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}}$$

One can verify that it satisfies Equation 7.12, and that

$$\lim_{t\to 0} p_1(y, t) = \delta(y)$$

These choices define a non-stationary Markov process.

Exercise 7.4.1. *Show that the Poisson process defined by*

$$p_{1|1}(n_2t_2|n_1t_1) = \frac{(t_2 - t_1)^{n_2-n_1}}{(n_2 - n_1)!} e^{-(t_2-t_1)}, \qquad t_2 > t_1$$

and initial condition

$$p_1(n, 0) = \delta_{n,0} \quad \Rightarrow \quad p_1(n, t) = \frac{t^n}{n!} e^{-t}$$

is a non-stationary Markov process with discrete state space.

Exercise 7.4.2. *Show that for $x, y = \pm 1$ the function*

$$p_{1|1}(xt|yt') = \frac{1}{2}\left[1 + e^{-2\gamma(t-t')}\right]\delta_{x,y} + \frac{1}{2}\left[1 - e^{-2\gamma(t-t')}\right]\delta_{x,-y}$$

satisfies the Chapman-Kolmogorov identity in Equation 7.11.

Exercise 7.4.3. *Show that the process defined by*

$$p_1(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$p_{1|1}(xt|y, t') = \frac{1}{\sqrt{2\pi(1 - e^{-2|t-t'|})}} \exp\left[-\frac{(x - ye^{-|t-t'|})^2}{2(1 - e^{-2|t-t'|})}\right]$$

is a stationary Markov process. This is called the ***Ornstein-Uhlenbeck process****.*

Chapter 8

Path Integrals

Ref. Techniques and Applications of Path Integration by L.S. Schulman.

We now introduce the connection between Markov processes and path integrals. This connection is useful because it lets us reinterpret stochastic processes using tools familiar from physics – such as action and Lagrangian – which appear naturally once we rewrite propagators as integrals over paths.

8.1 The Wiener measure and the Wiener path integral

In general, path integrals are difficult to handle rigorously, but Brownian motion provides a case where the construction is clean and closely parallels the free particle in quantum mechanics.

Since the Brownian propagator satisfies the Chapman-Kolmogorov identity – because Brownian motion is a Markov process – we may combine Equation 7.13 and 7.11 and write:

$$w(x_2, t_2|x_0, t_0) = \int_{-\infty}^{\infty} w(x_2, t_2|x_1, t_1) w(x_1, t_1|x_0, t_0)\, dx_1 \tag{8.1}$$

Iterating Equation 8.1 n times for $t_n > t_{n-1} > \ldots > t_0$ yields

$$\begin{aligned} w(x_n, t_n|x_0, t_0) = \int_{-\infty}^{\infty} w(x_n, t_n|x_{n-1}, t_{n-1}) w(x_{n-1}, t_{n-1}|x_{n-2}, t_{n-2}) \ldots \\ \ldots w(x_1, t_1|x_0, t_0)\, dx_{n-1} \ldots dx_1 \end{aligned} \tag{8.2}$$

Obs. This is fully analogous to the identity used in quantum mechanics

$$\langle x_2 t_2|x_0 t_0\rangle = \int dx_1\ \langle x_2 t_2|x_1 t_1\rangle \langle x_1 t_1|x_0 t_0\rangle$$

obtained by inserting the resolution identity $\int dx_1\, |x_1 t_1\rangle \langle x_1 t_1| = 1$.

The expression above suggests how to compute the probability of finding a Brownian particle in prescribed sets at prescribed times. Define the **Wiener measure** at times $t_n > t_{n-1} > \ldots > t_1 > t_0$ as:

$$\mathbb{Q}_n[dx] = \prod_{i=1}^{n} w(x_i t_i|x_{i-1} t_{i-1})\, dx_i$$

Given n measurable sets $A_1, \ldots, A_n$, the probability that $x(t_1) \in A_1$, $x(t_2) \in A_2$, ..., $x(t_n) \in A_n$, is

$$\mathbb{Q}_n(\{A\}) = \int_{A_n} \int_{A_{n-1}} \ldots \int_{A_1} w(x_n, t_n | x_{n-1}, t_{n-1}) \ldots w(x_1, t_1 | x_0, t_0)\, dx_n \ldots dx_1 \tag{8.3}$$

In other words, we integrate the product of n propagators over the sets $A_1, \ldots, A_n$. The result gives the probability that a Brownian particle, starting from x_0 at time t_0, lies in a region A_1 at time t_1, A_2 at time t_2, and so on.

In Figure 8.1 a trajectory of a Brownian particle is shown together with a selection of intervals A_i in the state space. The Wiener measure allows us to compute the probability that the process lies inside the chosen set A_i at each time t_i. In this way, we can describe the behavior of the entire family of trajectories consistent wit these conditions.

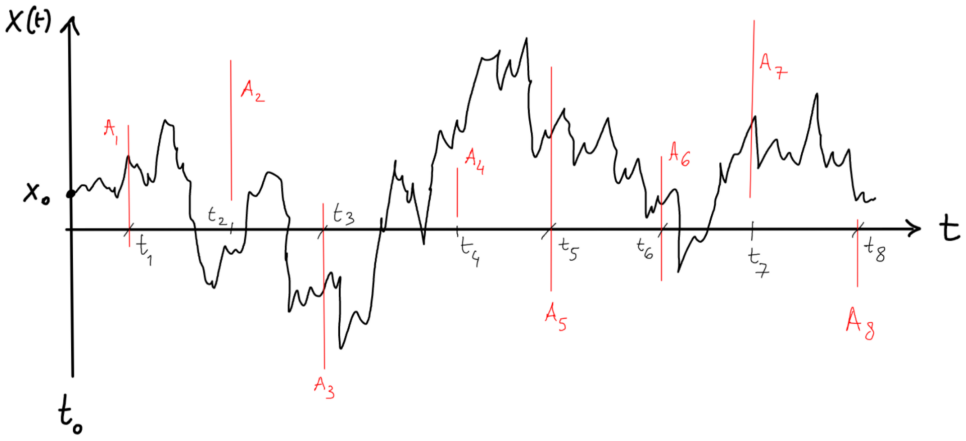


Figure 8.1

Using the Gaussian form of the Brownian propagator, the Wiener measure becomes:

$$\mathbb{Q}_n[dx] = e^{-\frac{1}{2D}\sum_{i=1}^{n} \frac{(x_i - x_{i-1})^2}{\Delta t_i}} \prod_{i=1}^{n} \frac{dx_i}{\sqrt{2\pi D \Delta t_i}} \tag{8.4}$$

where $\Delta t_i = t_i - t_{i-1} > 0$.

This formula is valid for fixed $n \in \mathbb{N}$. The measure extends naturally to the σ-algebra generated by the cylindrical sets $A = \{\, x(t) : x(t_1) \in A_1, \ldots, x(t_n) \in A_n \,\}$.

As $n \to \infty$ with $\max\{\Delta t_i\} \to 0$, we obtain the so called **Wiener path integral**. In this case, the exponent can be written as

$$\sum_{i=1}^{n} \frac{(x_i - x_{i-1})^2}{\Delta t_i} = \sum_{i=1}^{n} \left(\frac{x_i - x_{i-1}}{\Delta t_i} \right)^2 \Delta t_i \xrightarrow[\sum_i \Delta t_i = T]{\substack{n\to\infty \\ \max(\Delta t_i)\to 0}} \int_0^T (\dot{x}(s))^2\, ds$$

This leads to the heuristic expression

$$\mathbb{Q}_n[\mathcal{D}x] = N e^{-\frac{1}{2D}\int_0^T ((\dot{x}))^2\, ds} \mathcal{D}x(\tau) \tag{8.5}$$

where $\dot{x}(t)$ does not actually exist. In fact, this expression is only formal. Since $x(t)$ is nowhere differentiable, $\dot{x}(t)$ is meaningless. However, it is very useful for calculating averages of functionals of the entire trajectory.

Another way to look at Equation 8.2 is the following. Fix $x_n = x$ and $t_n = T$. Then all the successive integrations over $x_{n-1}, \dots, x_1$ at the corresponding times $t_{n-1}, \dots, t_1$ show that in the limit $n \to \infty$ (with fixed initial conditions x, T; x_0, t_0) the propagator w can be written as

$$w(x, T|x_0, t_0) = \int_{x(0)=0}^{x(T)=x} \mathcal{D}x(\tau)\, e^{-\frac{1}{2D}\int_0^T \dot{x}(s)^2\, ds} = \int_{x_0}^{x(T)} \mathcal{D}x\, e^{-S[x(s)]} \tag{8.6}$$

That is, the propagator can be written as an integral over all trajectories joining the initial point (t_0, x_0) and (T, x). Mathematically, this is a functional integral; in physics, its meaning becomes precise only after discretizing time.

In practice, if the process starts at x_0 and ends at x at time T, the path integral sums over all possible trajectories connecting these two points, as illustrated in Figure 8.2.

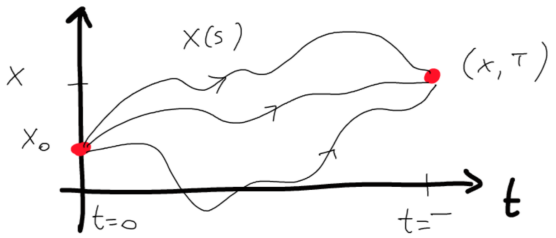


Figure 8.2

Here $S[x(s)]$ is the **action**, which can be written as

$$S = \int_0^T L[x]\, ds$$

where L is the **Lagrangian** of the system.

For a Brownian motion $L = \frac{1}{2D}\left(\frac{\mathrm{d}x}{\mathrm{d}s}\right)^2$ which is formally analogous to the kinetic energy of a free particle: the derivative plays the role of velocity, while $1/D$ behaves like an effective mass.

8.1.1 Analogy with quantum mechanics

If we perform the analytic continuation, by changing variable $s = \frac{it'}{\hbar}$, we identify $D = \frac{\hbar^2}{m}$ and introduce the imaginary time $\tilde{t} = -i\hbar T$, then the Brownian propagator Equation 8.6 becomes

$$w(x, \tilde{t}|x_0, 0) = \int_{x(0)=0}^{x(\tilde{t})=x} \mathcal{D}x\, e^{\frac{i}{\hbar}\int_0^{\tilde{t}} \frac{1}{2}m\left(\frac{\mathrm{d}x}{\mathrm{d}t'}\right)^2 dt'} \overset{7.13}{=} \sqrt{\frac{m}{2\pi\hbar i\tilde{t}}}\, e^{\frac{im}{2\hbar\tilde{t}}(x-x_0)^2} \tag{8.7}$$

which is the quantum-mechanical propagator of a free particle. In particular, this is the propagator (or **Green's function**) of the Schrödinger equation:

$$i\hbar\frac{\partial\Psi}{\partial\tilde{t}} = \hat{H}\Psi, \qquad \hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

Thus, the propagator can be written compactly in the position basis as

$$w(x, \tilde{t}|x_0, 0) = \langle x|\, e^{-\frac{i}{\hbar}\tilde{t}\hat{H}}\, |x_0\rangle$$

Correspondingly, in the momentum basis we have:

$$\hat{H}\,|k\rangle = \frac{\hbar^2}{2m}k^2\,|k\rangle = \frac{Dk^2}{2}\,|k\rangle$$

and the propagator is:

$$\begin{aligned} w(x,T|x_0,0) &= \langle x|\,e^{-TH}\,|x\rangle = \int dk \int dk'\ \langle x|k'\rangle\,\langle k'|\,e^{-TH}\,|k\rangle\,\langle k|x\rangle \\ &= \frac{1}{2\pi}\int dk\, e^{-\frac{D}{2}Tk^2}e^{ik(x-x_0)} = \frac{1}{\sqrt{2\pi DT}}e^{-\frac{(x-x_0)^2}{2DT}} \end{aligned} \tag{8.8}$$

This construction generalizes to a quantum Hamiltonian containing a potential

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

(see Schulman, ch. 9 and ch. 26)

8.1.2 Two-point correlation function

We now compute the two-point correlation function of a Brownian particle by using the Wiener measure. Using Equation 8.4, we want to evaluate

$$\langle x(t_1)x(t_2)\rangle\,, \qquad t_0 < t_1 < t_2$$

averaged across all trajectories, assuming the particle starts at $x_0 = x(t_0)$.

Geometrically, we have many possible trajectories (Figure 8.3), and we select the values at times t_1 and t_2 and average their product over all possible stochastic paths.

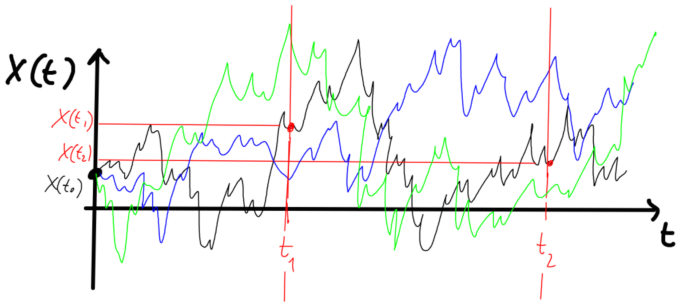


Figure 8.3

Because the Wiener process is Markovian, the joint pdf factorizes and using the Gaussian form of the propagatos we have:

$$\langle x(t_1)x(t_2)\rangle_w \overset{8.4}{=} \int_{\mathbb{R}} dx_2 \int_{\mathbb{R}} dx_1 \underbrace{\frac{e^{-\frac{1}{2D}\frac{(x_2-x_1)^2}{t_2-t_1}}}{\sqrt{2\pi D(t_2-t_1)}}\frac{e^{-\frac{1}{2D}\frac{(x_1-x_0)^2}{t_1-t_0}}}{\sqrt{2\pi D(t_1-t_0)}}}_{\text{Wiener measure with 2 points}} x_1x_2 \tag{8.9}$$

By changing variables

$$x = x_1 - x_0, \qquad y = x_2 - x_1$$

we get

$$x_2 = x_0 + x + y, \qquad x_1 = x + x_0$$

The Jacobian is

$$J = \begin{pmatrix} \frac{\partial x_1}{\partial x} & \frac{\partial x_1}{\partial y} \\ \frac{\partial x_2}{\partial x} & \frac{\partial x_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad \Rightarrow \quad |J| = 1$$

Thus, the average becomes

$$\langle x(t_1)x(t_2)\rangle_w = \int dx \int dy \, \frac{e^{-\frac{1}{2D}\frac{y^2}{t_2-t_1}}}{\sqrt{2\pi D(t_2-t_1)}} \frac{e^{-\frac{1}{2D}\frac{x^2}{t_1-t_0}}}{\sqrt{2\pi D(t_1-t_0)}} (x_0 + x + y)(x + x_0) =$$

Expanding the product insider the integral, we have:

$$(x_0 + x + y)(x + x_0) = 2x_0x + x_0^2 + x^2 + yx_0 + xy$$

By using the Gaussian properties of averages, we get that some are zero:

$$\underset{=0}{2x_0 \langle x\rangle} + x_0^2 + \langle x^2\rangle + \underset{=0}{\langle y\rangle x_0} + \underset{=0}{\langle x\rangle \langle y\rangle}$$

Therefore, the smaller time controls the correlation

$$= \int dx \, \frac{e^{-\frac{x^2}{2D(t_1-t_0)}}}{\sqrt{2\pi D(t_1-t_0)}} x^2 + x_0^2 = D(t_1 - t_0) + x_0^2$$

Thus, for $t_1, t_2 > t_0$, we get

$$\langle x(t_1)x(t_2)\rangle_w = D \min\{t_1, t_2\} + x_0^2 \tag{8.10}$$

Exercise 8.1.1. *Calculate* $\langle x(t_1)x(t_2)\rangle_w$ *for a general i.c.* $g(x_0)$*, where* $\int g(x_0)\,dx_0 = 1$*.*

Chapter 9

Introduction to Stochastic Differential Equations

Up to now, we have examined some fundamental properties of Brownian motion: it is the most important continuous stochastic process, tightly connected to the diffusion equation. It is a Markov process with continuous paths, yet extremely irregular ones, as it is nowhere differentiable. We also know how to simulate it, and how to use it to define a path integral, which – although not rigorously defined in full generality – can, at least in its discretized form, be employed to compute physically relevant functionals.

We now want to use Brownian motion as a building block for constructing new continuous stochastic processes. We will do this by introducing stochastic differential equations (SDEs). This requires great care, because continuous stochastic processes come with a variety of subtleties. In what follows, we will focus on the main features. We will begin with additive noise, the simplest case, and then move to multiplicative noise, which is more challenging.

In particular, we will encounter a typical issue of SDEs that does not arise in deterministic differential equations: the problem of discretizing the stochastic path. Since Brownian motion is nowhere differentiable, one must be careful in discretizing it and in deciding which terms of an expansion should be retained. The key message is that:

- with additive noise, physical observables do not depend on the discretization scheme;
- with multiplicative noise, observables do depend on the chosen discretization.

This means that, in the multiplicative case, a specific prescription must be selected. The convention we will adopt is the Itô prescription, which leads naturally to a modified form of calculus – Itô calculus.

9.1 White Gaussian noise

We have already encountered a simple SDE in the form Equation 7.4 and 7.5. This was easy to integrate numerically. In the deterministic case, an ODE has the form

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x,t) \tag{9.1}$$

The question is how to generalize this when we want to include noise.

One might be tempted to write

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \mu + \sigma \dot{B}(t)$$

However, since $B(t)$ is a Brownian process, $\dot{B}(t)$ does not exist as an ordinary function because Brownian motion is continuous but nowhere differentiable. What does exist are the increments $\Delta B = B(t + \Delta t) - B(t)$. Therefore, the only meaningful representation is:

$$dx = \mu \, dt + \sigma \, dB$$

To give a *formal* meaning to $\dot{B}$, we discretize time and examine the behavior of the discrete derivative. Define:

$$\xi_{\Delta,i} = \frac{\Delta B_i}{\Delta t}$$

with $\Delta B_i = B(t_i + \Delta t) - B(t_i)$.

Clearly, $\langle \xi_{\Delta,i} \rangle = 0$. We want to compute the correlation $\langle \xi_{\Delta,i}, \xi_{\Delta,j} \rangle$. To do this, we contract it with a test function $g(t)$, evaluated at discrete times t_i, We denote $g_i \equiv g(t_i)$. Then, we compute

$$\begin{aligned} \sum_j g_j \, \langle \xi_{\Delta,i}, \xi_{\Delta,j} \rangle \, \Delta t = \sum_j g_j \left\langle \frac{\Delta B_i}{\Delta t} \frac{\Delta B_j}{\Delta t} \right\rangle \Delta t = \\ = \sum_j g_j \left\langle \frac{\Delta B_i \Delta B_j}{\Delta t} \right\rangle = \sum_j g_j \delta_{ij} = g_i \end{aligned}$$

where we used $\langle \Delta B_i \Delta B_j \rangle = \delta_{ij} \Delta t$.

Passing to the continuum limit $\Delta t \to 0$ gives

$$\int g(t') \left\langle \xi(t) \xi(t') \right\rangle \, dt' = g(t)$$

Since this holds for any test function g, we can recognize the correlation function, which is

$$\left\langle \xi(t) \xi(t') \right\rangle = \delta(t - t')$$

with zero mean $\langle \xi(t) \rangle = 0$. This process is called **Gaussian white noise** or **δ-correlated noise** and it is the formal interpretation of $\dot{B}$.

Thus, when we write

$$\dot{x}(t) = \mu + \sigma \xi(t) \quad \overset{\text{equiv}}{\Longleftrightarrow} \quad dx = \mu \, dt + \sigma \, dB(t) \tag{9.2}$$

The first expression is the physicists' notation, where ξ is Gaussian white noise with δ-correlation, and the second is the mathematicians' notation, where $B(t)$ is a standard Brownian motion with covariance $\langle B(t) B(s) \rangle = t \wedge s$. Here, the wedge denotes the minimum between t and s.

9.2 Langevin equation

We now want to generalize what we have seen so far. Starting from a deterministic differential equation such as

$$dx = f(x, t) \, dt$$

there are, in principle, many ways of introducing noise. One might try to define an SDE of the form

$$dx = \tilde{f}(x, t, dB)$$

where $\tilde{f}$ is some non-linear smooth function that depends explicitly on the Brownian increment. In practice, however, equations written in this fully non-linear form are extremely difficult – and often impossible – to define rigorously. Even the meaning of solution may not be well posed.

A formulation that does work in a wide range of situations, including higher-dimensional cases, is to introduce noise additively. This leads to SDEs of the form

$$dx = \mu(x, t)\, dt + \sigma\, dB(t), \qquad \sigma > 0$$

For this class of equations, we can develop a consistent theory: existence and uniqueness of solutions can be established under reasonable assumptions, and many analytical tools become available.

This structure is also well motivated physically. For example, if $x(t)$ denotes the position of a large particle suspended in a fluid, the term $\sigma\, dB(t)$ represents the aggregate effect of collisions with the surrounding microscopic particles at temperatures T. In this picture, the coefficient σ depends on T and quantifies the strength of the random kicks. An SDE of this form is known as the **Langevin equation with additive noise**.

9.3 The Onsager-Machlup measure

We now introduce the Onsager-Machlup measure for additive noise SDEs. It plays the role of a path-integral representation of the process, extending the Wiener measure by incorporating the drift and the Jacobian that arises from the change of variables.

Consider the SDE

$$dx_t = \mu(x_t)\, dt + dB_t \tag{9.3}$$

where $B_t \equiv B(t)$ is a Brownian motion and $\mu : \mathbb{R} \to \mathbb{R}$ is sufficiently regular (at least C^1). We discretize the time interval $[t_i, t_f]$ into M steps

$$\Delta t = \frac{t_f - t_i}{M}, \qquad t_j = t_i + j\Delta t$$

and write $x_j = x_{t_j} \equiv x(t_j)$. We adopt the α-point discretization

$$x_{\alpha,j} = (1-\alpha)x_{j-1} + \alpha x_j = x_{j-1} + \alpha \Delta x_j, \qquad \alpha \in [0, 1] \tag{9.4}$$

with $\Delta x_j = x_j - x_{j-1}$. The α-discretized version of the SDE 9.3 becomes

$$x_j = x_{j-1} + \mu(x_{\alpha,j})\Delta t + \sqrt{\Delta t}\xi_j, \qquad j = 1, 2, \ldots, M \tag{9.5}$$

where $\{\, \xi_j \,\}_{j=1}^{M}$ are i.i.d. standard normal random variables $\xi_j \sim \mathcal{N}(0, 1)$. Solving Equation 9.5 for ξ_j:

$$\xi_j = \frac{x_j - x_{j-1} - \mu(x_{\alpha,j})\Delta t}{\sqrt{\Delta t}} = \frac{\Delta x_j - \mu(x_{\alpha,j})\Delta t}{\sqrt{\Delta t}} \tag{9.6}$$

We take as reference the Wiener measure in the variables $\{\, \xi_j \,\}$:

$$\mathbb{Q}[d\xi] = \prod_{j=1}^{M} \frac{d\xi_j}{\sqrt{2\pi}} e^{-\frac{1}{2}\sum_j \xi_j^2} \tag{9.7}$$

To obtain the measure induced on the process $\{ x_j \}_{j=1}^M$, we compute the Jacobian of the transformation $\{ \xi_j \}_{i=1}^M \to \{ x_j \}_{i=1}^M$, which induces a change from $\mathbb{Q}[d\xi]$ to $\mathbb{P}_\alpha[dx]$ (in general, it may depend on α, this is why we introduced it).

The Jacobian matrix is given by

$$J_{jk} = \frac{\partial \xi_j}{\partial x_k}, \qquad 1 \le j, k \le M$$

Because the scheme is causal, the Jacobian matrix is lower triangular: each x_j depends only on the noise variables $\xi_1, \dots, \xi_j$, and never on future noise values. This causality implies that $\partial \xi_i / \partial x_j = 0$ whenever $j > i$, so the Jacobian has non-zero entries only and below the diagonal. As a consequence, its determinant is simply the product of the diagonal elements.

For $k = j$, the element of the Jacobian is:

$$\frac{\partial \xi_j}{\partial x_j} = \frac{1 - \alpha \mu'(x_{\alpha,j}) \Delta t}{\sqrt{\Delta t}} \tag{9.8}$$

while for $k = j - 1$ we have another term:

$$\frac{\partial \xi_j}{\partial x_{j-1}} = \frac{-1 - (1-\alpha) \mu'(x_{\alpha,j}) \Delta t}{\sqrt{\Delta t}} \tag{9.9}$$

and all other terms are zero. Therefore, we have:

$$\det J = \prod_{j=1}^M \frac{\partial \xi_j}{\partial x_j} = \left(\frac{1}{\sqrt{\Delta t}} \right)^M \prod_{j=1}^M [1 - \alpha \mu'(x_{\alpha,j}) \Delta t] \tag{9.10}$$

By taking the logarithm, we obtain

$$\log \det J = -\frac{M}{2} \log (\Delta t) + \sum_{j=1}^M \log [1 - \alpha \mu'(x_{\alpha,j}) \Delta t] \tag{9.11}$$

For small Δt, we can expand it as

$$\log \det J \simeq -\frac{M}{2} \log (\Delta t) + \sum_j -\alpha \mu'(x_{\alpha,j}) \Delta t \tag{9.12}$$

so, up to an irrelevant constant, which is independent of the path, we can write

$$\log \det J = -\alpha \sum_{j=1}^M \mu'(x_{\alpha,j}) \Delta t + o(\Delta t) \tag{9.13}$$

and

$$\det J \propto e^{-\alpha \sum_j \mu'(x_{\alpha,j}) \Delta t} \tag{9.14}$$

Substituting the expression for ξ_j and combining it with the Jacobian, the path weight in the x_j-variables becomes

$$\mathbb{P}_\alpha[dx] \propto \prod_j dx_j \, (\det J) e^{-\frac{1}{2} \sum_j \xi_j^2} \tag{9.15}$$

The factor $(\det J) e^{-\frac{1}{2} \sum_j \xi_j^2}$ is the **discrete Onsager-Machlup measure**.

In the stochastic framework, the standard procedure is to start from the SDE, discretize it, perform all calculations at the discrete level, and only at the end return to the continuous formulation by taking the appropriate limit. We follow exactly the same logic here: having carried out the discrete calculation, we now take the continuum limit.

From Equation 9.6, we get

$$\frac{1}{2}\xi_j^2 = \frac{1}{2}\frac{(\Delta x_j - \mu(x_{\alpha,j})\Delta t)^2}{\Delta t} \tag{9.16}$$

By taking the sum

$$-\frac{1}{2}\sum_j \xi_j^2 = -\sum_j \frac{(\Delta x_j)^2}{2\Delta t} + \sum_j \Delta x_j \mu(x_{\alpha,j}) - \frac{1}{2}\sum_j \mu(x_{\alpha,j})^2 \Delta t \tag{9.17}$$

By combining Equation 9.14 with 9.17 in 9.15, we obtain

$$\begin{aligned}\mathbb{P}_\alpha[dx] \propto \prod_{j=1}^{M} dx_j \, \exp\Bigg\{ &-\sum_j \frac{(\Delta x_j)^2}{2\Delta t} + \sum_j \Delta x_j \mu(x_{\alpha,j}) - \\ &- \frac{1}{2}\sum_j \mu(x_{\alpha,j})^2 \Delta t - \alpha \sum_j \mu'(x_{\alpha,j})\Delta t \Bigg\} = \prod_j dx_j \, e^{-S_\alpha[x]}\end{aligned} \tag{9.18}$$

where $S_\alpha[x]$ is the **discrete Onsager-Machlup action**:

$$S_\alpha[x] = \sum_j \frac{(\Delta x_j)^2}{2\Delta t} - \sum_j \Delta x_j \mu(x_{\alpha,j}) + \frac{1}{2}\sum_j \mu(x_{\alpha,j})^2 \Delta t + \alpha \sum_j \mu'(x_{\alpha,j})\Delta t \tag{9.19}$$

which also has the form

$$S_\alpha[x] = \sum_j \left[\frac{(\Delta x_j - \mu(x_{\alpha,j})\Delta t)^2}{2\Delta t} + \alpha \mu'(x_{\alpha,j})\Delta t \right] \tag{9.20}$$

Taking the continuum limit $\Delta t \to 0$

$$\frac{\Delta x_j}{\Delta t} \to \dot{x}(t_j), \qquad x_{\alpha,j} \to x(t_j) \tag{9.21}$$

we obtain the formal Onsager-Machlup action

$$S_\alpha[x] \to \int_{t_i}^{t_f} \left[\frac{1}{2}(\dot{x}(t) - \mu(x(t)))^2 + \alpha \mu'(x(t)) \right] dt \tag{9.22}$$

and the corresponding **Onsager-Machlup Lagrangian**

$$\mathcal{L}_\alpha(x, \dot{x}) = \frac{1}{2}(\dot{x} - \mu(x))^2 + \alpha \mu'(x) \tag{9.23}$$

The Lagrangian depends explicitly on the discretization parameter α. For additive noise, this dependence does not affect observables of the continuous SDE: all α-schemes converge to the same process as $\Delta t \to 0$. The α-dependence reflects the discretization convention inside the path-integral representation, not a physical ambiguity.

In contrast, for multiplicative noise the choice of discretization becomes physically meaningful, and one must specify whether the SDE is interpreted in the Itô, Stratonovich, or anti-Itô sense.

We have the following discretization conventions:

- $\alpha = 0$ – **prepoint** or **Itô prescription**
- $\alpha = \frac{1}{2}$ – **midpoint** or **Stratonovich prescription**
- $\alpha = 1$ – **postpoint** or **Anti-Itô prescription**

Remark. At the level of the formal Onsager-Machlup Lagrangian, the choice of discretization scheme does matter: different values of α lead to different expression for the Lagrangian, because the derivation explicitly depends on how the drift term is evaluated along the discrete path.

However, the continuous-time SDE 9.3 defines a unique stochastic process (under the crucial assumption of additive noise). In this setting, the various α-schemes correspond only to different numerical discretizations that all converge to the same process as $\Delta t \to 0$.

Thus, the α-dependence in Equation 9.23 should be interpreted as a feature of the path-integral representation (analogous to a gauge choice), not of the underlying stochastic dynamics.

In fact, one can show that for additive noise the expectation of any functional of the path – such as averages, correlations, or response functions – computed with the α-Onsager-Machlup measure is independent of α. This ceases to be true for multiplicative noise, where the discretization scheme becomes physically meaningful and the SDE must be explicitly interpreted (Itô, Stratonovich, etc).

9.4 Numerical scheme in additive noise case

Here we show that, for SDEs with additive noise, the choice of numerical discretization scheme becomes irrelevant as $\Delta t \to 0$. In fact, all schemes differ only by higher-order terms. Consider the Langevin Equation 9.3 with additive noise:

$$dx_t = \mu(x_t)\, dt + dB_t \tag{9.24}$$

where dB_t is a standard Brownian motion and $\mu : \mathbb{R} \to \mathbb{R}$ is a smooth drift. Fix $\Delta t > 0$, and define $t_j = t_0 + j\Delta t$. A generic λ-scheme discretization from Equation 9.24 is

$$x_j = x_{j-1} + \mu(x_{\lambda,j})\Delta t + \Delta B_j \tag{9.25}$$

with

$$x_{\lambda,j} \coloneqq (1-\lambda)x_{j-1} + \lambda x_j, \qquad \Delta B_j \coloneqq B_{t_j} - B_{t_{j-1}}, \qquad \Delta x_j \coloneqq x_j - x_{j-1}$$

We expand the drift μ around x_{j-1}:

$$\mu(x_{\lambda,j}) = \mu(x_{j-1}) + \lambda\mu'(x_{j-1})\Delta x_j + o((\Delta x_j)^2) \tag{9.26}$$

Substituting this into the scheme gives:

$$\Delta x_j = \mu(x_{j-1})\Delta t + \lambda\mu'(x_{j-1})\Delta x_j \Delta t + o(\Delta x_j^2 \Delta t) + \Delta B_j \tag{9.27}$$

Since increments of Brownian motion satisfy $\Delta B_j = O(\sqrt{\Delta t})$, we also have $\Delta x_j = P(\sqrt{\Delta t})$. Therefore,

$$\Delta x_j \Delta t = O(\Delta t^{3/2}) \qquad \text{and} \qquad (\Delta x_j)^2 \Delta t = O(\Delta t^2)$$

and at leading order we obtain

$$\Delta x_j = \mu(x_{j-1})\Delta t + \Delta B_j + o(\Delta t^{3/2})$$

Thus, at leading order, there is no dependence on λ: all schemes differ only by terms of order $\Delta t^{3/2}$ or smaller. The leading-order increment:

$$\Delta x_j = \mu(x_{j-1})\Delta t + \Delta B_j$$

is the **Euler-Maruyama update** for additive noise (algorithm).

Therefore, for sufficiently small time steps, the discretization choice is irrelevant: the smoothness of μ ensures that scheme-dependent contributions appear only in higher-order terms.

This is not the case for multiplicative noise, where the increment $\sigma(x)\Delta B_j$ requires special care, and different discretization schemes yield different limits.

9.5 Multiplicative Langevin equation

We now show where numerical and conceptual problems arise when dealing with multiplicative noise. Let us start, for instance, with the simple linear SDE

$$dx = -\mu x\, dt + \sigma\, dB$$

with $\mu > 0$ and $\sigma > 0$. The mean is easy to compute

$$\langle dx \rangle = -\mu \langle x \rangle\, dt$$

because $\langle dB \rangle = 0$. Therefore, the derivative of the average is

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle x \rangle = -\mu \langle x \rangle$$

whose solution is $\langle x(t) \rangle = x_0 e^{-\mu t}$.

The difficulty appears when we look at fluctuations. We want an equation for $\langle x^2 \rangle$, because it give a first hint of the order of the fluctuations. Ordinary calculus would suggest:

$$\frac{\mathrm{d}x^2}{\mathrm{d}t} = 2x\frac{\mathrm{d}x}{\mathrm{d}t}$$

and multiplying the SDE by $2x$ gives

$$2x\, dx = -2\mu x^2\, dt + 2\sigma x\, dB$$

Taking the expectation:

$$2\langle x\, dx \rangle = -2\mu \langle x^2 \rangle\, dt + 2\sigma \langle x\, dB \rangle$$

The problem lies in the last term. One might be tempted to write $\langle x\, dB \rangle = 0$, but this is unjustified: $x(t)$ depends on the Brownian path, so x and dB are generally correlated. If one (incorrectly) assumes $\langle x\, dB \rangle = 0$, then

$$\frac{\mathrm{d}\langle x^2 \rangle}{\mathrm{d}t} = -2\mu \langle x^2 \rangle$$

yielding $\langle x^2 \rangle = x_0^2 e^{-2\mu t}$ and the variance becomes

$$\mathrm{Var}(x(t)) = \langle x^2 \rangle - \langle x \rangle^2 = x_0^2 e^{-2\mu t} - x_0^2 e^{-2\mu t} = 0$$

This is clearly impossible: a SDE cannot have zero variance.

This contradiction tells us exactly what went wrong:

- either one incorrectly assumed that $\langle x\, dB \rangle = 0$,
- or one tried to apply ordinary calculus to objects involving dB, which is illegal because Brownian paths are nowhere differentiable.

Thus, either $\langle x\, dB \rangle = 0$ but ordinary calculus cannot be used, or $\langle x\, dB \rangle \neq 0$ – which means they are correlated – but we don't know how to calculate it.

This is a major problem when we consider the **multiplicative SDE** or **multiplicative Langevin equation**:

$$dx = \mu(x,t)\, dt + \sigma(x,t)\, dB \tag{9.28}$$

This equation needs care.

Another way to see the problem is to look at the formal solution of Equation 9.28, which can be written as:

$$x(t) = x(0) + \int_0^t \mu(x(s), s)\, ds + \int_0^t \sigma(x(s), s)\, dB(s) \tag{9.29}$$

Here $x(t)$ is a functional of the entire Brownian path $B(s)$, $0 \leq s \leq t$. The difficulty lies in the last term

$$\int_0^t \sigma(x(s))\, dB(s)$$

We ask the meaning of this, since it involves a nowhere-differentiable increment $dB(s)$, multiplied by a function $\sigma(x(s))$, where $x(s)$ itself depends on the Brownian motion. This object is, therefore, a **stochastic integral**, and it cannot be interpreted using ordinary Riemann calculus. A precise limiting prescription is required.

Let's see another example. Let's consider the SDE

$$dx = B\, dB, \qquad x(0) = 0$$

with $\mu = 0$ and $\sigma(x) = x$. Formally, the solution is:

$$x(t) = \int_0^t B(s)\, dB(s)$$

But what is the meaning of this integral? A naive application of ordinary calculus would wrongly suggest:

$$x(t) = \frac{B(t)^2}{2}$$

Then one would conclude $\langle x \rangle = \frac{t}{2} \neq 0$, which is non-zero. However, this is not correct. As we will see, this is true only if we use the Stratonovich interpretation.

9.5.1 Quadratic finite variation for Brownian motion lemma

We now introduce a theorem that is essential for understanding how stochastic integrals must be defined and why ordinary calculus cannot be applied to Brownian paths.

First, we define the following:

Definition 9.5.1. A sequence x_n converges in **mean square** to x, $x_n \xrightarrow{\text{ms}} x$, if and only if:

$$\lim_{n\to\infty} \mathbb{E}\left[(x_n - x)^2\right] = 0$$

This is known as **mean square limit** and can be write as:

$$\underset{n\to\infty}{\text{ms-lim}}\ x_n = x$$

Now we introduce the following lemma, which establishes that Brownian motion has finite quadratic variation.

Lemma 9.5.2 (Finite quadratic variation of Brownian motion). *Let $t > 0$. Consider the interval $[0, t]$ and partitions $\mathbb{P}_n$ of the interval*

$$\mathbb{P}_n = \{\, t_0 = 0 < t_1 < t_2 < \ldots < t_n = t \,\}$$

whose mesh

$$|\mathbb{P}_n| \equiv \max_i |t_i - t_{i-1}|$$

satisfies $|\mathbb{P}_n| \to 0$ as $n \to \infty$. Then

$$\underset{\substack{n\to\infty \\ |\mathcal{P}_n|\to 0}}{\text{ms-lim}} \sum_{i=0}^{n-1} \left[B(t_{i+1}) - B(t_i)\right]^2 = t \tag{9.30}$$

This means that although for each finite n the sum is a random variable, its fluctuations shrink as the partition is refined, and the limit becomes a deterministic number.

Proof. Define

$$Q_n = \sum_{i=0}^{n-1} [B(t_{i+1}) - B(t_i)]^2$$

We study the mean-square deviation

$$Q_n - t = \sum_i \left\{ [B(t_{i+1}) - B(t_i)]^2 - (t_{i+1} - t_i) \right\} \tag{9.31}$$

Using $B(t_i) \equiv B_i$, we want to evaluate $\mathbb{E}[(Q_n - t)^2]$. Thus

$$\mathbb{E}[(Q_n - t)^2] = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \mathbb{E}\left[\{(B_{i+1} - B_i)^2 - (t_{i+1} - t_i)\} \{(B_{j+1} - B_j)^2 - ((t_{j+1} - t_j)\} \right]$$

$\boxed{\textbf{case } i \neq j}$ Independence of increments gives

$$\begin{aligned} &\mathbb{E}\left[\{(B_{i+1} - B_i)^2 - (t_{i+1} - t_i)\} \{(B_{j+1} - B_j)^2 - (t_{j+1} - t_j)\} \right] = \\ &= \mathbb{E}\left[\{(B_{i+1} - B_i)^2 - (t_{i+1} - t_i)\} \right] \mathbb{E}\left[\{(B_{j+1} - B_j)^2 - (t_{j+1} - t_j)\} \right] = 0 \end{aligned}$$

We used the property that Brownian increments satisfy $\mathbb{E}[(B_{i+1} - B_i)^2] = t_{i+1} - t_i$, that is, the variance of each increment scales linearly with the time step.

In this way, we remove one summation and Equation 9.31 becomes

$$\begin{aligned}
\mathbb{E}[(Q_n - t)^2] &= \sum_{i=0}^{n-1} \mathbb{E}\left[\{(B_{i+1} - B_i)^2 - (t_{i+1} - t_i)\}^2\right] = \\
&= \sum_i \left\{\mathbb{E}\left[(B_{i+1} - B_i)^4\right] - 2\mathbb{E}\left[(B_{i+1} - B_i)^2\right](t_{i+1} - t_i) + (t_{i+1} - t_i)^2\right\} = \\
&= \sum_i \left\{3(t_{i+1} - t_i)^2 - 2(t_{i+1} - t_i)^2 + (t_{i+1} - t_i)^2\right\} = \\
&= 2\sum_i (t_{i+1} - t_i)^2 \leq 2\sum_i (t_{i+1} - t_i) \max_i (t_{i+1} - t_i) = \\
&= 2|P_n|t \to 0 \text{ as } n \to \infty
\end{aligned}$$

Here, we used the fact that the $\mathbb{E}\left[(B_{i+1} - B_i)^4\right]$ is the 4-th moment of the Gaussian $\mathcal{N}(0, t_{i+1} - t_i)$. Thus, we obtain $3(t_{i+1} - t_i)^2$. Check as an exercise that

$$\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x^4 e^{-\frac{x^2}{2\sigma}}\, dx = 3\sigma^2 \tag{9.32}$$

Finally, we used the definition of partitions, and since t is fixed, the last term goes to zero as $n \to \infty$. □

One can actually show that the convergence is even stronger: the quadratic variation converges almost surely. This is quite remarkable, because although Q_n is a random variable for any finite partition n, in the limit $n \to \infty$ the randomness disappears and it collapses to the deterministic value t.

The lemma confirms and *partially* justifies the heuristic rule $|dB| \simeq \sqrt{dt}$, because squaring increments and summing them yields something of order t, not zero. This property is the key reason stochastic integrals behave differently from classic ones.

9.5.2 Riemann stochastic sum theorem and solutions of the stochastic integral

We now use the result above to define the stochastic integral of B with respect to itself.

Theorem 9.5.3 (Stochastic Riemann integration). *Let's consider the interval* $[0, t]$ *and partitions*

$$\mathbb{P}_n = \{\, t_0 = 0 < t_1 < \ldots < t_n = t \,\} \quad s.t. \quad |\mathbb{P}_n| \equiv \max_i |t_i - t_{i-1}| \to 0 \quad as \quad n \to \infty$$

as before, and choose a point inside each interval:

$$s_i = (1 - \lambda)t_i + \lambda t_{i+1}, \qquad 0 \leq \lambda \leq 1$$

Define the Riemann sum:

$$x_n \equiv \sum_{i=0}^{n-1} B(s_i)[B(t_{i+1}) - B(t_i)]$$

This is the discrete version of

$$\int_0^t B(s)\, dB(s)$$

Thus,

$$\underset{\substack{n\to\infty \\ \mathcal{P}_n\to 0}}{\text{ms-lim}}\ x_n = \frac{[B(t)]^2}{2} + \left(\lambda - \frac{1}{2}\right) t \tag{9.33}$$

This dependence on the sampling point has no analog in classical calculus, and is the central feature that forces the choice of a stochastic convention. In fact, we have the following:

- $\lambda = 1/2$ gives
$$\int_0^t B\, dB = \frac{1}{2}B(t)^2 \qquad \text{(Stratonovich integral)}$$
matching the classical chain rule.

- $\lambda = 0$ gives
$$\int_0^t B\, dB = \frac{1}{2}B(t)^2 - \frac{t}{2} \qquad \text{(Itô integral)}$$

- $\lambda = 1$ gives
$$\int_0^t B\, dB = \frac{1}{2}B(t)^2 + \frac{t}{2} \qquad \text{(anti-Itô integral)}$$

Obs. The quantity x_n is precisely the Riemann-type sum associated with the stochastic integral $x(t) = \int_0^t B(s)\, dB(s)$, where the value of the integrand is evaluated at an arbitrary intermediate point of each interval $[t_i, t_{i+1}]$. In ordinary calculus, Riemann integrals are insensitive to the choice of this intermediate point. In contrast, for stochastic integrals this choice is unavoidable and leads to different limits. This dependence on the discretization point is one of the key structural features of stochastic calculus built on Brownian motion.

Proof. We must prove Equation 9.33. Let's introduce the notation

$$B_i^\lambda \equiv B(\lambda t_{i+1} + (1-\lambda)t_i), \qquad B_i \equiv B(t_i)$$

so that

$$x_n = \sum_i B_i^\lambda (B_{i+1} - B_i)$$

To simplify the computation, it is convenient rewrite the product using increments. We insert and subtract intermediate terms and expand:

$$\begin{aligned} B_i^\lambda(B_{i+1} - B_i) &= (B_i^\lambda - B_i + B_i)(B_{i+1} - B_i^\lambda + B_i^\lambda - B_i) = \\ &= (B_i^\lambda - B_i)(B_{i+1} - B_i^\lambda) + (B_i^\lambda - B_i)^2 + B_i(B_{i+1} - B_i^\lambda) + B_i(B_i^\lambda - B_i) = \\ &= (B_i^\lambda - B_i)(B_{i+1} - B_i^\lambda) + (B_i^\lambda - B_i)^2 + B_i B_{i+1} - B_i^2 \end{aligned}$$

We now rewrite the cross-term $B_i B_{i+1}$ in a useful way:

$$B_i B_{i+1} = -\frac{1}{2}(B_{i+1} - B_i)^2 + \frac{1}{2}B_{i+1}^2 + \frac{1}{2}B_i^2$$

Putting everything together gives the decomposition

$$B_i^\lambda(B_{i+1}-B_i) = \underbrace{(B_i^\lambda - B_i)(B_{i+1}-B_i^\lambda)}_{\text{D}} + \underbrace{(B_i^\lambda - B_i)^2}_{\text{C}} - \underbrace{\frac{1}{2}(B_{i+1}-B_i)^2}_{\text{B}} + \underbrace{\frac{1}{2}(B_{i+1}^2 - B_i^2)}_{\text{A}}$$

We now analyze the contributions of each term in the sum.

- Term A We have

$$\sum_{i=0}^{n-1}\left(\frac{B_{i+1}^2}{2} - \frac{B_i^2}{2}\right) = \frac{1}{2}(B_n^2 - B_{n-1}^2 + B_{n-1}^2 - B_{n-2}^2 + \ldots + B_1^2 - B_0^2) = \\ = \frac{1}{2}(B_n^2 - B_0^2) = \frac{1}{2}B^2(t)$$

  where $B_n = B(t)$ and $B_0 = 0$.

- Term B By quadratic variation of Brownian motion (Theorem 9.5.2), we have:

$$-\frac{1}{2}\sum_i (B_{i+1}-B_i)^2 \quad \xrightarrow{\text{ms-lim}} \quad -\frac{t}{2}$$

- Term C Again, by applying the same Theorem 9.5.2

$$\sum_i (B_i^\lambda - B_i)^2 \quad \xrightarrow{\text{ms-lim}} \quad \lambda t$$

- Term D Finally, we show that the mixed term vanishes in mean-square. By independence of increments on disjoint intervals, we have:

$$\mathbb{E}\left[\left(\sum_{i=0}^{n-1}(B_i^\lambda - B_i)(B_{i+1}-B_i^\lambda)\right)^2\right] \overset{\text{ind. of incr.}}{=} \sum_{i=0}^{n-1}\mathbb{E}\left[(B_i^\lambda - B_i)^2\right]\mathbb{E}\left[(B_{i+1}-B_i^\lambda)^2\right] = \\ = \sum_i \lambda(t_{i+1}-t_i)(1-\lambda)(t_{i+1}-t_i) \leq \\ \leq \lambda(1-\lambda)\max_j(t_{j+1}-t_j)\sum_{i=0}^{n-1}(t_{i+1}-t_i) \leq \\ \leq \lambda(1-\lambda)|\mathbb{P}_n|t \quad \xrightarrow{n\to\infty} \quad 0$$

  Since $|\mathbb{P}_n| \to 0$.

□

In the following, we will always use the Itô prescription ($\lambda = 0$) when dealing with stochastic integrals arising from SDEs of the form Equation 9.28 (i.e., with multiplicative noise).

Obs. Notice that for SDEs with additive noise ($\sigma = \text{const.}$), there is no need to introduce any discretization prescription: all schemes lead to the same continuous-time limit. When the noise is multiplicative, however, the choice of prescription becomes essential.

In general, instead of the Brownian motion B itself (as in Equation 9.33), we may have a functional of the Brownian path. Suppose we want to interpret an integral of the form

$$\int \sigma(x_s)\, dB(s)$$

as appears in Equation 9.29. Here, $\sigma(x_s)$ depends on the past of the process and hence indirectly on the Brownian motion $B(s)$ for $s \leq t$. Importantly, at time t_i we cannot allow σ to depend on future increments $B(t_{i+1}) - B(t_i)$, because this would violate causality.

Functions $\sigma(x_s)$ that depend only on the information available up to time s, and never on future increments, are called **non-anticipating** (or, in probabilistic terms, they are adapted to the natural filtration of Brownian motion).

If $\sigma(x_s)$ is non-anticipating, the Itô stochastic integral corresponding to the choice $\lambda = 0$ (i.e., selecting the left endpoint t_i) is defined as the mean-square limit

$$\int_0^t \sigma(B(s))\, dB(s) \coloneqq \underset{n\to\infty}{\text{ms-lim}} \sum_{i=0}^{n-1} \sigma(B(t_i))[B(t_{i+1}) - B(t_i)]$$

where $\sigma(B(s))$ may be a functional.

Using this definition, one can show the general relation between different λ-prescriptions

$$\int_0^t \sigma(B(s))\, d_\lambda B(s) = \underbrace{\int_0^t \sigma(B(s))\, d_0 B(s)}_{\text{Itô stoch. int}} + \lambda \int_0^t \sigma'(B(s))\, ds$$

This formula highlights explicitly how the choice of discretization affects the value of the stochastic integral when the noise is multiplicative.

Comments: The advantage of choosing the Stratonovich prescription ($\lambda = \frac{1}{2}$) is that one may use ordinary calculus. However, in this case we have

$$\mathbb{E}\left[\left(\int_0^t B(s)\, dB(s)\right)\right]\Bigg|_{\text{Strat.}} = \frac{t}{2}$$

This already shows that the Brownian motion and its future increments are correlated when the midpoint rule is used.

From the Theorem 9.5.3, we get

$$\int_0^t B(s)\, dB(s) = \frac{B^2(s)}{2} + \left(\lambda - \frac{1}{2}\right) t, \qquad \text{(mean-square limit)}$$

Therefore,

- In the Itô case, this is

$$\int_0^t B(s)\, dB(s) = \frac{B^2(t)}{2} - \frac{t}{2}, \qquad \text{Itô integral } (\lambda = 0) \tag{9.34}$$

- In the Stratonovich case, it is

$$\int_0^t B(s)\, dB(s) = \frac{B^2(t)}{2}, \qquad \text{Strato. integ. } (\lambda = \frac{1}{2}) \tag{9.35}$$

We may ask why should we use $\lambda = 0$? Because at time t_i we do not know the value that B will take in the next interval $[t_i, t_{i+1}]$. The only available information is $B(t_i)$. Choosing the left endpoint is therefore the causal choice and leads to the Itô formula Equation 9.34.

Moreover,

$$\mathbb{E}\left[\int_0^t B(s)\, dB(s)\right] = 0$$

which is extremely convenient when dealing with multiplicative noise. In fact, one can show that, in general,

$$\mathbb{E}\left[\int \sigma(s)\, dB(s)\right] = \int \mathbb{E}\left[\sigma(s)\right] \underbrace{\mathbb{E}\left[dB(s)\right]}_{0} = 0$$

for Itô integrals.

One can show that averages (observables) do depend on the chosen prescription. As an example, let us compute $\mathbb{E}\left[\int_0^t B(s)\, dB(s)\right]$. Using the discretized definition of the λ-stochastic integral:

$$\begin{aligned}
&\mathbb{E}\left[\sum_{i=0}^{n-1} B(t_i + \lambda(t_{i+1} - t_i))(B(t_{i+1}) - B(t_i))\right] = \\
&= \mathbb{E}\left[\sum_{i=0}^{n-1} B(t_i + \lambda(t_{i+1} - t_i))B(t_{i+1})\right] - \mathbb{E}\left[\sum_{i=0}^{n-1} B(t_i + \lambda(t_{i+1} - t_i))B(t_i))\right] = \\
&= \sum_i (t_i + \lambda(t_{i+1} - t_i) - t_i) = \sum_i \lambda \Delta t \overset{n\to\infty}{=} \lambda t
\end{aligned}$$

where we used the Brownian covariance:

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=0}^{n-1} B(t_i + \lambda(t_{i+1} - t_i))B(t_{i+1})\right] &= \min\left[t_i + \lambda(t_{i+1} - t_i), t_{i+1}\right] = t_i + \lambda(t_{i+1} - t_i) \\
\mathbb{E}\left[\sum_{i=0}^{n-1} (B(t_i + \lambda(t_{i+1} - t_i)B(t_i))\right] &= \min\left[t_i + \lambda(t_{i+1} - t_i), t_i\right] = t_i
\end{aligned}$$

This example is useful because, in general, when one has to calculate an integral and the definition of a stochastic integral is unclear, one must discretize, compute the expectation of the Riemann sum, and then take the limit.

Exercise 9.5.1. *Show the Itô isometry. Let $g(t)$ be a non-anticipating stochastic process adapted to the Brownian motion $B(s)$ for any $s < t$. Prove that*

$$\mathbb{E}\left[\left(\int_0^t g(B(s))\, dB(s)\right)^2\right] = \mathbb{E}\left[\int_0^t g^2(B(s))\, ds\right]$$

if we use the Itô prescription. Hints:

1. *Discretize using the Itô rule:*

$$I_n = \sum_{i=0}^{n-1} g(B_i)[B_{i+1} - B_i]$$

2. *Compute* $\mathbb{E}\left[I_n^2\right]$ *and use independence of increments to show that only diagonal terms contribute:*

$$\mathbb{E}\left[I_n^2\right] = \textit{(diag. terms)} + \textit{(off-diag. terms)} =$$
$$= \ldots = \mathbb{E}\left[\sum_i g_i^2 \Delta t_i\right] \quad \to \quad \textit{take limit} \quad n \to \infty$$

9.6 Itô calculus

We have seen that Brownian motion forces a modification of the usual rules of calculus. In the following, we present the rules of stochastic differentiation and integration using the Itô prescription. Thanks to the results established earlier, we now have the tools to justify these rules.

9.6.1 Itô differential rule

We consider a function $u(B(t), t)$, that is $C^2([0, T])$ in both variables. We want to compute the increment

$$\begin{aligned}
\Delta u(B) &\equiv u(B(t+\Delta t), t+\Delta t) - u(B(t), t) \overset{\text{Taylor exp. up to 2nd order}}{=} \\
&= \frac{\partial u}{\partial t}\Delta t + \frac{\partial u}{\partial B}\Delta B + \frac{1}{2}\frac{\partial^2 u}{\partial B^2}(\Delta B)^2 + \text{h.o.t.} \overset{\Delta B^2 = O(\Delta t)}{=} \\
&= \frac{\partial u}{\partial t}\Delta t + \frac{\partial u}{\partial B}\Delta B + \frac{1}{2}\frac{\partial^2 u}{\partial B^2}\Delta t + \text{h.o.t.} = \\
&= \left(\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial^2 u}{\partial B^2}\right)\Delta t + \frac{\partial u}{\partial B}\Delta B + \text{h.o.t.}
\end{aligned}$$

The second term inside the parentheses is not present in ordinary calculus. It is there because Brownian increments satisfy $(\Delta B)^2 = O(\Delta t)$ and then $\Delta B \simeq \sqrt{\Delta t}$. Passing to the infinitesimal limit yields the **Itô differential rule**

$$du = \left(\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial^2 u}{\partial B^2}\right) dt + \frac{\partial u}{\partial B}\, dB \tag{9.36}$$

This means that in the presence of Brownian motion, a Taylor expansion must be carried out up to second order in ΔB, because the term $(\Delta B)^2$ is of order Δt and therefore gives a non-negligible contribution in the limit $\Delta t \to 0$.

Example 9.6.1

Let $u(B) = B^2$, then $dB^2 = dt + 2B\, dB$. Integrating both sides between t_1 and t_2

$$\int_{t_1}^{t_2} d(B^2) = B^2(t_2) - B^2(t_1) = t_2 - t_1 + 2\int_{t_1}^{t_2} B(s)\, dB(s)$$

Therefore,

$$\int_{t_1}^{t_2} B(s)\, dB(s) = \frac{B^2(t_2) - B^2(t_1)}{2} - \frac{t_2 - t_1}{2}$$

in agreement with the Itô result in Equation 9.34.

Thus, the practical rule is: to compute a stochastic integral, first compute the Itô differential and then integrate.

Exercise 9.6.1. *Calculate the Itô differential of* $[B(t)]^n \equiv B_t^n$ *and show:*

$$dB_t^n = nB_t^{n-1}\,dB_t + \frac{n(n-1)}{2}B_t^{n-2}\,dt$$

Then integrate to obtain:

$$\int_0^t B_s^n\,dB = \frac{B_t^{n+1}}{n+1} - \frac{n}{2}\int_0^t B_s^{n-1}\,ds$$

9.6.2 Itô's chain rule (Itô's formula)

We now apply Itô calculus to Itô processes. In practice, these processes are defined by Langevin-type SDEs that may include multiplicative noise. As we have seen, in this case, we must choose a prescription to interpret the SDE; here, we work with the Itô prescription.

Assume that $x(t)$ is a stochastic process satisfying the SDE

$$dx(t) = \mu(x,t)\,dt + \sigma(x,t)\,dB(t)$$

interpreted in the Itô sense ($\lambda = 0$). Let $u(x,t)$ be a function of class C_t^1 and C_x^2. We want to derive the SDE satisfied by the transformed process $u(x(t),t)$. In other words, we perform a change of variables: starting from a process x defined by an SDE with multiplicative noise, we construct a new process $u(x,t)$ and ask how its dynamics look (moments, etc).

We use the Itô differential up to the second order in x:

$$du = \frac{\partial u}{\partial t}\,dt + \frac{\partial u}{\partial x}\,dx + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\,(dx)^2 + \text{h.o.t.}$$

We now use the Itô rules

$$dt\,dB = O(dt^{3/2}), \qquad (dB)^2 \sim dt$$

which imply that products involving $dt\,dB$ and dt^2 are negligible, while $(dB)^2$ must be kept. Therefore,

$$(dx)^2 = (\mu\,dt + \sigma\,dB)^2 = \mu^2(dt)^2 + \sigma^2(dB)^2 + \mu\sigma\,dt\,dB = \sigma^2\,dt$$

where the other terms are neglected because they are higher order in dt. Substituting into the expansion for du gives

$$du = \frac{\partial u}{\partial t}\,dt + \frac{\partial u}{\partial x}(\mu\,dt + \sigma\,dB) + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\sigma^2\,dt$$

so that, collecting the drift and noise terms, we obtain

$$du = \left[\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\mu(x,t) + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\sigma^2(x,t)\right]dt + \frac{\partial u}{\partial x}\sigma(x,t)\,dB(t) \tag{9.37}$$

This is **Itô's formula**. The term proportional to σ^2 comes from the Itô correction and reflects the fact that we are dealing with a Brownian term $B \propto \sqrt{t}$: when we transform from x to $u(x,t)$, second derivatives with respect to x contribute to the drift. The term proportional to σ is the stochastic contribution (the noise term). This is why it is crucial that u be twice differentiable in x.

Exercise 9.6.2. *The stochastic process $x(t)$ satisfies the Itô SDE*

$$\begin{cases} dx = \frac{x}{2}\,dt + x\,dB \\ x(0) = 1 \end{cases}$$

This process is called ***geometric Brownian motion****. Show that the function $y(x) = \ln x$ satisfies the simpler Itô SDE*

$$\begin{cases} dy = dB \\ y(0) = 0 \end{cases}$$

and deduce that the solution of the original SDE is $x(t) = e^{B(t)}$. This illustrates how Itôis formula can be used to find explicit solution of SDE.

Exercise 9.6.3. *Itô formula for two variables. Assume that two processes $x(t)$ and $y(t)$ satisfy the Itô SDEs:*

$$dx = \mu_x\,dt + \sigma_x\,dB, \qquad dy = \mu_y\,dt + \sigma_y\,dB$$

driven by the same Brownian motion $B(t)$. Then, for any $u(x, y) \in C^2$, one can show that:

$$du = \left(\frac{\partial u}{\partial x}\mu_x + \frac{\partial u}{\partial y}\mu_y + \frac{1}{2}\frac{\partial^2 u}{\partial x^2}\sigma_x^2 + \frac{1}{2}\frac{\partial^2 u}{\partial y^2}\sigma_y^2 + \frac{\partial^2 u}{\partial x \partial y}\sigma_x\sigma_y\right)dt + \left(\frac{\partial u}{\partial x}\sigma_x + \frac{\partial u}{\partial y}\sigma_y\right)dB$$

9.7 Ornstein-Uhlenbeck process

Formally, the Ornstein-Uhlenbeck process is defined by the SDE:

$$\begin{cases} dx = -\mu x\,dt + \sigma\,dB \\ x(0) = x_0 \end{cases} \tag{9.38}$$

where $\mu, \sigma > 0$ are constants.

To solve this SDE, we perform the change of variables $y(t) = e^{\mu t}x(t)$. Using Itô's rule, one finds $dy = \sigma e^{\mu t}\,dB$. Thus, with this trick, the drift has been removed. Integrating both sides, we get

$$y(t) = y(0) + \sigma \int_0^t e^{\mu s}\,dB(s)$$

where the integrand is deterministic. Returning to $x(t) = e^{-\mu t}y(t)$, we obtain the explicit solution

$$x(t) = e^{-\mu t}\left(x_0 + \sigma \int_0^t e^{\mu s}\,dB(s)\right) \tag{9.39}$$

The solution consists of a deterministic exponentially decaying term and a stochastic term representing the accumulated effect of the noise.

A more general Ornstein-Uhlenbeck process includes a constant term β

$$\begin{cases} dx = (\beta - \mu x)\,dt + \sigma\,dB \\ x(0) = x_0 \end{cases}$$

This model describes relaxation toward a non-zero equilibrium. Its solution can be obtained by the same exponential change of variables, and its mean is easily computed by taking expectations and using $\mathbb{E}[dB] = 0$. One finds $\mathbb{E}[x(t)] = x_0 e^{-\mu t}$.

9.7.1 Variance

An essential feature of a stochastic process is the behavior of its variance, which captures how its fluctuations evolve in time. To compute $\mathrm{Var}(x(t))$ for the Ornstein-Uhlenbeck process, we first need the second moment $\langle x^2(t)\rangle$. This can be obtained by deriving the SDE for x^2. Using Itô's rule

$$\begin{aligned} d(x^2) &= 2x\,dx + (dx)^2 = \\ &\overset{9.38}{=} 2x(-\mu x\,dt + \sigma\,dB) + (-\mu x\,dt + \sigma\,dB)^2 = \\ &= -2\mu x^2\,dt + 2\sigma x\,dB + \sigma^2\,dt = \\ &= (-2\mu x^2 + \sigma^2)dt + 2\sigma x\,dB \end{aligned}$$

where all higher-order terms were discarded. The first contribution is deterministic, whereas the second contains multiplicative noise. Since we work with the Itô prescription, we may safely write $\mathbb{E}\left[x\,dB\right] = 0$.

Thus, the equation for the 2nd moment becomes:

$$d\left\langle x^2\right\rangle = \left(-2\mu\left\langle x^2\right\rangle + \sigma^2\right)\,dt + 2\sigma\underbrace{\left\langle x\,dB\right\rangle}_{=0\,(\text{Itô})}$$

where the last term is zero because of Itô. In this way, we obtain a deterministic equation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle x^2\right\rangle = -2\mu\left\langle x^2\right\rangle + \sigma^2$$

This is a linear ODE whose solution is

$$\left\langle x^2(t)\right\rangle = \frac{\sigma^2}{2\mu} + \left(x_0^2 - \frac{\sigma^2}{2\mu}\right)e^{-2\mu t}$$

Of course, this result is highly non trivial as it tells us that as $t \to \infty$, the fluctuations converge to the finite value $\mathbb{E}\left[x_\infty^2\right]\sigma^2/2\mu$.

Finally, the variance is

$$\mathrm{Var}(x(t)) = \left\langle x^2(t)\right\rangle - \left\langle x(t)\right\rangle^2 = \frac{\sigma^2}{2\mu}\left(1 - e^{-2\mu t}\right) \tag{9.40}$$

Even though all sample paths (trajectories) start at the same initial value x_0, and the mean decays to zero, Equation 9.40 shows that the fluctuations do not vanish. Instead, they converge to a constant stationary level $\sigma^2/2\mu$ characteristic of this process.

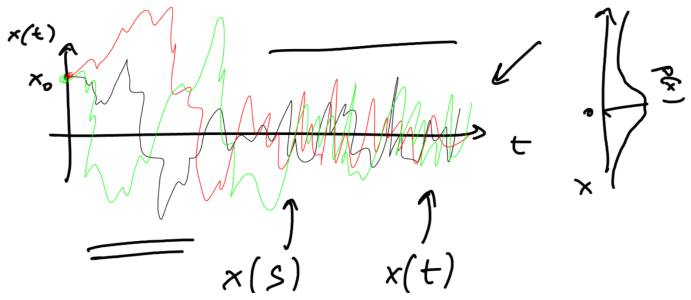


Figure 9.1

In Figure 9.1 we show some realizations of the process, all starting from the same initial condition x_0. As time evolves and the system reaches stationarity, these trajectories display non-trivial fluctuations around the mean. We will prove in the next chapter that, at stationarity, the distribution of these fluctuations is Gaussian.

9.7.2 Autocorrelation function

Another important quantity we can study is the correlation between the values of the process at two different times, $x(s)$ and $x(t)$. We define the **autocorrelation function** of $x(t)$ as:

$$c(t,s) = \langle x(t)x(s)\rangle - \langle x(t)\rangle \langle x(s)\rangle$$

To compute it, we need to compute the product of the solution 9.39 with itself at two different times. This requires averaging products of stochastic integrals, so we first establish a general identity.

Consider two continuous, non-anticipating functions G and H. We want to compute

$$\mathbb{E}\left[\int_0^t G(t')\,dB(t')\int_0^s H(s')\,dB(s')\right]$$

For simplicity, we take $t = s$ (it can be simply generalized to any $t \neq s$). By using the Itô discretization, we write

$$\mathbb{E}\left[\sum_{i=0}^{m-1} G_i(B_{i+1}-B_i)\sum_{j=0}^{m-1} H_j(B_{j+1}-B_j)\right] = \\ = \mathbb{E}\left[\sum_{i=0}^{m-1} G_iH_i(\Delta B_i)^2\right] + \mathbb{E}\left[\sum_{i>j}(G_iH_j + G_jH_i)\Delta B_j\Delta B_i\right]$$

where we split the sum into diagonal and off-diagonal contributions. We also used the notation $B_{i+1} - B_i \equiv \Delta B_i$ and $B_{j+1} - B_j \equiv \Delta B_j$.

We now focus on the second term. Since G and H are non-anticipating and increments of Brownian motion are independent, all off-diagonal terms vanish:

$$\mathbb{E}\left[\sum_{i>j}(G_iH_j + G_jH_i)\Delta B_j\Delta B_i\right] = \mathbb{E}\left[\sum_{i>j}(\ldots)\Delta B_j\right]\mathbb{E}\left[\Delta B_i\right] = 0$$

Only the diagonal terms remain:

$$\mathbb{E}\left[\sum_{i=0}^{m-1} G_iH_i\Delta t\right] \quad \xrightarrow{\text{in the limit}} \quad \mathbb{E}\left[\int_0^t G(s')H(s')\,ds'\right]$$

Thus, for general t, s, one can show

$$\mathbb{E}\left[\int_0^t G(t')\,dB(t')\int_0^s H(s')\,dB(s')\right] = \int_0^{t\wedge s}\mathbb{E}\left[G(s')H(s')\right]\,ds' \tag{9.41}$$

We now use Equation 9.41 to compute the autocorrelation of the Ornstein-Uhlenbeck process.

Starting from the explicit solution, we write:

$$\begin{aligned}
&\mathbb{E}\left[x(t)x(s)\right] = \mathbb{E}\left[e^{-\mu t}\left(x_0 + \sigma\int_0^t e^{\mu t'}\,dB(t')\right)e^{-\mu s}\left(x_0 + \sigma\int_0^s e^{\mu s'}\,dB(s')\right)\right] = \\
&= x_0^2 e^{-\mu(t+s)} + x_0\sigma e^{-\mu(t+s)}\int_0^s e^{\mu s'}\underbrace{\mathbb{E}\left[dB(s')\right]}_{=0} + \\
&\quad + x_0\sigma e^{-\mu(t+s)}\int_0^t e^{\mu t'}\underbrace{\mathbb{E}\left[dB(t')\right]}_{=0} + \sigma^2 e^{-\mu(t+s)}\mathbb{E}\left[\int_0^t e^{\mu t'}\,dB(t')\int_0^s e^{\mu s'}dB(s')\right] = \\
&\overset{9.41}{=} x_0^2 e^{-\mu(t+s)} + \sigma^2 e^{-\mu(t+s)}\int_0^{t\wedge s} e^{2\mu s'}\,ds' = \\
&= x_0^2 e^{-\mu(t+s)} + \frac{\sigma^2}{2\mu}e^{-\mu(t+s)}\left(e^{2\mu(t\wedge s)} - 1\right) = \\
&= x_0^2 e^{-\mu(t+s)} + \frac{\sigma^2}{2\mu}\left(e^{-\mu|t-s|} - e^{-\mu(t+s)}\right)
\end{aligned}$$

Therefore,

$$\begin{aligned}
c(t,s) &= x_0^2 e^{-\mu(t+s)} + \frac{\sigma^2}{2\mu}\left(e^{-\mu|t-s|} - e^{-\mu(t+s)}\right) - x_0^2 e^{-\mu(t+s)} = \\
&= \frac{\sigma^2}{2\mu}\left(e^{-\mu|t-s|} - e^{-\mu(t+s)}\right)
\end{aligned} \tag{9.42}$$

This equation tells us how the process is correlated to itself in time.

If the initial condition is distributed with variance $\mathrm{Var}(x_0)$, the general result becomes:

$$c(t,s) = \mathrm{Var}(x_0)e^{-\mu(t+s)} + \frac{\sigma^2}{2\mu}\left(e^{-\mu|t-s|} - e^{-\mu(t+s)}\right)$$

where we have an initial term that decays as t or s becomes larger and larger.

Notice that, when $t, s \to \infty$ while $t - s$ remains fixed, the transient terms vanish:

$$c_{\mathrm{stat}}(|t-s|) = \frac{\sigma^2}{2\mu}e^{-\mu|t-s|} \tag{9.43}$$

which depends only on differences of time $|t - s|$. This is called **stationary autocorrelation**. The fact that the autocorrelation depends only on the time difference, not on the absolute times, is a general feature of stationary Markov processes.

As shown in Figure 9.1, we can see that the transient depends on both t and s, but once stationarity is reached, correlations decay exponentially and the process effectively loses memory over a timescale of order $1/\mu$.

Chapter 10

Fokker-Planck Equation

So far, we have focused on continuous-time stochastic processes described by Langevin equations (i.e., SDEs) and on properties of individual trajectories. We now shift perspective and study the statistics of these processes, meaning their PDF at a given time t.

Given an SDE, we want to answer questions such as: what is the distribution $p(x,t)$ at time t? Does $p(x,t)$ converge to a stationary (equilibrium) distribution as $t \to \infty$? If yes, how can we compute it? For Itô diffusion processes, the time evolution of the PDF is governed by the Fokker-Planck equation.

10.1 Numerical solution

To connect the Langevin equation to the Fokker-Planck equation, it is useful to start from a numerical viewpoint.

Consider the Itô SDE (mathematical notation)

$$dx(t) = \mu(x,t)\,dt + \sigma(x,t)\,dB(t) \tag{10.1}$$

with initial condition $x(0) = x_0$. In physics notation, we can also write:

$$\dot{x} = \mu(x,t) + \sigma(x,t)\xi(t) \tag{10.2}$$

where $\xi(t)$ is Gaussian white noise with mean $\langle \xi \rangle = 0$ and delta correlation $\langle \xi(t')\xi(t) \rangle = \delta(t'-t)$. This notation is shorthand: $\xi(t)$ is not an ordinary function, so the equation is to be interpreted in the Itô sense through the SDE above.

With Equation 10.1 we can simulate N independent trajectories of the process $x(t)$ over the interval $[0,T]$. How can we estimate the probability that at time t the process lies in the interval $[x, x+\Delta x]$?

Numerically, we discretize the x-axis into bins of width Δx and count how many trajectories randomly fall into each bin at time t (Figure 10.1). Thus, we have to count how many paths fall in the interval $[x, x+\Delta x]$ at time $t = t_1$ as x varies within the domain of definition of the process. Note that the partitioning is Δx for each point.

Let's suppose we have done N simulations. Let x_j denote the left endpoint of the j-th bin, e.g., $x_j = x_{\min} + j\Delta x$. We introduce the **indicator function** I

$$I(a,A) = \begin{cases} 1 & a \in A \\ 0 & a \notin A \end{cases}$$

where a is a number and A is a set. In our case, a will be the j-th simulation at time t, denoted as $x^j(t)$ (trajectory, stochastic process).

Then the probability mass $p(x = x_j, t = t_1)$ in the bin $[x_j, x_j + \Delta x)$ can be estimated as the fraction of the trajectories that fall into that bin:

$$p(x = x_j, t = t_1)\Delta x = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} I\left(x^i(t_1), [x_j, x_j + \Delta x)\right) \tag{10.3}$$

for any x_j and t, and $x_j = j\Delta x$ with $j = j_{\min}, \ldots, j_{\max}$.

If we take $x_j = x$ at $t = t_j$ and then take the limit $\Delta x \to 0$, the indicator function converges to a Dirac delta, and we obtain a representation of the probability density

$$p(x, t) = \lim_{N\to\infty} \frac{1}{N} \sum_{i} \delta(x - x_i(t)) \tag{10.4}$$

because

$$\lim_{\Delta x\to 0} \frac{I(x^i(t), [x, x + \Delta x))}{\Delta x} = \delta(x - x_i(t))$$

Equivalently, the PDF can be written as the ensemble average

$$p(x, t) = \langle \delta(x - x(t)) \rangle$$

where the average is taken over all realizations of the process.

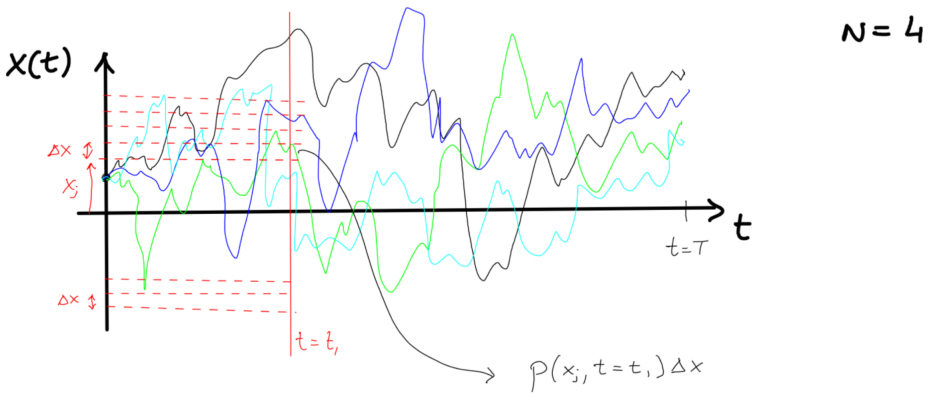


Figure 10.1

For example, in the case of Brownian motion, a histogram built from a sufficiently large number of trajectories converges to a Gaussian distribution centered at zero, with variance proportional to t.

10.2 Derivation of the Fokker-Planck equation for a general Langevin equation in the Ito prescription

We now move from the numerical approach to a fully theoretical derivation of the Fokker-Planck equation. While a Langevin equation defines individual realizations of a stochastic process through a SDE, in physics the primary object of interest is not a single trajectory but the statistical behavior of the process: moments, correlations, and, ultimately, its probability distribution.

From a theoretical perspective, our goal is therefore to derive an evolution equation for the PDF $p(x,t)$, or more precisely for the conditional density propagator $p(x,t|x_0t_0)$. Once this quantity is known, all statistical properties of the process can be obtained.

To this end, we introduce a smooth test function $f(x)$ with compact support on $\mathbb{R}$ and twice-differentiable, $f \in C_c^2(\mathbb{R})$. We want to compute the expectation value of $f(x(t))$, defined as the average over an ensemble of independent realizations of the stochastic process. In the continuum limit, this expectation can be written as

$$\begin{aligned}\langle f(x(t))\rangle &= \lim_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N} f(x_i(t)) = \lim_{N\to\infty}\frac{1}{N}\sum_i \int dx\, f(x)\delta(x-x_i(t)) = \\ &= \int dx\, f(x)\lim_{N\to\infty}\frac{1}{N}\sum_i \delta(x-x_i(t)) \overset{10.4}{=} \int dx\, f(x)p(x,t)\end{aligned} \tag{10.5}$$

where we used the definition of Dirac delta in the second equality and the definition of $p(x,t)$ in the last equality. Notice that we integrate over x but the time dependence is on p. This expresses the fact that averages over trajectories can equivalently be computed as averages with respect to the PDF.

Taking the time derivative of this expression gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f(x(t))\rangle \equiv \int \dot{p}(x,t)f(x)\,dx \tag{10.6}$$

which motivates the derivation of an explicit evolution equation for $p(x,t)$.

We now discretize time and Taylor-expand the function f around $x(t)$. Because the process is defined through an Itô SDE, terms up to second order in the increment Δx must be retained:

$$f(x(t+\Delta t)) = f(x(t)) + \Delta x f'(x(t)) + \frac{\Delta x^2}{2}f''(x(t)) + \text{h.o.t.} \tag{10.7}$$

The stochastic increment is determined by the Itô SDE Equation 9.28, then

$$\Delta x \equiv x(t+\Delta t) - x(t) = \mu(x(t),t)\Delta t + \sigma(x(t),t)\Delta B(t) \tag{10.8}$$

where $\Delta B(t)$ is the increment of a Wiener process.

Define

$$\Delta f \equiv f(x(t+\Delta t)) - f(x(t))$$

Substituting this expression into the Taylor expansion 10.7 and keeping only terms that survive in the limit $\Delta t \to 0$, one finds

$$\begin{aligned}\Delta f &= f'\cdot(\mu\Delta t + \sigma\Delta B) + \frac{1}{2}f''\cdot(\mu\Delta t + \sigma\Delta B)^2 + \text{h.o.t.} = \\ &= f'\cdot(\mu\Delta t + \sigma\Delta B) + \frac{1}{2}f''\cdot(\sigma^2\Delta B^2) + \text{h.o.t.}\end{aligned}$$

where we neglected the terms $(\mu\Delta t)^2 \sim O(\Delta t^2)$ and $2\mu\sigma\Delta t\Delta B \sim (\Delta t^{3/2})$.

Taking expectations, we get

$$\langle\Delta f\rangle = \langle f'\mu\Delta t\rangle + \langle f'\sigma\Delta B\rangle + \frac{1}{2}\langle f''\sigma^2\,dt\rangle$$

In the Itô assumption, we get

$$\left\langle f'\sigma\Delta B\right\rangle = \left\langle f'\sigma\right\rangle \left\langle \Delta B\right\rangle = 0 \tag{10.9}$$

which yields

$$\begin{aligned}\lim_{\Delta t\to 0}\left\langle \frac{\Delta f}{\Delta t}\right\rangle &= \left\langle f'(x)\mu(x,t)\right\rangle + \frac{1}{2}\left\langle f''(x)\sigma^2(x,t)\right\rangle = \\ &= \int dx\, p(x,t)\left[f'(x)\mu(x,t) + \frac{1}{2}f''(x)\sigma^2(x,t)\right] = \frac{\mathrm{d}}{\mathrm{d}t}\left\langle f\right\rangle\end{aligned} \tag{10.10}$$

Since $f \in C_c^2(\mathbb{R})$, which is our test function, has compact support, we can integrate by parts and assume that boundary terms vanish, that is $f, f', f'' \to 0$ as $|x| \to \infty$. Thus, we have

$$\begin{aligned}\int dx\, p(x,t)\mu\frac{\partial f}{\partial x} &= -\int dx\, f(x)\frac{\partial}{\partial x}(p\mu) \\ \int dx\, p(x,t)\frac{\sigma^2}{2}\frac{\partial^2 f}{\partial x^2} &= \frac{1}{2}\int dx\, f(x)\frac{\partial^2}{\partial x^2}(\sigma^2 p)\end{aligned} \tag{10.11}$$

By comparing these expressions with the definition

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle f(x)\right\rangle = \int dx\, f(x)\frac{\partial p}{\partial t}$$

we get

$$\int dx\, f(x)\left[\frac{\partial p}{\partial t} - \left(-\frac{\partial}{\partial x}(\mu p) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(\sigma^2 p)\right)\right] = 0$$

and using the arbitrariness of the test function f, we arrive at the **forward Fokker-Planck equation**:

$$\frac{\partial}{\partial t}p(x,t) = -\frac{\partial}{\partial x}\left[\mu(x,t)p(x,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial x^2}\left[\sigma^2(x,t)p(x,t)\right] \tag{10.12}$$

This equation governs the time evolution of the probability density associated with the Itô Langevin equation. It provides a complete statistical description of the process and allows one to compute moments and other observables directly from the distribution. For example, for the mean we multiply by x and integrate over x.

The Fokker-Planck equation is a linear, deterministic PDE, which can be used to derive the propagator $p(x,t|x_0,t_0)$ of the process $x(t)$ with the initial condition $p(x,t_0) = \delta(x - x_0)$. The formalism generalizes straightforwardly to multidimensional processes, where μ becomes a vector field and σ a positive definite diffusion matrix.

Equation 10.12 can also be written in continuity form,

$$\frac{\partial p}{\partial t} = -\frac{\partial J(x,t)}{\partial x}$$

which has the form of a continuity equation and makes explicit the conservation of probability. In this representation, we have introduced the probability flux

$$J(x,t) \equiv \mu(x,t)p(x,t) - \frac{1}{2}\frac{\partial}{\partial x}\sigma^2(x,t)p(x,t) \tag{10.13}$$

This form highlights that variations in the probability contained in a domain $\mathcal{D} \subseteq \mathbb{R}$ are entirely due to the net probability fluxes across its boundaries. Hence

$$\frac{\partial}{\partial t} \int_{\mathcal{D}} p(x,t)\, dx = -\int_{\mathcal{D}} \frac{\partial J}{\partial x}\, dx = -J\big|_{x\in\partial\mathcal{D}}$$

where $\partial\mathcal{D}$ is the boundary of the domain $\mathcal{D}$. If there is no leakage of probability through the boundary , that is, if the probability flux vanishes on $\partial\mathcal{D}$ ($J\big|_{x\in\partial\mathcal{D}} = 0$), then the total probability inside the domain is conserved:

$$\int_{\mathcal{D}} p(x,t)\, dx = 1$$

for all t (conservation of probability).

When the stochastic process is defined on a restricted domain $\mathcal{D} \subseteq \mathbb{R}$, Equation 10.12 must be equipped with boundary conditions.

We list two different boundary situations:

- Absorbing boundary conditions require $p(0,t) = 0$ (or $p(a,t) = 0$) if a is a boundary of the domain. In general,
$$p(x,t)\big|_{x\in\partial\mathcal{D}} = 0,\ \forall t$$
- Reflecting boundary conditions require:
$$J(x,t)\big|_{x\in\partial\mathcal{D}} = 0,\ \forall t$$

Remark. These expressions hold when $\sigma(x,t)\big|_{x\in\partial\mathcal{D}} > 0\ \forall t$ (the process at the bound must be non-degenerate).

10.3 Equilibrium solution of the Fokker-Planck equation

We now study the long-time behavior of the Fokker-Planck equation and determine the shape of the distribution at stationarity, that is, in the regime where $\partial_t p(x,t) = 0$.

We assume that the drift and diffusion coefficients are time-independent, $\mu(x,t) = \mu(x)$ and $\sigma(x,t) = \sigma(x)$, and we denote by $p^{\mathrm{st}}(x)$ a **stationary distribution**. Under suitable conditions, the propagator associated with the Fokker-Planck equation 10.12 converges at long times to this stationary density,

$$\lim_{t\to\infty} p(x,t|x_0,t_0) = p^{\mathrm{st}}(x)$$

which no longer depends on the initial condition.

Setting $\partial_t p^{\mathrm{st}}(x) = 0$ in Equation 10.12 yields

$$-\frac{\partial}{\partial x}\left[\mu(x)p^{\mathrm{st}}(x) - \frac{1}{2}\frac{\partial}{\partial x}\left(\sigma^2(x)p^{\mathrm{st}}(x)\right)\right] = 0$$

This implies that the stationary probability flux

$$J^{\mathrm{st}}(x) = \mu(x)p^{\mathrm{st}}(x) - \frac{1}{2}\frac{\partial}{\partial x}\left(\sigma^2(x)p^{\mathrm{st}}(x)\right) = \mathrm{const}$$

which is constant throughout the domain.

A particularly important situation arises when there is no probability current anywhere in the domain $x \in \mathcal{D}$, that is, when $J^{\text{st}}(x) = 0$. In this case the system is said to be in equilibrium. This condition is different than stationarity: while stationarity only requires $\dot{p} = 0$, equilibrium additionally requires the absence of probability flux. Imposing $J^{\text{st}}(x) = 0$ leads to the first-order differential equation

$$\mu(x)p^{\text{st}} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}x}(\sigma^2(x)p^{\text{st}}(x))$$

Rewriting it in separable form,

$$2\frac{\mu}{\sigma^2}(\sigma^2 p^{\text{st}}) = \frac{\mathrm{d}}{\mathrm{d}x}(\sigma^2 p^{\text{st}})$$

one readily obtains the equilibrium distribution

$$p^{\text{st}}(x) = \frac{1}{z}\frac{1}{\sigma^2(x)}\exp\left\{2\int_{x_m}^{x}\frac{\mu(y)}{\sigma^2(y)}\,dy\right\}, \qquad x_m \le x \le x_M \tag{10.14}$$

where z is a normalization constant chosen so that is finite and

$$\int_{x_m}^{x_M} p^{\text{st}}(x)\,dx = 1$$

while the interval $[x_m, x_M]$ denotes the domain of definition of the process.

It is important to note that the existence of a stationarity solution as 10.14 is not guaranteed. In particular, if the formal solution is not normalizable, the system does not admit a stationary distribution. Moreover, even when a stationary solution exists, its accessibility depends on the initial condition and on the boundary behavior of the process; stationarity alone does not ensure convergence from arbitrary initial states.

10.4 Fokker-Planck equation for Ornstein-Uhlenbeck process

We have already studied the Ornstein-Uhlenbeck process, defined by the Itô SDE

$$\begin{cases} dx = -\mu x\,dt + \sigma\,dB(t) \\ x(0) = x_0 \end{cases} \tag{10.15}$$

with constants $\mu, \sigma > 0$. The associated Fokker-Planck equation for the PDF $p(x,t)$ is

$$\partial_t p(x,t) = -\partial_x(-\mu x p(x,t)) + \frac{\sigma^2}{2}\partial_x^2 p(x,t) \tag{10.16}$$

Because the coefficients are time-independent, one can look for a stationary density $p^{\text{st}}(x)$ satisfying $\partial_t p^{\text{st}} = 0$. The equilibrium distribution of Equation 10.16 (from Equation 10.14, zero stationary flux) is

$$p^{\text{st}}(x) \propto e^{-\frac{2}{\sigma^2}\int^x \mu y\,dy} = e^{-\frac{\mu}{\sigma^2}x^2}$$

By normalizing it, we get:

$$p^{\text{st}}(x) = \sqrt{\frac{\mu}{\pi\sigma^2}}\exp\left\{-\frac{\mu}{\sigma^2}x^2\right\} \tag{10.17}$$

which is a Gaussian distribution with zero-mean and variance $\sigma^2/2\mu$. Notice that the stationary distribution does not depend on the initial conditions. Regardless the initial probability distribution, the law of the process converges at long times to the same Gaussian stationary distribution with fluctuations around zero.

Qualitatively, the OU process differs from standard Brownian motion because it contains a restoring drift $-\mu x$ that pulls the process back toward the origin. The noise term $\sigma\, dB$ continuously drives fluctuations, while the drift acts as a negative feedback. The interplay of these two effects produces a stationary distribution concentrated around $x = 0$. By contrast, Brownian motion has no restoring force and its variance grows without bound, so it admits no normalizable stationary density on $\mathbb{R}$.

For this process is possible to find the full time-dependence solution. By taking the Fourier transform of Equation 10.16 (exercise, it gives the evolution of the characteristic function), one finds the propagator of the Ornstein-Ulhenbeck process. The result is:

$$p(x,t|x_0,s) = \sqrt{\frac{\mu}{\pi\sigma^2(1-e^{-\mu(t-s)})}}\exp\left\{-\frac{\mu}{\sigma^2}\frac{(x-x_0e^{-\mu(t-s)})^2}{1-e^{-2\mu(t-s)}}\right\} \tag{10.18}$$

This satisfies the initial condition

$$p(x,s|x_0,s) = \delta(x-x_0)$$

If instead the initial state at time s is distributed according to some density $q(x_0)$, then the density at time t is obtained by convolving the propagator with q:

$$p(x,t) = \int_{\mathbb{R}} p(x,t|x_0,s)q(x_0)\,dx_0$$

As $t-s \to \infty$, the mean decays to zero and the variance approaches $\sigma^2/(2\mu)$, so the propagator converges to the stationary Gaussian Equation 10.17

A simple generalization adds a constant drift term:

$$dx = (\alpha - \mu x)\,dt + \sigma\, dB_t$$

leading to

$$\dot{p} = -\partial_x[(\alpha-\mu x)p] + \frac{\sigma^2}{2}\partial_x^2 p$$

In this case the stationary distribution is still Gaussian, but centered at the deterministic fixed point α/μ rather than at zero. For the multi-dimensional Ornstein-Ulhenbeck process see Gardiner, p 105.

Exercise 10.4.1. *Show that the equation for the variance that one gets from Equation 10.15 is the same that one obtains Equation 10.16 (they are the same). It is possible to prove that the statistic from the two equations is the same.*

Chapter 11

Classical Mechanics Applications

We now establish a link between statistical mechanics and the stochastic processes introduced earlier. The connection goes through the fluctuation-dissipation theorem. Our goal is to obtain an effective equation of motion for a mesoscopic particle immersed in a thermal environment, of the form

$$m\ddot{x}(t) = -\partial_x U(x) - \gamma\dot{x}(t) + \sqrt{2\gamma k_B T}\xi(t)$$

where U is an external potential, $-\gamma\dot{x}$ is a viscous (damping) force, and the stochastic forcing term models the random microscopic impacts from the surrounding medium, a *heat bath*, at temperature T. Here $\xi(t)$ is a Gaussian white noise with $\langle\xi(t)\rangle = 0$ and $\langle\xi(t)\xi(t')\rangle = \delta(t-t')$. The noise amplitude is fixed by the fluctuation-dissipation relation and ensures that the system thermalizes at temperature T.

Although the position process $x(t)$ alone is not Markovian because the dynamics is second order in time, the pair $(x(t), v(t))$ with $v = \dot{x}$ is Markovian. In the overdamped limit $m/\gamma \to 0$, inertia becomes negligible on the time scales of interest and one recovers the first-order (overdamped) Langevin equation.

A particularly important solvable case is the harmonic potential $U(x) = \frac{1}{2}kx^2$, which corresponds to an Ornstein-Uhlenbeck dynamics for velocity (and, in the overdamped regime, for position). Because the model is linear with Gaussian noise, it can be solved exactly.

11.1 Particle in a large medium

Consider a particle of mass m suspended in a large fluid. The medium contains an enormous number of microscopic constituents, so large that it can be treated as a heat bath in thermal equilibrium at constant temperature T, and – at the macroscopic level – as homogeneous and isotropic.

If $\vec{x}(t)$ is the position of the suspended particle and $\vec{x}_i(t)$ are the position of the medium particles, the Newton equation is

$$m\ddot{\vec{x}}(t) = \vec{F}_{\text{ext}}(\vec{x}) + \sum_{i=1}^{N} \vec{F}(\vec{x}(t) - \vec{x}_i(t)) \tag{11.1}$$

where F_{ext} is any imposed external force and the second contribution is given by all forces exerted by the other particles that surround the suspended one. Thus, $F(\vec{x} - \vec{x}_i)$ is the interaction force exerted by the i-th bath particle.

Since $N \sim O(N_A)$, Equation 11.1 is a *huge* ODE of coupled differential equations and directly integrating is unpractical. The key modeling step is therefore to replace the complicated microscopic force exerted by the bath with an effective decomposition:

$$\sum_i \vec{F}(\vec{x}(t) - \vec{x}_i(t)) \simeq \vec{F}_{\text{avg}} + \vec{F}_{\text{noise}}$$

The average contribution represents systematic friction due to collisions with the fluid (viscous term), while the fluctuating contribution represents the random (in direction and velocity), rapidly varying microscopic forces (noise term); this is a statistical contribution. In this way, we reduced a huge deterministic problem (N coupled ODEs) in one stochastic equation.

We now want to understand the form of the two forces of the previous approximation.

For moderate velocities, the friction is well approximated by a linear damping term,

$$\vec{F}_{\text{avg}}(t) = -\gamma \dot{\vec{x}}(t)$$

where $\gamma > 0$ is a damping coefficient (for a spherical particle in a viscous fluid at low Reynolds number one recovers Stokes' law $\gamma = 6\pi\eta R$, with η the viscosity and R the particle radius).

To model the fluctuating force $\vec{F}_{\text{noise}}$, one assumes a separation of time scales between microscopic dynamics and macroscopic observation. Microscopic collisions between the suspended particle and the surrounding fluid particles occur on a very short correlation time τ_c whereas observations are made on much longer time scales.

One can therefore introduce an intermediate time interval Δt such that $\tau_c \ll \Delta t \ll \Delta T_{\text{obs}}$. On these intermediate scales, the detailed microscopic dynamics cannot be resolved, but successive random impulses acting on the mesoscopic particle can be treated as statistically independent.

This approximation relies on two distinct sources of independence. First, the particles of the medium are assumed to be at thermal equilibrium and to move independently of one another, so that there are no correlations between forces exerted by different particles. Second, due to the very large number of rapid thermal collisions, the force exerted by the bath on the suspended particle loses memory of its past values over times of order τ_c; as a result, the force acting on the same particle at sufficiently separated times can also be treated as independent. Under these assumptions, increments of the force over different time intervals are uncorrelated.

Moreover, since the heat bath is homogeneous and isotropic, there are no preferred spatial directions. This symmetry implies that the average fluctuating force vanishes, while its correlations are local in time. Consequently, the random force can be modeled as Gaussian white noise:

$$\vec{F}_{\text{noise}}(t) = \sqrt{2\gamma k_B T}\vec{\xi}(t), \qquad \langle \xi(t) \rangle = 0, \qquad \left\langle \xi(t)\xi(t') \right\rangle = \delta(t - t')$$

In this representation, the noise is delta-correlated in time and statistically independent at different instants, reflecting the absence of memory on the relevant time scales. The stochastic force is thus proportional to a Brownian process (more precisely, to the time derivative of a Brownian motion).

This resulting equation of motion is the Langevin equation for a Brownian particle. Importantly, the amplitude $\sqrt{2\gamma k_B T}$ is not arbitrary: it is fixed by the fluctuation-dissipation theorem and ensures that the particle relaxes toward thermal equilibrium, with a stationary distribution given by the Maxwell-Boltzmann law at temperature T.

11.1.1 Connection with the diffusion process

Notice the close parallel with the stochastic processes discussed earlier and with the Brownian motion derived from Markovian dynamics. These descriptions are not independent of one another: rather, they represent different levels of approximation of the same underlying physical phenomenon.

To study how the variance of the position of a mesoscopic particle evolves in time, we need to determine the time derivative of the mean square displacement $\langle x(t)^2 \rangle$. To this end, we focus on the one-dimensional case in the absence of external forces. The systematic and fluctuating forces acting on the particle are modeled as follows:

$$F_{\text{avg}} = -\gamma \dot{x}, \qquad F_{\text{noise}} = \sigma \xi(t)$$

where $\xi(t)$ is a Gaussian white noise with zero mean and unit variance. From Equation 11.1, the equation of motion becomes:

$$m\ddot{x}(t) = -\gamma \dot{x}(t) + \sigma \xi(t) \tag{11.2}$$

with

$$\langle \xi \rangle = 0, \qquad \left\langle \xi(t)\xi(t') \right\rangle = \delta(t - t')$$

While Equation 11.1 is a deterministic Newton equation with $O(N_A)$ dof, Equation 11.2 replaces the microscopic interaction with an effective force composed of two contributions, with just one dof, namely the position of the suspended particle. The price to pay for this simplification is that the parameters γ and σ are not known a priori and must be fixed using physical arguments.

Equation 11.2 is a linear SDE of second order in time. While the position process $x(t)$ alone is not Markovian, the joint process $(x(t), \dot{x}(t))$ is Markovian. To extract physical information from this equation, we focus on the evolution of the mean square displacement.

We multiply both sides of Equation 11.2 by $x(t)$:

$$mx\ddot{x} = -\gamma x\dot{x} + \sigma x \xi(t)$$

Using the identity

$$\frac{\mathrm{d}^2 x^2}{\mathrm{d}t^2} = 2(\dot{x}^2 + x\ddot{x})$$

we can rewrite the term $x\ddot{x}$ as

$$x\ddot{x} = \frac{1}{2}\frac{\mathrm{d}^2 x^2}{\mathrm{d}t^2} - \dot{x}^2$$

Substituting this expression into the previous equation gives

$$\frac{m}{2}\frac{\mathrm{d}^2 x^2}{\mathrm{d}t^2} - m\dot{x}^2 = -\frac{\gamma}{2}\frac{\mathrm{d}x^2}{\mathrm{d}t} + \sigma x \xi(t)$$

We now take the average of both sides. Using the Itô prescription, the mixed term $\langle x\xi(t) \rangle$ vanishes, since the noise is uncorrelated with the position at the same time. We therefore obtain:

$$\frac{m}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left\langle x^2 \right\rangle - m\left\langle \dot{x}^2 \right\rangle = -\frac{\gamma}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left\langle x^2 \right\rangle$$

The term $m\langle \dot{x}^2 \rangle$ is the average kinetic energy. In thermal equilibrium, the equipartition theorem gives

$$m\left\langle \dot{x}^2 \right\rangle = k_B T$$

Using this result, we obtain the closed equation

$$m\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left\langle x^2\right\rangle + \gamma\frac{\mathrm{d}}{\mathrm{d}t}\left\langle x^2\right\rangle = 2k_BT \tag{11.3}$$

which is a deterministic equation for the mean square displacement and follows from the Langevin dynamics under the physical assumptions introduced above. Introducing the variable

$$y(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left\langle x^2\right\rangle$$

Equation 11.3 reduces to

$$m\dot{y} + \gamma y = 2k_BT$$

Its solution is:

$$y(t) = C\exp\left\{-\frac{\gamma}{m}t\right\} + \frac{2k_BT}{\gamma} \tag{11.4}$$

where C is a constant determined by the initial condition. The solution contains a transient exponentially decaying term and a constant stationary contribution. The characteristic relaxation time is m/γ.

For typical suspended particles, one has $m/\gamma \simeq 10^{-8}\,\mathrm{s}$, which is much smaller than experimental observation times and this goes to stationarity very quickly. Therefore, for times $t \gg m/\gamma$, the system rapidly reaches the stationary regime and

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\langle x^2\right\rangle \simeq \frac{2k_BT}{\gamma}$$

Integrating in time yields

$$\left\langle x(t)^2\right\rangle - \left\langle x_0^2\right\rangle = \frac{2k_BT}{\gamma}t, \qquad \text{for } t \gg 10^{-8} \text{ sec.} \tag{11.5}$$

This result shows that the mean square displacement grows linearly with time, which is the defining signature of Brownian motion. While the microscopic dynamics at very short time scales is complex and inertia plays an important role, at sufficiently large time scales the motion of the particle is effectively diffusive.

We now make the connection with the Brownian process. In the overdamped regime, where intertial effects can be neglected on the relevant time scales, the dynamics of the particle position is effectively described by the SDE:

$$\dot{x} = \sqrt{2D}\xi_t \qquad (dx = \sqrt{2D}\,dB) \tag{11.6}$$

where $B(t)$ is the standard Brownian motion and D is the diffusion coefficient. The probability density $p(x,t)$ associated with this process satisfies the heat (diffusion) equation

$$\partial_t p(x,t) = D\partial_x^2 p(x,t)$$

Integrating Equation 11.6 yields

$$x(t) = x_0 + \sqrt{2D}B(t)$$

From this expression, we can compute the mean square displacement:

$$\left\langle x^2(t)\right\rangle = \left\langle x_0^2 + 2x_0\sqrt{2D}B(t) + 2DB^2(t)\right\rangle = x_0^2 + 2Dt \tag{11.7}$$

where we used $\langle B(t)\rangle = 0$ and $\left\langle B^2(t)\right\rangle = t$. Equation 11.7 is known as the **diffusion law**.

11.1.2 Einstein relation

By comparing Equation 11.7 with the large-time behavior of the Langevin dynamics derived in Equation 11.5, we identify the diffusion coefficient as

$$D = \frac{k_B T}{\gamma} \tag{11.8}$$

This relation ensures that the Brownian process reproduces the correct physical behavior of the mesoscopic particle at long time scales. Equation 11.8 is the **Einstein relation** and represents the simplest example of the fluctuation-dissipation theorem: in thermal equilibrium, the intensity of thermal fluctuations is directly linked to the strength of dissipative forces and the two must balance.

A few observations:

1. Equation 11.8 tells us how the diffusion coefficient must be chosen in order to give a correct physical interpretation of the mesoscopic particle as an effective free particle. Although microscopically the particle interacts with a very large number of surrounding particles, at long time scales its dynamics can be approximated by that of a free Brownian particle in a heat bath, provided that the diffusivity is fixed according to the Einstein relation. In this sense, the Einstein relation connects dissipation, thermal fluctuations, and macroscopic diffusion in a single framework.

2. The diffusion coefficient D does not depend on the initial conditions of the particle nor on the detailed form of the microscopic interactions. This reflects a universal behavior: at long time scales, the dynamics becomes insensitive to microscopic details, which are effectively washed out by the interaction with the heat bath.

3. The Einstein relation can be used to estimate fundamental constant, such as the Boltzmann constant k_B or Avogadro's number N_A. This was historically one of the most important achievements of Einstein's theory of Brownian motion.

4. Finally, this interpretation allows us to fix the amplitude of the noise term in the Langevin equation. Choosing $\sigma = \sqrt{2\gamma k_B T}$ in Equation 11.2 and dividing the equation by γ, one can take the limit $m/\gamma \to 0$. In this limit, inertia becomes negligible and the Langevin equation reduces to a first-order stochastic equation for the position, which coincides wit the equation of Brownian motion.

$$\frac{m}{\gamma}\ddot{x} = -\dot{x} + \frac{\sqrt{2\gamma k_B T}}{\gamma}\xi_t \quad \xrightarrow{m/\gamma \to 0} \quad \dot{x} = \sqrt{\frac{2k_B T}{\gamma}}\xi_t$$

   This procedure is known as the overdamped Langevin limit and the resulting equation is called the overdamped Langevin equation. In this regime, the motion of the particle is therefore well described by Brownian motion. We'll see this in the next section.

11.2 Stochastic harmonic oscillator in a heat bath

11.2.1 Connection with statistical mechanics: the Boltzmann distribution

From statistical mechanics, we know the equilibrium PDF for a particle with position $\vec{x}$ and momentum $\vec{p}$, subject to an external potential $U(\vec{x})$ and in contact with a heat bath

at temperature T. This joint distribution is given by

$$\mathcal{P}(\vec{x},\vec{p}) = \frac{1}{z} e^{-\beta(\frac{\vec{p}^2}{2m}+U(\vec{x}))} \tag{11.9}$$

where $\beta = 1/k_B T$ is the **Boltzmann's weight**, m is the mass of the particle, and z is the total partition function

$$z = \int d\vec{p} \int d\vec{x}\, \mathcal{P}(\vec{x},\vec{p}) = (2\pi m k_B T)^{3/2} z_0$$

where z_0 is the reduced partition function. Integrating Equation 11.9 over momenta yields the equilibrium distribution of the position alone,

$$w(\vec{x}) = \frac{1}{z_0} e^{-\beta U(\vec{x})} \tag{11.10}$$

which is the Boltzmann distribution in configuration space.

A natural question is whether this equilibrium result can be recovered from the stochastic dynamics developed so far. The answer is yes. For simplicity, we restrict ourselves to one spatial dimension, noting that the generalization to higher dimensions is straightforward.

We consider the system described by Equation 11.2 adding an external conservative force derived from a potential $U(x)$,

$$m\ddot{x}(t) = -\gamma \dot{x(t)} - \frac{\mathrm{d}U}{\mathrm{d}x} + \sigma\xi(t) \tag{11.11}$$

In the overdamped limit $m/\gamma \to 0$, inertia becomes negligible and the equation reduces to:

$$\frac{m}{\gamma}\ddot{x} = -\dot{x} - \frac{1}{\gamma}\frac{\mathrm{d}U}{\mathrm{d}x} + \frac{\sigma}{\gamma}\xi(t) \quad \xrightarrow{m/\gamma \to 0} \quad -\dot{x} - \frac{1}{\gamma}\frac{\mathrm{d}U}{\mathrm{d}x} + \frac{\sigma}{\gamma}\xi(t) = 0 \tag{11.12}$$

At thermal equilibrium, we can write $\xi(t)$ using Equation 11.6 as $\sqrt{2D}$. Using Einstein relation 11.8, we have:

$$\frac{\sigma}{\gamma} = \sqrt{\frac{2k_B T}{\gamma}} \tag{11.13}$$

By inserting this in Equation 11.12, we obtain the **overdamped Langevin equation** in stochastic differential form,

$$dx = -\frac{1}{\gamma}\frac{\mathrm{d}U}{\mathrm{d}x}\, dt + \sqrt{\frac{2k_B T}{\gamma}}\, dB(t)$$

We can identify two terms: the drift and diffusion coefficients, which are therefore

$$\mu(x) = -\frac{1}{\gamma}\frac{\mathrm{d}U}{\mathrm{d}x}, \qquad D = \frac{k_B T}{\gamma}$$

The corresponding Fokker-Planck equation at equilibrium, comparing Equation 10.14, is given by

$$p_{\mathrm{eq}}(x) \propto \exp\left\{2\int_0^x \left(-\frac{1}{\gamma}\frac{\mathrm{d}U}{\mathrm{d}x'}\right)\cdot \frac{\gamma}{2k_B T}\, dx'\right\} = \exp\left\{-\frac{1}{k_B T}\int_0^x \frac{\mathrm{d}U}{\mathrm{d}x'}\, dx'\right\}$$

that is,

$$p_{\text{eq}}(x) \propto e^{-\frac{1}{k_B T}U(x)} \tag{11.14}$$

where we have chosen for simplicity $U(0) = 0$.

This result coincides exactly with the Boltzmann distribution obtained from equilibrium statistical mechanics. The stochastic dynamics therefore reproduces the correct thermodynamic equilibrium, showing the consistency between the Langevin description and statistical mechanics. In this overdamped framework, only the positional dof appears explicitly, while the kinetic contribution has been integrated out.

As an important example, consider a harmonic potential $U(x) = \frac{1}{2}kx^2$. In this case, the overdamped Langevin equation becomes

$$dx = -\frac{k}{\gamma}x\,dt + \sqrt{\frac{2k_B T}{\gamma}}dB(t)$$

which defines the Ornstein-Uhlenbeck process. This process describes a stochastic harmonic oscillator in contact with a heat bath and can be solved exactly, which is what we will see.

11.2.2 Brownian particle in a heat bath under a harmonic potential

We now want to solve Equation 11.11 without taking the overdamped limit, i.e. keeping the inertial second-order term. For simplicity we choose a harmonic potential $U(x) = \frac{1}{2}kx^2$, which leads to what is usually called the inertial (or damped) Brownian motion.

Since $dU/dx = kx$ and, at equilibrium, the fluctuation-dissipation relation (11.13) fixes the noise amplitude to $\sigma = \sqrt{2\gamma k_B T}$, the Langevin equation becomes

$$\ddot{x}(t) + \frac{\gamma}{m}\dot{x}(t) + \frac{k}{m}x(t) = \frac{\sqrt{2\gamma k_B T}}{m}\xi(t) \tag{11.15}$$

where $\xi(t)$ is Gaussian white noise. This is a second-order linear non-homogeneous differential equation (stochastically forced). The standard strategy is to first solve the associated homogeneous equation

$$\ddot{x}(t) + \frac{\gamma}{m}\dot{x}(t) + \frac{k}{m}x(t) = 0 \tag{11.16}$$

and then incorporating the forcing term.

To solve the homogeneous equation we look for exponential solutions of the form $x(t) = e^{\lambda t}$. Substituting into Equation 11.16 gives the characteristic polynomial

$$\lambda^2 + \frac{\gamma}{m}\lambda + \frac{k}{m} = 0$$

whose roots are

$$\lambda_{\pm} = -\frac{\gamma}{2m} \pm \sqrt{\frac{\gamma^2}{4m^2} - \frac{k}{m}}$$

It is convenient to introduce the reduced friction $\gamma_r = \frac{\gamma}{2m}$ and the natural frequency $\omega_0 = \sqrt{\frac{k}{m}}$. In these terms,

$$\lambda_{\pm} = -\gamma_r \pm \sqrt{\gamma_r^2 - \omega_0^2} = -\gamma_r \pm \sqrt{\Omega} \tag{11.17}$$

where we defined the discriminant $\Omega = \gamma_r^2 - \omega_0^2$. The qualitative behavior depends on the sign of Ω, leading to three regimes:

1. **Overdamped regime**: ($\Omega > 0$, i.e. $\gamma_r^2 > \omega_0^2$)

   The two roots are real and distinct. The homogeneous solution is a linear combination of two decaying exponentials:

$$x^{\mathrm{h}}(t) = Ae^{(-\gamma_r - \sqrt{\Omega})t} + Be^{(-\gamma_r + \sqrt{\Omega})t} = \left(Ae^{-\sqrt{\Omega}t} + Be^{\sqrt{\Omega}t}\right) e^{-\gamma_r t}$$

2. **Critical damping**: ($\Omega = 0$, i.e. $\gamma_r^2 = \omega_0^2$)

   The two roots coincides at $\lambda = -\gamma_r$. In this case the homogeneous solution is not just a single exponential: the second independent solution picks up a factor of t, so

$$x^{\mathrm{h}}(t) = (A + Bt)\, e^{-\gamma_r t}$$

3. **Underdamped regime**: ($\Omega < 0$, i.e. $\gamma_r^2 < \omega_0^2$

   The roots are complex conjugates. Writing $\omega_d = \sqrt{\omega_0^2 - \gamma_r^2}$ (the damped oscillation frequency), we have

$$\lambda_{\pm} = -\gamma_r \pm i\omega_d =$$

   Therefore, the homogeneous solution is an exponentially damped oscillation

$$x^{\mathrm{h}}(t) = \left[A\cos\left(\omega_d t\right) + B\sin\left(\omega_d t\right)\right] e^{-\gamma_r t}$$

To solve the full equation we must now take into account the stochastic forcing term, represented by the noise $\xi(t)$. Because the equation is linear, the most convenient approach is the **Green function** method.

We rewrite the Langevin equation in operator form as

$$\mathcal{L}\left[x(t)\right] = \frac{\sqrt{2\gamma k_B T}}{m}\xi(t)$$

where the linear differential operator is

$$\mathcal{L} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + \frac{\gamma}{m}\frac{\mathrm{d}}{\mathrm{d}t} + \frac{k}{m}$$

The noise term can be formally expressed using the Dirac delta function as

$$\xi(t) = \int_{-\infty}^{\infty} \xi(t')\delta(t - t')\, dt'$$

Restricting the dynamics to the causal interval $[0, t]$, this becomes

$$\mathcal{L}\left[x(t)\right] = \frac{\sqrt{2\gamma k_B T}}{m}\int_0^t \xi(t')\delta(t - t')\, dt'$$

We now introduce the Green function $G(t, t')$, defined as the solution of

$$\mathcal{L}[G(t, t')] = \delta(t - t')$$

with the causality condition $G(t, t') = 0$ for $t < t'$.

Using linearity of the operator, we can write

$$\mathcal{L}\left[x(t)\right] = \mathcal{L}\left[\frac{\sqrt{2\gamma k_B T}}{m}\int_0^t \xi(t')G(t, t')\, dt'\right]$$

Therefore, the general solution of the stochastic equation is

$$x(t) = x^{\mathrm{h}}(t) + \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t \xi(t') G(t,t')\, dt'$$

where $x^{\mathrm{h}}(t)$ is the solution of the homogeneous equation and $\mathcal{L}[x^{\mathrm{h}}(t)] = 0$ by definition.

Since white noise is the formal time derivative of Brownian motion,$\xi(t)\, dt = dB(t)$, the stochastic solution can be written in the Itô form as

$$x(t) = x^{\mathrm{h}}(t) + x^{s}(t) = x^{\mathrm{h}}(t) + \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t G(t,t')\, dB(t') \tag{11.18}$$

Because the Green functions for the damped harmonic oscillator are known explicitly, we can write the solution in the three damping regimes.

1. **Overdamped regime** ($\gamma_r^2 > \omega_0^2$):

   The causal Green function is

   $$G(t,t') = \Theta(t-t') e^{-\gamma_r (t-t')} \frac{\sinh[\omega_d (t-t')]}{\omega_d}$$

   Since $t > t'$, then $\Theta(t-t') = 1$. Therefore, the full solution splits into

   - a deterministic part composed of two decaying exponentials;
   - a stochastic part given by an exponentially damped convolution with the noise.

   These are:

   $$\begin{cases} x^{\mathrm{h}}(t) = \left(A e^{-\omega_d t} + B e^{\omega_d t}\right) e^{-\gamma_r t} \\ x^{s}(t) = \frac{1}{m} \frac{\sqrt{2\gamma k_B T}}{\omega_d} \int_0^t e^{-\gamma_r (t-t')} \sinh\left[\omega_d (t-t')\right] dB(t') \end{cases}$$

   Although the stochastic contribution involves a hyperbolic sine, the overall behavior remains exponentially decaying because it is multiplied by the damping factor $e^{-\gamma_r (t-t')}$. As a result, the system relaxes monotonically to equilibrium without oscillations.

2. **Critical damping** ($\gamma_r^2 = \omega_0^2$):

   $$G(t,t') = \Theta(t-t') e^{-\gamma_r (t-t')} (t-t')$$

   Since $t > t'$, then $\Theta(t-t') = 1$.

   Here, the homogeneous solution decays as $(A+Bt)e^{-\gamma_r t}$, while the stochastic contribution involves an additional factor of $(t-t')$, which accelerates the decay. This regime corresponds to the fastest relaxation to equilibrium without oscillatory behavior.

   $$\begin{cases} x^{\mathrm{h}}(t) = (A + Bt)\, e^{-\gamma_r t} \\ x^{s}(t) = \frac{\sqrt{2\gamma k_B T}}{m} \int_0^t e^{-\gamma_r (t-t')} (t-t')\, dB(t') \end{cases}$$

3. **Underdamped regime** ($\gamma_r^2 < \omega_0^2$):

   $$G(t,t') = \Theta(t-t') e^{-\gamma_r (t-t')} \frac{\sin[\omega_d (t-t')]}{\omega_d}$$

Since $t > t'$, then $\Theta(t - t') = 1$. In this regime the homogeneous solution is an exponentially damped oscillation, and the stochastic contribution inherits the same oscillatory structure.

$$\begin{cases} x^{\mathrm{h}}(t) = \left[A\cos\left(\omega_d t\right) + B\sin\left(\omega_d t\right)\right] e^{-\gamma_r t} \\ x^{s}(t) = \frac{1}{m}\frac{\sqrt{2\gamma k_B T}}{\omega_d}\int_0^t e^{-\gamma_r (t-t')}\sin\left[\omega_d (t-t')\right] dB(t') \end{cases}$$

The particle approaches equilibrium through damped oscillations, with random fluctuations superimposed by the thermal noise.

The physical interpretation is that in all three cases, the green function enforces causality and encodes the characteristic relaxation time scales of the system. The deterministic part reflects how the system would relax in the absence of noise, while the stochastic integral accounts for continuous random kicks from the heat bath. Despite the different transient behaviors (monotonic decay versus damped oscillations), the long-time stationary state is the same in all regimes and is entirely fixed by the fluctuation-dissipation relation.

Finally, we compute the two-point correlation function $\langle x(t)x(s)\rangle$, which quantifies how the position of the oscillator at two different times is correlated.

From the Green-function representation, we write the solution as the sum of a homogeneous (deterministic) part and a stochastic part. The two point correlation function then reads

$$\langle x(t)x(s)\rangle = \left\langle [x^{\mathrm{h}}(t) + x^{s}(t)][x^{\mathrm{h}}(s) + x^{s}(s)]\right\rangle \tag{11.19}$$

where

$$x^{\mathrm{s}}(t) = \frac{\sqrt{2\gamma k_B T}}{m}\int_0^t G(t,t')\, dB(t')$$

Expanding Equation 11.19, we obtain

$$\langle x^{\mathrm{h}}(t)x^{\mathrm{h}}(s)\rangle + \langle x^{\mathrm{h}}(t)x^{\mathrm{s}}(s)\rangle + \langle x^{\mathrm{s}}(t)x^{\mathrm{h}}(s)\rangle + \langle x^{\mathrm{s}}(t)x^{\mathrm{s}}(s)\rangle$$

Assuming deterministic (or noise-independent) initial conditions, the homogeneous solution $x^{\mathrm{h}}(t)$ is non-random. Since the Itô integral has zero mean,

$$\langle x^{\mathrm{s}}(t)\rangle = \frac{\sqrt{2\gamma k_B T}}{m}\left\langle \int_0^t G(t,t')\, dB(t')\right\rangle = 0$$

the mixed terms vanish (independence of increments):

$$\langle x^{\mathrm{h}}(t)x^{\mathrm{s}}(s)\rangle = \frac{\sqrt{2\gamma k_B T}}{m}\langle x^{\mathrm{h}}(t)\rangle \int_0^s \left\langle G(s,t')\right\rangle \left\langle dB(t')\right\rangle = 0$$

analogously for $\langle x^{\mathrm{h}}(s)x^{\mathrm{s}}(t)\rangle$.

Therefore,

$$\langle x(t)x(s)\rangle = \langle x^{\mathrm{h}}(t)x^{\mathrm{h}}(s)\rangle + \langle x^{\mathrm{s}}(t)x^{\mathrm{s}}(s)\rangle$$

Using Itô's isometry (or equivalently the white-noise correlation ($\langle \xi(t')\xi(t'')\rangle = \delta(t'-t'')$), we obtain

$$\begin{aligned} \langle x^{\mathrm{s}}(t)x^{\mathrm{s}}(s)\rangle &= \frac{2\gamma k_B T}{m^2}\int_0^t\int_0^s \left\langle G(t,t')G(s,t'')\right\rangle \left\langle \xi(t')\xi(t'')\right\rangle \, dt'dt'' = \\ &= \frac{2\gamma k_B T}{m^2}\int_0^t\int_0^s \left\langle G(t,t')G(s,t'')\right\rangle \delta(t'-t'')\, dt'dt'' = \\ &= \frac{2\gamma k_B T}{m^2}\int_0^{\min(t,s)} \left\langle G(t,t')G(s,t')\right\rangle \, dt' \end{aligned}$$

Hence, the general expression for the two-point correlation function is:

$$\langle x(t)x(s)\rangle = x^{\mathrm{h}}(t)x^{\mathrm{h}}(s) + \frac{2\gamma k_B T}{m^2}\int_0^{\min(t,s)} \left\langle G(t,t')G(s,t')\right\rangle \, dt'$$

This formula is valid in all damping regimes; the only difference lies in the explicit form of the Green function $G(t,t')$.

We now analyze the three cases.

1. **Overdamped case** $\omega_d^2 = \gamma_r^2 - \omega_0^2 > 0$.

   The Green function here is

   $$G(t,t') = \Theta(t-t')e^{-\gamma_r(t-t')}\frac{\sinh\left(\omega_d(t-t')\right)}{\omega_d}$$

   A convenient homogeneous solution is

   $$x^{\mathrm{h}}(t) = Ae^{(-\gamma_r-\omega_d)t} + Be^{(-\gamma_r+\omega_d)t}$$

   Then

   $$\begin{aligned}\langle x^{\mathrm{h}}(t)x^{\mathrm{h}}(s)\rangle &= \left(Ae^{-\omega_d t} + Be^{\omega_d t}\right)\left(Ae^{-\omega_d s} + Be^{\omega_d s}\right)e^{-\gamma_r(t+s)} = \\ &= \left[A^2e^{-\omega_d(t+s)} + 2AB\cosh\left(\omega_d(t-s)\right) + B^2e^{\omega_d(t+s)}\right]e^{-\gamma_r(t+s)}\end{aligned}$$

   For the stochastic part,

   $$\langle x^{\mathrm{s}}(t)x^{\mathrm{s}}(s)\rangle = \frac{2\gamma k_B T}{m^2\omega_d^2}\int_0^{\min(t,s)} e^{-\gamma_r(t+s-2t')}\sinh\left(\omega_d(t-t')\right)\sinh\left(\omega_d(s-t')\right)dt'$$

   Thus, the variance is ($t=s$):

   $$\left\langle x(t)^2\right\rangle = \left(x^{\mathrm{h}}(t)\right)^2 + \frac{2\gamma k_B T}{m^2\omega_d^2}\int_0^t e^{-2\gamma_r(t-t')}\sinh^2\left(\omega_d(t-t')\right)dt'$$

   Setting $\tau = t - t'$ gives

   $$\left\langle x(t)^2\right\rangle = \left(x^{\mathrm{h}}(t)\right)^2 + \frac{2\gamma k_B T}{m^2\omega_d^2}\int_0^t e^{-2\gamma_r\tau}\sinh^2\left(\omega_d\tau\right)dt'$$

   As $t\to\infty$, the homogeneous term decays to zero (since $\gamma_r > \omega_d$). The integral converges and one finds the stationary variance

   $$\lim_{t\to\infty}\left\langle x^2(t)\right\rangle = \frac{k_B T}{k}$$

   Therefore, the stationary variance scales linearly with the temperature of the bath and inversely with the stiffness k of the harmonic potential, reflecting the balance between thermal fluctuations and restoring forces.

2. **Critical damping** $\omega_d^2 = \gamma_r^2 - \omega_0^2 = 0$.

   The Green function for critical damping is

   $$G(t,t') = \Theta(t-t')e^{-\gamma_r(t-t')}(t-t')$$

The homogeneous solution (double root) is

$$x^{\mathrm{h}}(t) = (A + Bt)e^{-\gamma_r t}$$

Hence,

$$\begin{aligned}\langle x^{\mathrm{h}}(t)x^{\mathrm{h}}(s)\rangle &= (A + Bt)(A + Bs)e^{-\gamma_r(t+s)} = \\ &= \left[A^2 + AB(t + s) + B^2ts\right] e^{-\gamma_r(t+s)}\end{aligned}$$

For the stochastic contribution,

$$\langle x^{\mathrm{s}}(t)x^{\mathrm{s}}(s)\rangle = \frac{2\gamma k_B T}{m^2}\int_0^{\min(t,s)} e^{-\gamma_r(t+s-2t')}(t - t')(s - t')\,dt'$$

Therefore, the variance ($t = s$) is

$$\left\langle x(t)^2\right\rangle = (A + Bt)^2\, e^{-2\gamma_r t} + \frac{2\gamma k_B T}{m^2}\int_0^t e^{-2\gamma_r(t-t')}(t - t')^2\,dt'$$

With $\tau = t - t'$,

$$\left\langle x^2(t)\right\rangle = (A + Bt)^2 e^{-2\gamma_r t} + \frac{2\gamma k_B T}{m^2}\int_0^t e^{-2\gamma_r\tau}\tau^2\,d\tau$$

Using $\int_0^\infty e^{-a\tau}\tau^2\,d\tau = 2/a^3$ (for $a > 0$), with $a = 2\gamma_r$, we get

$$\lim_{t\to\infty}\int_0^t e^{-2\gamma_r\tau}\tau^2\,d\tau = \frac{1}{4\gamma_r^3}$$

Therefore

$$\lim_{t\to\infty}\left\langle x^2(t)\right\rangle = \frac{2\gamma k_B T}{m^2}\cdot\frac{1}{4\gamma_r^3} = \frac{\gamma k_B T}{2m^2\gamma_r^3}$$

Since $\gamma_r = \gamma/2m$

$$\lim_{t\to\infty}\left\langle x^2(t)\right\rangle = \frac{k_B T}{k}$$

3. **Underdamped case** $\omega_d^2 = \gamma_r^2 - \omega_0^2 < 0$.

The Green function is

$$G(t, t') = \Theta(t - t')e^{-\gamma_r(t-t')}\frac{\sin(\omega_d(t - t'))}{\omega_d}$$

A standard homogeneous solution is

$$x^{\mathrm{h}}(t) = e^{-\gamma_r t}\left(A\cos(\omega_d t) + B\sin(\omega_d t)\right)$$

Thus

$$\langle x^{\mathrm{h}}(t)x^{\mathrm{h}}(s)\rangle = \left[A\cos(\omega_d t) + B\sin(\omega_d t)\right]\left[A\cos(\omega_d s) + B\sin(\omega_d s)\right] e^{-\gamma_r(t+s)}$$

For the stochastic part,

$$\langle x^{\mathrm{s}}(t)x^{\mathrm{s}}(s)\rangle = \frac{2\gamma k_B T}{m^2\omega_d^2}\int_0^{\min(t,s)} e^{-\gamma_r(t+s-2t')}\sin\left(\omega_d(t - t')\right)\sin\left(\omega_d(s - t')\right)\,dt'$$

and the variance ($t = s$) becomes

$$\left\langle x^2(t) \right\rangle = \left(x^{\mathrm{h}}(t) \right)^2 + \frac{2\gamma k_B T}{m^2 \omega_d^2} \int_0^{\min(t,s)} e^{-2\gamma_r (t-t')} \sin^2 \left(\omega_d (t-t') \right) dt'$$

With $\tau = t - t'$,

$$\left\langle x^2(t) \right\rangle = \left(x^{\mathrm{h}}(t) \right)^2 + \frac{2\gamma k_B T}{m^2 \omega_d^2} \int_0^{\min(t,s)} e^{-2\gamma_r \tau} \sin^2 \left(\omega_d \tau \right) d\tau$$

The homogeneous term decays to zero as $t \to \infty$. The integral converges and one obtains again

$$\lim_{t \to \infty} \left\langle x^2(t) \right\rangle = \frac{k_B T}{k}$$

which is the same result of the overdamped case.

Hint: to compute the integral of the homogeneous correlation term, use the following trigonometric identities

$$\begin{cases} \cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)] \\ \cos(\alpha)\sin(\beta) + \sin(\alpha)\cos(\beta) = \sin(\alpha + \beta) \\ \sin(\alpha)\sin(\beta) = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)] \end{cases}$$

while for the second integral in the variance, use

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

Chapter 12

Introduction to Birth and Death Markov Processes

We now discuss the last class of Markov processes considered in this course: discrete-state Markov processes, in which the set of possible states is countable and can be labeled with integers. Since the state space is no longer continuous, stochastic calculus does not apply. Instead, the time evolution of the system is described by the master question, which governs the probabilities of occupying the discrete states.

In this framework, the fundamental object of interest is the probability $p_n(t)$ that the system is in state n at time t. The master equation provides a balance equation for these probabilities, accounting for transitions into and out of each state.

Among discrete Markov processes, we focus in particular on birth and death processes, which constitute one of the most tractable and physically relevant subclasses. These processes are widely used to model systems in which populations increase through creation events (birth event) and decrease through annihilation or removal events (death event).

12.1 Introduction

We consider a population-based Markov process whose state is labeled by an integer n. From a given state, the system can jump either to $n+1$ or to $n-1$. While more general jump processes are possible (for instance, transitions involving multiple individuals at once), we focus on the simplest and most tractable case: **birth and death processes**, in which transitions always occur by a single unit and the rates depend on the state they start.

We assume discrete times $t = 0, \Delta t, 2\Delta t, \ldots$ and a discrete state space $n = 0, \pm 1, \pm 2, \ldots$, with transitions allowed only between nearest neighbors, from n to $n \pm 1$. For sufficiently small Δt, we introduce the **birth rate** $b_n \geq 0$, defined such that $b_n \Delta t$ is the probability of a transition from state n to $n+1$ during the interval $[t, t+\Delta t]$, given that the system is in state n at time t.

The birth rate therefore represents the probability per unit time of a single birth event. Transitions involving more than one birth within the same time step are of higher order in Δt and are neglected. Unlike a simple random walk, where transition rates are constant, here the rates depend explicitly on the current state n. In our case, the rates are assumed to be time-independent; when this is the case, the process is said to be **homogeneous**. Analogously, we define the **death rate** $d_n \geq 0$, such that $d_n \Delta t$ is the probability of a transition from state n to state $n-1$ during the interval $[t, t+\Delta t]$.

Our objective is to compute the propagator $p(n, t|n_0, t_0)$, that is, the probability that the system is found in state n at time t, given that it started in state n_0 at time t_0.

By conditioning on the state of the system at time t, the probability at time $t + \Delta t$ can be written as

$$p(n, t + \Delta t) = \underbrace{b_{n-1}\Delta t\, p(n-1, t)}_{1} + \underbrace{d_{n+1}\Delta t\, p(n+1, t)}_{2} + \underbrace{[1 - (b_n + d_n)\Delta t]p(n, t)}_{3}$$

The three contributions correspond respectively to reaching state n from $n-1$ through a birth, from $n+1$ through a death, or remaining in state n with no transition occurring (Figure 12.1).

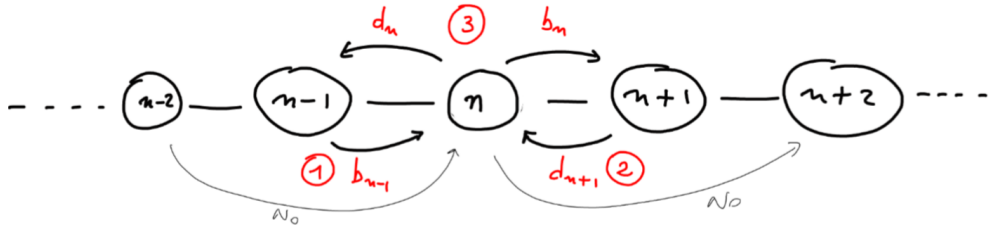


Figure 12.1

Assuming $\Delta t \ll 1$, we expand the left-hand side to first order in time

$$p_n(t) + \dot{p}_n(t)\Delta t = p_n + b_n\Delta t\, p_{n-1} + d_{n+1}\Delta t\, p_{n+1} - (b_n + d_n)\Delta t\, p_n$$

Then, by taking the limit $\Delta t \to 0$, we get the **master equation of the birth-death process**

$$\frac{\partial p_n}{\partial t} = b_{n-1}p_{n-1}(t) + d_{n+1}p_{n+1}(t) - (b_n + d_n)p_n(t) \tag{12.1}$$

Let's consider the following observations:

1. The master equation has the structure of a gain-loss balance: probability flows into state n from neighboring states and flows out due to transitions originating from n. As a result, total probability is conserved.

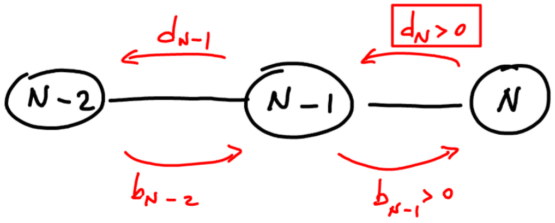
2. To determine a unique solution, the master equation also needs an initial condition. Since $p_n(t) \equiv p(nt|n_0t_0)$, the natural choice is:

$$\lim_{t \to t_0} p_n = \delta_{n,n_0}$$

3. Boundary conditions are also required when the state space is restricted, as in population models with a maximum population size N. These boundary conditions are not dictated by the master equation itself, which governs only the bulk dynamics.

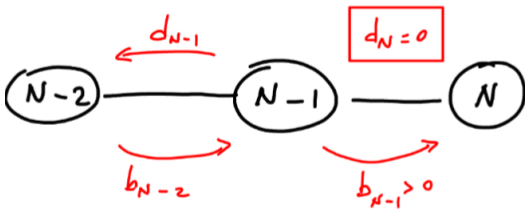
   We define the following boundary conditions:

   - If N is a **reflecting boundary**, we have:



In this case the system may reach state N but cannot transition beyond it. Thus, birth events at N are forbidden, while death events remain allowed. In this case, we have $b_n = 0$ for $n \geq N$ and $d_n = 0$ for $n > N$ and $d_N > 0$.

- If instead N is an **absorbing boundary**, the system can reach state N but cannot leave it once it arrives ($b_N = 0$). Then we set $d_N = 0$.



WARNING! Because of the presence of boundaries, the master equation 12.1 must be applied with care as it holds in its standard form only for states that are not boundary states (we repeat that it governs the states in the bulk).

4. Equation 12.1 is linear and deterministic, despite describing an underlying stochastic process. It plays a role analogous to that of the Fokker-Planck equation, but for discrete state spaces. In contrast, it is fundamentally different from a Langevin equation, which governs individual stochastic trajectories.

   The master equation can be written in compact matrix form. Defining a transition-rate matrix $\mathbf{W}$ with elements

$$\mathbf{W}_{n,n'} = d_{n'}\delta_{n,n'-1} + b_{n'}\delta_{n,n'+1} - (d_n + b_n)\delta_{n,n'} \tag{12.2}$$

   and introducing the probability vector $[\vec{p}(t)]_n \equiv p_n(t)$, the evolution equation 12.1 becomes

$$\begin{cases} \dot{\vec{p}}(t) = \mathbf{W}\vec{p}(t) \\ \vec{p}(t_0) = \vec{p}_0 \end{cases} \tag{12.3}$$

   Formally, the solution is given by $\vec{p}(t) = e^{\mathbf{W}(t-t_0)}\vec{p}_0$.

   This matrix formulation applies to general discrete Markov processes, not only to birth-death systems, provided that the transition matrix $\mathbf{W}$ satisfies two conditions:

   (a) off-diagonal elements must be non-negative: $\mathbf{W}_{n,n'} \geq 0$ for $n \neq n'$;

   (b) each column must sum to zero, ensuring conservation of probability in the absence of absorbing states: $\sum_n \mathbf{W}_{n,n'} = 0$ for each n'.

12.2 Equations for the mean and variance

Let's define the mean population size (first moment) as

$$\langle n \rangle \equiv \sum_{-\infty}^{+\infty} n p_n(t)$$

where the sum runs over the admissible states. In a population model with a hard boundary at $n = 0$, the correct range is $n = 0, 1, 2, \ldots$ (try to calculate it).

Differentiating with respect to time and inserting the master equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle n\rangle = \sum_{n=-\infty}^{+\infty} (nb_{n-1}p_{n-1} - nb_np_n + nd_{n+1}p_{n+1} - nd_np_n)$$

To simplify this expression, we shift indices in the *inflow* terms so that everything is written with the same index n. The birth inflow term becomes $\sum_n (n+1)b_np_n$, and the death inflow term becomes $\sum_n (n-1)d_np_n$. After cancellation with the outflow term, one obtains

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\langle n\rangle &= \sum_n [(n+1)b_np_n - nb_np_n + (n-1)d_np_n - nd_np_n] = \\ &= \sum_n (b_n - d_n)p_n\end{aligned}$$

Since the right-hand side is simply an expectation with respect to $p_n(t)$, this can be written compactly as

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle n\rangle = \langle b_n\rangle - \langle d_n\rangle \tag{12.4}$$

meaning $\langle b_n\rangle = \sum_n b_np_n(t)$ and similarly for d_n.

Exercise 12.2.1. *An analogous computation yields the evolution of the second moment* $\langle n^2\rangle$. *Starting from the master equation, multiplying by* n^2, *and summing over* n, *find that*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle n^2\rangle = 2\langle n(b_n - d_n)\rangle + \langle (b_n + d_n)\rangle \tag{12.5}$$

From $\langle n\rangle$ *and* $\langle n^2\rangle$, *the variance follows as* $\langle n^2\rangle - \langle n\rangle^2$.

12.3 The equilibrium distribution of the b/d master equation

We now study the equilibrium (stationary) distribution of the birth-death master equation 12.1, i.e., the distribution p^s for which $\dot{p}_n = 0$ for all n. For a one-step birth-death process this stationary distribution can be obtained in closed form, whereas for more general discrete Markov chains an explicit stationary measure is not always available.

We consider a birth-death process on the half-line $n = 0, 1, 2, \ldots$, with a reflecting boundary at $n = 0$ (so the process cannot jump to negative states). A useful way to analyze stationarity is to introduce the probability flux across each bond $(n-1) \to n$. Define the stationary flux from $n-1$ to n (Figure 12.2) as

$$J^s_{n-1} \equiv b_{n-1}p^s_{n-1} - d_np^s_n$$

This expression represents the net probability current: the rate of transitions $n-1 \to n$ minus the rate of transitions $n \to n-1$, both evaluated at stationarity. At equilibrium, it must go to zero.

At stationarity, the master equation 12.1 for state n can be written as a balance of two neighboring currents:

$$0 = \underbrace{b_{n-1}p^s_{n-1} - d_np^s_n}_{J_{n-1}} + \underbrace{d_{n+1}p^s_{n+1} - b_np^s_n}_{-J_n}$$

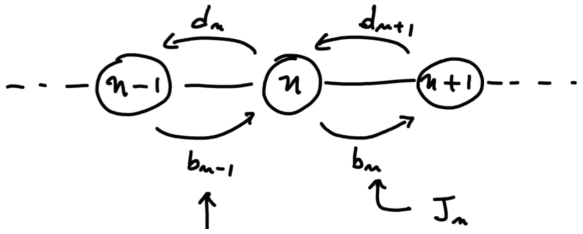


Figure 12.2

Hence, we can write

$$J_n = J_{n-1} = \ldots = J_0 \equiv b_{-1}p^s_{-1} - d_0 p^s_0 = 0$$

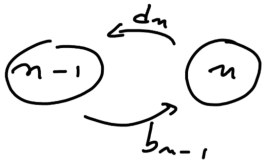
or, briefly, $J^s_n = J^s_{n-1}$ for all n, meaning that the stationary current is independent of n: it is a constant current through the chain. Note that the term $b_{-1}p^s_{-1}$ is zero because it does not exist.

The boundary condition at $n = 0$ fixes this constant. Reflecting at $n = 0$ means there is no probability flux leaving the state space at the lower boundary, which is equivalent to imposing $d_0 = 0$ (no transition $0 \to -1$). Therefore the current through the *nonexistent* bond $-1 \to 0$ must vanish, and the constant stationary current is $J^s_n \equiv 0$ for all n.

Setting $J^s_n = 0$ yields the recursive relation

$$J^s_{n-1} = b_{n-1}p^s_{n-1} - d_n p^s_n = 0 \quad \Rightarrow \quad b_{n-1}p^s_{n-1} = d_n p^s_n$$

for $n \geq 1$. This expresses the **detailed balance condition**: at equilibrium the probability flow from $n-1$ to n is exactly balanced, bond by bond, by the flow from n to $n-1$. In such a stationary state there are no circulating currents. Systems with such a property are called **equilibrium systems**.



Iterating the recursion relation gives the stationary distribution in product form:

$$p^s_n = \frac{b_{n-1}}{d_n}p^s_{n-1} = \frac{b_{n-1}}{d_n}\frac{b_{n-2}}{d_{n-1}}p^s_{n-2} = \ldots$$

Thus, in general

$$p^s_n = \prod_{i=1}^{n} \frac{b_{i-1}}{d_i} p^s_0 \qquad \text{for } n = 1, 2, \ldots \tag{12.6}$$

The remaining constant p^s_0 is fixed by normalization

$$\sum_{n=0}^{\infty} p^s_n = 1 \quad \Rightarrow$$

so

$$(p^s_0)^{-1} = 1 + \sum_{n=1}^{\infty}\prod_{i=1}^{n}\frac{b_{i-1}}{d_i}$$

A normalizable stationary distribution exists only if this series converges; equivalently, p_0^s must be finite.

The same approach of flux-based reasoning can be extended to compute p_n for a finite state space $n = 1, 2, \ldots, N$. With reflecting boundaries at both ends, the stationary current must vanish, and the stationary distribution is again determined by detailed balance and normalization.

12.4 Simple birth and death processes

We now look at simple but important examples of birth-death processes, both as applications of the master-equation formalism and as preparation for understanding how, in suitable limits, one can approximate a discrete master equation by a Fokker-Planck equation.

12.4.1 Poisson process

A fundamental example is the **Poisson process**. This is a pure-birth process on the state space $n = 0, 1, 2, \ldots$, defined by constant birth rates and no deaths:

$$b_n = \lambda, \qquad d_n = 0$$

(the rates are independent of n). The master equation is therefore

$$\begin{cases} \dot{p}_n(t) = \lambda(p_{n-1}(t) - p_n(t)) \\ p_n(0) = \delta_{n,0} \end{cases} \tag{12.7}$$

or more generally $p_n(0) = \delta_{n,n_0}$ if the process starts from n_0.

We can solve the master equation 12.7 with the generating function method. Define the generating function

$$g(z,t) \equiv \sum_{n=0}^{\infty} z^n p_n(t), \quad z \in \mathbb{C}$$

Multiplying Equation 12.7 by z^n and summing over $n \geq 0$, one obtains a closed PDE for g:

$$\begin{cases} \frac{\partial}{\partial t} g(z,t) = \lambda(z-1) g(z,t) \\ g(z,0) = 1 \end{cases} \tag{12.8}$$

where $(g(1,t) = \sum_n p_n(t) = 1$ expresses normalization of the probabilities. Solving gives

$$g(z,t) = e^{\lambda(z-1)t} = e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!} \stackrel{\text{def}}{=} \sum_{n=0}^{\infty} p_n(t) z^n$$

where we expanded the exponential in powers of z and compared term-by-term with the definition of the generating function. Thus, the probability distribution is:

$$p_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \tag{12.9}$$

Thus $n(t)$ is Poisson-distributed with parameter λt (or, if starting from n_0, $n(t) - n_0$ is Poisson with parameter λt). As a consequence, the variance equals the mean increment:

$$\mathrm{Var}(n(t)) = \lambda t$$

Show it as exercise.

Exercise 12.4.1. *Show that the mean satisfies the equation* $\frac{\mathrm{d}}{\mathrm{d}t}\langle n\rangle = \lambda$ *and so* $\langle n(t)\rangle = n_0 + \lambda t$ *if* $p_n(0) = \delta_{n,n_0}$.

A useful consistency check in generating-function methods is always to verify that $g(1,t) = 1$ for all t, since this is equivalent to probability conservation.

12.4.2 Radioactive decay

Consider a system that initially contains N_0 radioactive particles. Each particle decays independently at a constant rate γ. Let n be the number of particles that are still present (have not decayed) at time t. Over a short interval Δt, the probability that one decay occurs is proportional to the number of particles available to decay, so it is $\gamma n \Delta t$, while the probability of two or more decays in the same interval is of higher order in Δt and can be neglected in the continuous-time limit.

This is therefore a pure-death birth-death process with

$$d_n = \gamma n, \qquad b_n = 0$$

defined on the finite state space $n = 0, 1, 2, \ldots, N_0$. The master equation has only death terms: probability can flow only from $n+1$ to n, but not in the opposite direction

$$\begin{cases} \dot{p}_n(t) = \gamma(n+1)p_{n+1}(t) - \gamma n p_n(t), & n = 0, 1, \ldots, N_0 - 1 \\ \dot{p}_{N_0}(t) = -\gamma N_0 p_{N_0}(t) & \text{B.C.} \\ p_n(0) = \delta_{n,N_0} & \text{I.C} \end{cases}$$

where the initial condition is the number of particle at time $t = 0$. Notice that the number of particles that are still alive can only decrease and this sets the boundary conditions.

Exercise 12.4.2. *Show that from the general moment equation for birth-death processes, the mean satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle n(t)\rangle = \langle b_n - d_n\rangle = -\gamma\langle n(t)\rangle$$

so $\langle n(t)\rangle = N_0 e^{-\gamma t}$.

To solve the master equation and obtain the full distribution $p_n(t)$, it is convenient to use again the generating function method adapted to the finite state space:

$$g(z,t) = \sum_{n=0}^{N_0} z^n p_n(t)$$

Multiplying the master equation by z^n and summing over n yields a deterministic first-order PDE for g

$$\begin{cases} \partial_t g(z,t) = \gamma(1-z)\partial_z g(z,t) \\ g(1,t) = 1 \\ g(z,0) = z^{N_0} \end{cases} \tag{12.10}$$

together with the boundary condition and the initial condition, respectively.

This PDE can be solved by the method of characteristic (see PDE by L.C. Evans), but here it is simple enough to solve directly starting with an ansatz: the solution g has the form

$$h(z,t) = \varepsilon(t)(1-z) + 1$$

If we inser h into the PDE, we compute:

$$\partial_t h = \dot{\varepsilon}(t)(1-z), \qquad \partial_z h = -\varepsilon(t)$$

Requiring h to satisfy the same PDE as g, namely $\partial_t h = \gamma(1-z)\partial_z h$, gives

$$\dot{\varepsilon}(t)(1-z) = \gamma(1-z)(-\varepsilon(t)) \quad \Rightarrow \quad \dot{\varepsilon}(t) = -\gamma\varepsilon(t)$$

hence

$$\varepsilon(t) = \varepsilon_0 e^{-\gamma t}$$

By construction, $h(1,t) = 1$ for all t, so this automatically enforces the normalization constraint once g is taken as a function of h. However, $h(z,0) = \varepsilon_0(1-z)+1$ does not match the initial condition $g(z,0) = z^{N_0}$, so h alone cannot be the solution.

The key observation is that if h satisfies the PDE, then any composition $g(z,t) = f(h(z,t))$, with f differentiable, also satisfies it. Indeed, by the chain rule,

$$\partial_t g = f'(h)\partial_t h, \qquad \partial_z g = f'(h)\partial_z h$$

and substituting into $\partial_t g = \gamma(1-z)\partial_z g$ cancels the common factor $f'(h)$, leaving exactly the PDE for h.

We then choose f to enforce the initial condition. Taking $f(x) = x^{N_0}$ gives

$$g(z,t) = h(z,t)^{N_0} = \left(\varepsilon_0 e^{-\gamma t}(1-z)+1\right)^{N_0}$$

Now we impose the initial condition $g(z,0) = z^{N_0}$. At $t = 0$,

$$g(z,0) = \left(\varepsilon_0(1-z)+1\right)^{N_0}$$

To equal z^{N_0}, we need $\varepsilon_0 = -1$, so that $\varepsilon_0(1-z)+1 = z$. This yields the final generating function

$$g(z,t) = \left(e^{-\gamma t}(z-1)+1\right)^{N_0} \tag{12.11}$$

To recover the probabilities $p_n(t)$, we expand $g(z,t)$ as a polynomial in z:

$$g(z,t) = \left(e^{-\gamma t}z + \left(1-e^{-\gamma t}\right)\right)^{N_0} = \sum_{n=0}^{N_0} \binom{N_0}{n} \left(1-e^{-\gamma t}\right)^{N_0-n} (e^{-\gamma t})^n z^n$$

Comparing coefficients with $g(z,t) = \sum_{n=0}^{N_0} z^n p_n(t)$, we obtain

$$p_n(t) = \binom{N_0}{n} (e^{-\gamma t})^n \left(1-e^{-\gamma t}\right)^{N_0-n} \tag{12.12}$$

This is a binomial distribution and can be interpreted in the following way: each particle survives independently up to time t with probability $e^{-\gamma t}$, and decays by time t with probability $1-e^{-\gamma t}$. Consequently, the number of n surviving particles at time t is binomially distributed with parameters N_0 and $e^{-\gamma t}$. The factor $(e^{-\gamma t})^n$ accounts for the survival of a specific set of n particles, while $(1-e^{-\gamma t})^{N_0-n}$ accounts for the decay of the remaining N_0-n particles. The binomial coefficient $\binom{N_0}{n}$ reflects the number of distinct ways to choose which n particles survive, since the particles are indistinguishable.

12.4.3 Yule-Furry process

A simple idealized model for a particle cascade (such as the early stages of an electromagnetic shower) is a pure-birth branching process in which each existing particle independently *splits* at a constant rate. In the birth-death formalism, this corresponds to a pure birth process with state $n = 1, 2, \dots$ and rates

$$b_n = \gamma n, \qquad d_n = 0$$

so the total birth intensity is proportional to the current population size. The master equation is then

$$\begin{cases} \dot{p}_n(t) = \gamma(n-1)p_{n-1}(t) - \gamma n p_n(t) \\ p_n(0) = \delta_{n,1} \end{cases}$$

From the general moment equation for birth-death processes and with $\langle n(0)\rangle = 1$ we obtain

$$\langle n(t)\rangle = e^{\gamma t}$$

A standard calculation, e.g., via the generating function, yields the variance

$$\mathrm{Var}(n(t)) = e^{\gamma t}\left(e^{\gamma t} - 1\right)$$

The full distribution can be written explicitly. For one initial particle, the state probabilities are

$$p_n(t) = e^{-\gamma t}\left(1 - e^{-\gamma t}\right)^{n-1} \tag{12.13}$$

This is a geometric distribution on $\{1, 2, \dots\}$ with parameter $e^{-\gamma t}$: at fixed time t, the population size is geometrically distributed, and its mean and variance match the formulas above.

Interpretation: each particle independently branches at rate γ, so the population grows by repeated splitting events. The probability of observing exactly n particles at time t decays geometrically in n, with a parameter determined by the single-particle survival/branching dynamics encoded in $e^{-\gamma t}$.

Try to derive the previous results as exercise.

12.4.4 Contact process

We now consider an important model in non-equilibrium statistical mechanics known as **the contact process**. Despite its simple definition, this is a genuinely non-linear stochastic process and has been widely studied in physics, epidemiology, and related fields as a paradigmatic model of spreading phenomena.

We focus on a mean-field (fully connected) version of the model, where spatial structure is neglected and every individual interacts with every other one. This setting allows us to introduce the Kramers-Moyal approximation, which provides a systematic way to approximate a discrete stochastic process by a continuous one.

Consider a population of N individuals, each of whom can be either infected or susceptible. Let n denote the number of infected individuals. The model is of SI type: susceptible individuals can become infected, while infected individuals can recover.

A susceptible individual may become infected through contact with any of the n infected individuals. Since the system is well mixed, the probability of such contact is proportional

to the fraction of infected individuals. The transition rate from n to $n+1$ (birth rate) infected individuals is therefore

$$b_n = \tilde{\beta}\frac{N-n}{N}\frac{n}{N-1}$$

where $(N-n)/N$ is the fraction of susceptible individuals and $n/(N-1)$ is the probability that a randomly chosen contact involves an infected individual.

Infected individuals recover independently at rate γ, so the transition rate from n to $n-1$ (death rate) is

$$d_n = \gamma n$$

The absorbing state $n=0$ corresponds to the extinction of the infection ($b_0 = 0 = d_0$), while at $n = N$ no further infections are possible ($b_N = 0$, $d_N \neq 0$, reflecting state), though recovery still occurs, and we also have to set $d_{N+1} = 0$.

Since the birth rate depends on the ratio $x = n/N$, it is convenient to introduce the rescaled variable $x \in [0,1]$ and define

$$b_x = N\beta x(1-x), \qquad d_x = N\gamma x$$

where $\beta = \tilde{\beta}/(N-1)$. The system thus possesses a natural large parameter N, which sets the typical scale of fluctuations.

Here, the master equation is:

$$\dot{q}_n(t) = b_{\frac{n-1}{N}} q_{n-1} + d_{\frac{n+1}{N}} q_{n+1} - (b_{\frac{n}{N}} + d_{\frac{n}{N}}) q_n \tag{12.14}$$

However, under appropriate assumptions (formalized by **Kurtz's theorem**), the discrete process for n can be approximated by a continuous stochastic process for x in the limit of large N.

Writing $q_n(t) = \Delta x\, p(x,t)$ with $\Delta x = 1/N$ (of the order of $1/N$), the master equation can be written as

$$\dot{p}(x,t) = b_{x-\frac{1}{N}} p\left(x - \tfrac{1}{N}, t\right) + d_{x+\frac{1}{N}} p\left(x + \tfrac{1}{N}, t\right) - (b_x + d_x) p(x,t) \tag{12.15}$$

and under the assumption that $\frac{1}{N} \ll 1$, it can be expanded in powers of $1/N$:

$$b_{x-\frac{1}{N}} p\left(x - \tfrac{1}{N}\right) - b_x p(x) = -\frac{1}{N}\frac{\mathrm{d}}{\mathrm{d}x}(b_x p) + \frac{1}{2}\frac{1}{N^2}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(b_x p) + \text{h.o.t.}$$
$$d_{x+\frac{1}{N}} p\left(x + \tfrac{1}{N}\right) - d_x p(x) = \frac{1}{N}\frac{\mathrm{d}}{\mathrm{d}x}(d_x p) + \frac{1}{2}\frac{1}{N^2}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(d_x p) + \text{h.o.t.}$$

Therefore, truncating the expansion at second order yields

$$\dot{p}(x,t) = -\frac{1}{N}\frac{\partial}{\partial x}\left[(b_x - d_x)p(x,t)\right] + \frac{1}{2N^2}\frac{\partial^2}{\partial x^2}\left[(b_x + d_x)p(x,t)\right] + \text{h.o.t.} \tag{12.16}$$

Finally, as $b_x = N\beta x(1-x)$ and $d_x = N\gamma x$ we get the Fokker-Planck equation

$$\dot{p} = -\frac{\partial}{\partial x}\left[(\beta x(1-x) - \gamma x)p\right] + \frac{1}{2N}\frac{\partial^2}{\partial x^2}\left[(\beta x(1-x) + \gamma x)p\right] \tag{12.17}$$

which corresponds to the Itô SDE

$$dx = (\beta x(1-x) - \gamma x)\, dt + \sqrt{\frac{\beta x(1-x) + \gamma x}{N}}\, dB(t) \tag{12.18}$$

where fluctuations are $O(N^{-1/2})$.

Although this derivation is not fully rigorous, it captures the qualitative behavior of the system away from boundaries and for large N. This approximation is called **Kramers-Moyal expansion**. More systematic approximations include the van Kampen system-size expansion and WKB methods, which are more accurate and equally difficult as well.

We now want to compute the mean of infected individuals $\rho(t) \equiv \mathbb{E}[x(t)]$, which satisfies (see Equation 12.18)

$$\dot{\rho} = (\beta - \gamma)\rho - \beta \mathbb{E}\left[x(t)^2\right] \tag{12.19}$$

This equation is not closed, since it depends on the second moment, and it cannot be solved. Trying to solving it by computing the second moment, one finds that this depends on the third moment and so on. Thus, the non-linearity of the process generates an infinite hierarchy of coupled moment equations.

A standard mean-field closure assumes that fluctuations are small, so that

$$\mathbb{E}\left[x^2\right] \simeq \mathbb{E}[x]^2 = \rho^2$$

In this way, we are able to close the equation of the mean. Under this approximation, the mean density obeys the logistic equation (which we have seen at the beginning of the course):

$$\dot{\rho} = (\beta - \gamma)\rho - \beta\rho^2 \tag{12.20}$$

This equation admits two stationary solutions ($\dot{\rho} = 0$):

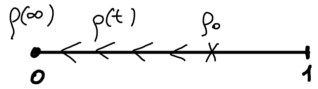
$$\rho^* = 0, \qquad \bar{\rho} = \frac{\beta - \gamma}{\beta}$$

Linear stability analysis shows that:

$$\frac{\mathrm{d}\dot{\rho}}{\mathrm{d}\rho} = \beta - \gamma - 2\beta\rho\Big|_{\rho=\rho^{\mathrm{st}}} = \begin{cases} \beta - \gamma & \rho = \rho^* = 0 \\ \gamma - \beta & \rho = \bar{\rho} = \frac{\beta-\gamma}{\beta} \end{cases}$$

Therefore, we have:

- for $\beta < \gamma$, the absorbing state $\rho = 0$ is stable; This means that starting from any point between 0 and 1, one goes back to zero as $t \to \infty$. Thus, the flow of ρ is

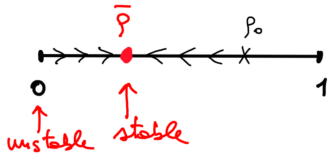


  and after some time, the equation is well approximated by

$$\dot{\rho} = (\beta - \gamma)\rho \quad \to \quad \rho(t) = \tilde{\rho}e^{-|\beta-\gamma|t}$$



- For $\beta > \gamma$, the active state $\rho = \bar{\rho}$ is stable and $\rho = 0$ is unstable. Thus, given the state $\bar{\rho}$, starting between 0 and $\bar{\rho}$ leads us to the right, and starting between $\bar{\rho}$ and 1 leads to the left.



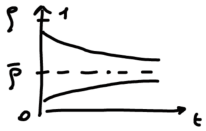
After some time, the equation is well approximated by

$$\rho(t) = \bar{\rho} + y(t)$$

($y \ll \bar{\rho}$). That is

$$\dot{\rho} = -(\beta - \gamma)(\bar{\rho} - \rho) \quad \rightarrow \quad \rho(t) = \bar{\rho} + \tilde{\rho} e^{-|\beta-\gamma|t}$$

The characteristic relaxation time scale is $|\beta - \gamma|^{-1}$ in both cases.



The full solution of the logistic equation can be summarized by

$$\rho(t) = \frac{\bar{\rho}}{1 + (\frac{\bar{\rho}}{\rho_0} - 1)e^{(\gamma-\beta)t}} = \frac{(\beta - \gamma)\rho_0}{\beta\rho_0 + (\beta - \gamma - \beta\rho_0)e^{(\gamma-\beta)t}} \tag{12.21}$$

where $0 < \rho_0 < 1$ is the initial density of infected individuals.

We can represent the behavior of the process through a phase diagram, where the control parameter $\beta - \gamma$ is plotted on the horizontal axis and the stationary value of the density ρ on the vertical axis. From the stability analysis, we find that $\beta - \gamma = 0$ marks a critical point. Below this value, the only stable stationary state is $\rho = 0$, while above it the system evolves toward a nonzero stationary density.

This change of behavior signals a phase transition at $\beta = \gamma$. At criticality, the mean-field equation reduces to

$$\dot{\rho} = -\beta\rho^2$$

whose solution is

$$\rho(t) = \frac{\rho_0}{1 + \beta t} \sim \frac{1}{t} \qquad \text{as } t \to \infty$$

(it follows a power law). Instead, way from criticality, i.e., for $\beta < \gamma$ or $\beta > \gamma$), the density ρ approaches its stationary value exponentially fast.

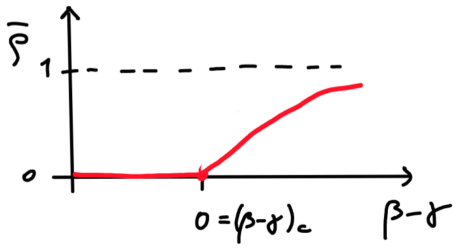


Figure 12.3

Exercise 12.4.3. *The discrete random walk in continuous time is governed by the master equation (symmetric random walk):*

$$\begin{aligned} \dot{p}_n &= p_{n+1} + p_{n-1} - 2p_n \\ p_n(0) &= \delta_{n,0} \end{aligned}$$

Show that the generating function

$$F(z,t) = \sum_{n=-\infty}^{+\infty} z^n p_n(t)$$

satisfies

$$\dot{F} = \left(z + \frac{1}{z} - 2 \right) F$$

So the solution is

$$F(z,t) = \exp\left[t(z + z^{-1} - 2) \right]$$

From this find

$$p_n(t) = e^{-t} \sum_{\substack{\ell \geq 0 \\ \ell + n \geq 0}} \frac{t^{2\ell+n}}{(\ell+n)!\ell!}$$

Find the Kramers-Moyal expansion of the master equation and find its solution.

12.5 Exact simulation of a birth/death process: the Gillespie's algorithm

The master equation 12.1

$$\dot{p}_n = b_{n-1} p_{n-1} + d_{n+1} p_{n+1} - (b_n + d_n) p_n$$

governs a birth-death process whose trajectories can be generated exactly, up to numerical round-off errors. Since the master equation fully specifies the stochastic dynamics, it contains all the information required to simulate individual realizations of the process, which is what we want to do now.

The process is Markovian, meaning that the state of the system at time t completely determines its future evolution. The algorithm introduced here applies to continuous-time Markov processes with a discrete state space, such as birth-death processes.

To generate a single trajectory, we need two answer two fundamental questions at each step:

1. When does the next event (birth or death) occur?

2. Which type of event occurs, given that an event takes place?

Once these two quantities are determined, the evolution of the path is completely specified.

12.5.1 Waiting-time distribution

Let $q_0(\tau|n)$ denote the probability that no events occurs in the time interval $[0, \tau)$, given that the system is in state n at time $t = 0$. Since the process is time-homogeneous, choosing $t = 0$ involves no loss of generality.

If T is the random time at which the next event occurs, then

$$q_0(\tau|n) = p(T > \tau|n)$$

To derive an equation for $q_0(\tau|n)$, we use both the Markov property and the fact that the process only allows birth and death transitions.

The probability that no event occurs in the interval $[0, \tau + \Delta\tau$ can be written as the probability that no event occurs in $[0, \tau)$ and no event occurs in $[\tau, \tau + \Delta\tau)$:

$$q_0(\tau + \Delta\tau|n) = \underbrace{q_0(\tau|n)}\underbrace{(1 - b_n\Delta\tau - d_n\Delta\tau)} \tag{12.22}$$

The first factor represents survival up to time τ; the second is the probability that neither a birth nor a death occurs in the short interval $\Delta\tau$.

Taking the limit $\Delta\tau \to 0$ and Taylor expanding yields the differential equation

$$\begin{cases} \frac{\mathrm{d}q_0}{\mathrm{d}\tau} = -(b_n + d_n)q_0 \\ q_0(0|n) = 1 \end{cases}$$

whose solution is

$$q_0(\tau|n) = e^{-(b_n+d_n)\tau} \tag{12.23}$$

This result shows that the waiting time until the next event is exponentially distributed, with rate $b_n + d_n$. The mean waiting time is therefore

$$\langle\tau(n)\rangle = \frac{1}{b_n + d_n}$$

provided that $b_n + d_n > 0$, i.e. that the state n is not absorbing.

This exponential waiting-time distribution is a key feature of continuous-time Markov processes and it can be generalized to processes which are not birth-death processes.

This provides the answer to the first question: given the current state n, compute b_n and d_n. Using 12.23, draw a waiting time τ from the exponential distribution with rate $b_n + d_n$; this τ is the time one has to wait until the next event occurs.

12.5.2 Type of event: birth or death

We now determine the probability that the next event is a birth or a death.

Let $q_+(\tau|n)\Delta\tau$ be the probability that a birth occurs in the interval $[\tau, \tau + \Delta\tau)$ given that the system was in state n at time 0. This probability can be written as

$$q_+(\tau|n)\Delta\tau = \underbrace{e^{-(b_n+d_n)\tau}}\underbrace{b_n\Delta\tau}$$

where the exponential factor accounts for the absence of events up to time τ, and the second factor is the probability of a birth in the short interval $\Delta\tau$.

Hence, the PDF for a birth event is

$$q_+(\tau|n) = b_n e^{-(b_n+d_n)\tau} \tag{12.24}$$

Similarly, the PDF for a death event is

$$q_-(\tau|n) = d_n e^{-(b_n+d_n)\tau} \tag{12.25}$$

The probability density that any event occurs at time τ is therefore

$$q_1(\tau|n) = (b_n + d_n)e^{-(b_n+d_n)\tau} \tag{12.26}$$

From Equation 13.22 we also get

$$\begin{aligned} q_+(\tau|n) &= \left(\begin{array}{c} \text{prob. of a birth event,} \\ \text{given that an event occurs} \\ \text{between } \tau \text{ and } \tau + \Delta\tau \end{array} \right) \left(\begin{array}{c} \text{prob. that an event occurs,} \\ \text{between } \tau \text{ and } \tau + \Delta\tau \end{array} \right) = \\ &= \frac{b_n}{b_n + d_n}(b_n + d_n)e^{-(b_n+d_n)\tau} \end{aligned} \tag{12.27}$$

12.5.3 Gillespie's algorithm

The results of Equation 12.27 leads directly to **Gillespie's algorithm** for birth-death processes from the master equation 12.1:

1. **Initialization**: set the initial time $t = 0$ and the initial state $n = n_0$.

2. **Check for absorption**: If $b_n + d_n > 0$, the state is absorbing and the trajectory terminates.

3. **Generate the waiting time**: draw a uniform number $r \in (0, 1)$ and set

   $$\tau = \frac{1}{b_n + d_n} \ln\left(\frac{1}{r}\right)$$

   (from Equation 12.26). Then, update the tie

   $$t_{\text{new}} = t_{\text{old}} + \tau$$

4. **Select the event type**: draw a second uniform random number $u \in (0, 1)$:

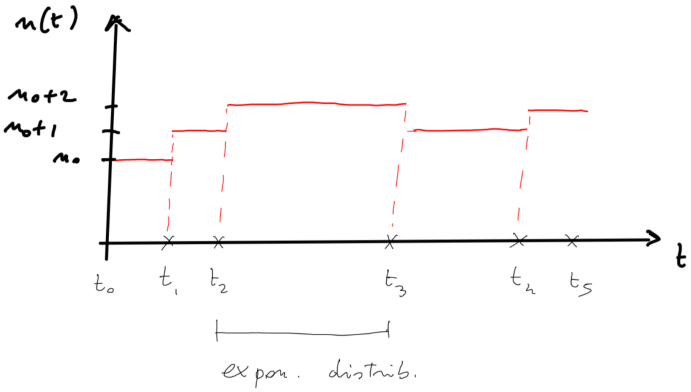
   - According to Equation 12.27, if

     $$u < \frac{b_n}{b_n + d_n}$$

     a birth occurs and $n \leftarrow n + 1$;
   - otherwise, a death occurs and $n \leftarrow n - 1$.

5. **Iterate**: repeat steps 2-4 until the desired final time is reached or an absorbing state is encountered ($b_n + d_n = 0$).

Interpretation: Gillespie's algorithm generates exact sample paths of the stochastic process whose ensemble statistics satisfy the master equation. No time discretization is involved: both the event times and their nature are sampled from the exact distributions implied by the Markov dynamics.



12.6 More general discrete Markov processes

It is straightforward to generalize the arguments developed for birth-death processes to continuous-time Markov processes on a discrete state space, where multiple transitions between arbitrary states are allowed. While the calculations become more involved, the underlying conceptual framework remains essentially the same.

The master equation can be derived from the Chapman-Kolmogorov equation, but it can also be obtained through a direct probabilistic argument, as we did previously.

Let $n \in E \subseteq \mathbb{Z}$ denote a generic state. We define $T(n'|n)\Delta t$ as the probability of a transition from state n to n' during the time interval $[t, t+\Delta t)$. The quantity $T(n'|n) \geq 0$ is the corresponding **transition rate**. In general, T may depend explicitly on time, but we restrict ourselves to the time-homogeneous case.

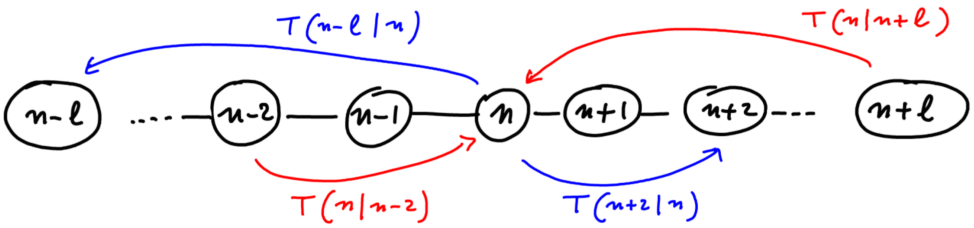


Figure 12.4

The evolution of the probability distribution is given by

$$p(n, t+\Delta t) = \underbrace{\sum_{m \neq n} T(n|m)\Delta t \; p(m,t)}_{} + \underbrace{\left(1 - \sum_{m \neq n} T(m|n)\Delta t\right)}_{} p(n,t)$$

where the first term represents transitions into state n from all other states $m \neq n$, while the second term accounts for the probability of remaining in state n.

Keeping only terms up to the first order in Δt and taking the limit $\Delta t \to 0$, we obtain the master equation

$$\dot{p}(n,t) = \sum_m [T(n|m)p(m,t) - T(m|n)p(n,t)] \tag{12.28}$$

where state $m = n$ has been included in the sum since $T(n|n) = 0$ and therefore contributes trivially. As usual, the equation is supplemented with the initial condition $p(n, t_0|n_0, t_0) = \delta_{n,n_0}$ together with appropriate boundary conditions when required.

As discussed earlier (Equation 12.2 and 12.3) the master equation can be written in matrix form by introducing the rate matrix $\mathbf{W}$ whose entries are

$$\mathbf{W}_{nm} = T(n|m) - \delta_{n,m} \sum_{\ell} T(\ell|n)$$

For $n \neq m$ one has $\mathbf{W}_{nm} \geq 0$ while probability conservation implies

$$\sum_{n} \mathbf{W}_{nm} = 0$$

The master equation 12.28 then reads

$$\dot{\vec{p}} = \mathbf{W}\vec{p} \quad \text{or} \quad \dot{p}_n = \sum_{m} \mathbf{W}_{mn} p_m$$

with diagonal entries $w_{ii} = -\sum_m T(m|i)$. Explicitly, the matrix has the form

$$\mathbf{W} = \begin{pmatrix} w_{11} & T(1|2) & T(1|3) & \dots \\ T(2|1) & w_{22} & T(2|3) & \dots \\ T(3|1) & T(3|2) & w_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where columns represent transitions out of a given state and rows transitions into it.

Stationary states are obtained by imposing $\dot{p}_n = 0$ in Equation 12.28, which leads to

$$\sum_{m} \left[T(n|m) p^{\text{st}}(m,t) - T(m|n) p^{\text{st}}(n,t) \right] = 0 \tag{12.29}$$

This condition is generally difficult to solve. A sufficient (but not necessary) condition is that each term in the sum vanishes individually, yielding

$$T(n|m) p^{\text{eq}}(m) = T(m|n) p^{\text{eq}}(n) \tag{12.30}$$

which is the detailed balance condition.

Detailed balance implies that probability currents between any pair of states vanish separately, meaning that all states are balanced pairwise. While this condition always guarantees stationarity, the converse is not true: a stationary state does not necessarily satisfy detailed balance.

In our case, the detailed balance condition satisfies Equation 12.29, but Equation 12.29 does not imply Equation 12.30. In general, one cannot impose Equation 12.30.

Finally, we connect what we have seen with statistical mechanics. Consider a system in thermal equilibrium with a heat bath at inverse temperature β. If the equilibrium distribution is of Boltzmann form,

$$p^{\text{eq}}(n) = \frac{1}{z} e^{-\beta E(n)}$$

where $E(n)$ is the energy of state n, then the detailed balance 12.29 is satisfied provided the transition rates obey

$$\frac{T(n|m)}{T(m|n)} = e^{-\beta[E(n) - E(m)]} \tag{12.31}$$

Although other choices of rates are possible, this condition ensure that the Markov dynamics is consistent with equilibrium statistical mechanics.

Chapter 13

Fluctuations in Chemical Reactions

We now turn to chemical reactions, with a particular focus on their stochastic fluctuations. Despite the name, chemical reactions represent a very general framework that extends far beyond chemistry. The same formalism applies naturally to a wide range of systems, including biological processes, epidemiology, population dynamics, gene expression, and even financial models. In this context, the term *chemical reaction* should be understood as a convenient abstraction for stochastic interaction processes, rather than as a reference to the microscopic structure of matter.

Our main goal is to understand how to describe the dynamics of such systems at two complementary levels. First, we will learn how to construct the deterministic equations governing the average behavior of the system, derived from the law of mass action. Second, we will study how fluctuations arise around this deterministic behavior in specific regimes, and how they can be characterized probabilistically. For example, the Gillespie's algorithm that we have seen in birth-death processes is an exact simulation of chemical reactions.

We will show how the formalism developed for discrete Markov processes can be systematically applied to chemical reaction systems, understanding how the master equation encodes the full stochastic evolution of reactions.

13.1 Chemical reactions

Chemical reactions provide a broad and flexible class of systems in which both deterministic and stochastic methods play a center role. In this framework, one considers collections of interacting entities that encounter each other randomly and undergo reactions. These entities, called **reactants**, interact according to prescribed rules and give rise to new entities, the **products**, which may differ in nature from the original reactants.

$$\text{reactant } 1 + \text{reactant } 2 \rightleftarrows \text{products}$$

Although the terminology originates from chemistry, this modeling framework applies to a wide variety of systems well beyond molecular reactions. Typical examples include:

1. Chemical systems, where atoms or molecules transform into different species as a result of collisions.

2. Population dynamics, where individuals may be born, die, reproduce, compete, migrate, or consume one another.

3. Epidemiological models, where diseases are transmitted through contacts between susceptible and infected individual.

4. Gene expression processes, where genes are transcribed into messenger RNA and translated into proteins, often regulated by the presence of specific transcription factors.

5. Other interacting systems (ecology, biophysics, etc).

What unifies all these examples is that the system evolves through discrete reaction events occurring at random times, while the state of the system is described by the number (or concentration) of each species involved.

These reactions may be **reversible** or not, according to whether the direct reaction can also occur in the backward direction.

We will now derive the deterministic ODEs governing the concentrations of reactants and products starting from the microscopic rules that define the reactions.

13.2 Binary and irreversible reactions

We begin with the simplest class of reactions, **binary and irreversible reactions**, which involve two different species. In this setting, a molecule of type X and a molecule of type Y (reactants) collide and react to produce a molecule of type Z (product). Schematically, the reaction is written as

$$\mathrm{X} + \mathrm{Y} \xrightarrow{\mathrm{k}} \mathrm{Z} \tag{13.1}$$

where the constant k is the **kinetic rate constant** of the reaction. This constant quantifies how frequently the reaction occurs when reactant molecules encounter each other, More precisely, k enters the definition of the **reaction rate** $r(x, y)$, which determines the number of reactions per unit time as a function of the concentrations of the reactants.

A concrete example of a real chemical reaction is

$$2\,\mathrm{H_2} + \mathrm{O_2} \xrightarrow{\mathrm{k}} 2\,\mathrm{H_2O}$$

Here, the integers appearing in front of the chemical species are the **stoichiometric coefficients**. They specify how many molecules of each species are consumed or produced during a single reaction event.

From this discussion, two key ingredients emerge as fundamental in the description of chemical reactions:

1. The identity of the reacting species, that is, which types of entities interact.

2. The stoichiometry of the reaction, namely, how many individuals of each species are consumed and produced in a single reaction event.

13.2.1 Calculation of the reaction rate

We consider a reaction taking place in a finite volume V, where the system is well stirred so that spatial correlations can be neglected. The concentrations of the species X, Y, and Z are assumed to be *small*, while the number of molecules of each species is *large*, so that fluctuations can initially be neglected and concentrations can be treated as continuous variables. We define

$$x = \frac{n_{\mathrm{X}}}{V}, \quad y = \frac{n_{\mathrm{Y}}}{V}, \quad z = \frac{n_{\mathrm{Z}}}{V}$$

Our goal is to estimate the **reaction rate** $r(x, y)$.

We focus on the **dilute limit**, where both x and y are small, In this regime, we can Taylor expand the reaction rate around $x = 0$ and $y = 0$:

$$r(x,y) \simeq r(0,0)+(\partial_x r_0)x+(\partial_y r_0)y+\left(\frac{1}{2}\partial^2_{xx} r_0\right)x^2+(\partial^2_{xy} r_0)xy+\left(\frac{1}{2}\partial^2_{yy} r_0\right)y^2+\text{h.o.t.} \quad (13.2)$$

Since the reaction requires the simultaneous presence of both reactants, it cannot occur if either X or Y is absent. Therefore, we must have

$$r(0, y) = 0 \quad \text{for all } y, \qquad r(x, 0) = 0 \quad \text{for all } x$$

Imposing these conditions on the Taylor expansion forces all constant, linear, and pure quadratic terms to vanish. In particular,

- $r(0, 0) = 0$
- $(\partial_x r)_0 = (\partial_y r)_0 = 0$
- $(\partial^2_{xx} r)_0 = (\partial^2_{yy} r)_0$

The leading non-vanishing contribution is therefore the mixed second-order term, yielding

$$r(x, y) = (\partial^2_{xy} r)_0 xy = kxy \quad (13.3)$$

This result is known as the **law of mass action**, which we will introduce now.

13.3 Law of mass action

The previous discussion motivates a simple but powerful approximation, which provides the deterministic equations governing the evolution of concentrations in chemical reactions: the **law of mass action**.

Theorem 13.3.1 (Law of Mass Action). *In a first approximation, the rate of a chemical reaction is proportional to the product of the concentrations of the reacting species. Each concentration appears raised to a power equal to the corresponding stoichiometric coefficient. The same coefficients determine, with appropriate signs, the rates of production or depletion of each species.*

This law states that, at the deterministic level, the temporal variation of the concentrations is obtained by multiplying the concentrations of the reactants involved in the reaction, taking into account their stoichiometric coefficients.

We now apply the law of mass action to the binary irreversible reaction introduced in Equation 13.1. Let $x(t)$, $y(t)$ and $z(t)$ denote the concentrations of species X, Y and Z at time t. Over as short time interval Δt, the law of mass action implies

$$\begin{cases} x(t+\Delta t) = x(t) - x(t)y(t)k\,\Delta t \\ y(t+\Delta t) = y(t) - x(t)y(t)k\Delta t \\ z(t+\Delta t) = z(t) + x(t)y(t)k\Delta t \end{cases} \xrightarrow{\Delta t \to 0} \begin{cases} \dot{x} = -kxy, & x(0) = x_0 \\ \dot{y} = -kxy, & y(0) = y_0 \\ \dot{z} = kxy, & z(0) = z_0 \end{cases} \quad (13.4)$$

The interpretation is straightforward. The decrease of x and y is proportional to the rate at which molecules of X and Y encounter each other and react. The constant k is a

rate constant, not a probability; the quantity $kxy\Delta t$ represents the expected fraction of reacting pairs in the time interval Δt. The concentration of Z increases by the same amount, reflecting the stoichiometry of the reaction.

Thus, to write down the deterministic evolution equations, one simply reads the reaction scheme and applies the law of mass action.

Because of the structure of the reaction, the system satisfies a conservation law. Subtracting the equations for x and y, we find

$$\dot{x} - \dot{y} = 0 \quad \Rightarrow \quad x(t) - y(t) = x_0 - y_0 = c$$

where x is a constant determined by the initial conditions. Using this constraint, we can eliminate y and write a closed equation for x:

$$\dot{x} = kxy = -kx(x - c) = kx(c - x) \tag{13.5}$$

This is the logistic equation, which can be solved exactly. The solution is

$$x(t) = \frac{cx_0 e^{ckt}}{c + x_0(e^{ckt} - 1)} = \begin{cases} c & \text{if } c > 0 \\ 0 & \text{if } c < 0 \end{cases} \tag{13.6}$$

From this expression, we see that the long-time behavior depends on the sign of $c = x_0 - y_0$. Thus, if $c > 0$, species X is initially in excess and its concentration approaches c. If $c < 0$, species X is depleted and $x(t) \to 0$.

13.4 Binary reversible reaction

We now consider the reversible binary version of Equation 13.1:

$$\mathrm{X} + \mathrm{Y} \underset{\mathrm{k}^-}{\overset{\mathrm{k}^+}{\rightleftharpoons}} \mathrm{Z}, \qquad k^{\pm} > 0 \tag{13.7}$$

Here k^+ is the forward reaction rate constant, while k^- is the backward reaction rate constant.

Applying the law of mass action as before, the deterministic evolution equations for the concentrations are

$$\begin{cases} \dot{x} = -k^+ xy + k^- z \\ \dot{y} = -k^+ xy + k^- z \\ \dot{z} = k^+ xy - k^- z \end{cases} \tag{13.8}$$

The forward reaction consumes one molecule of X and one of Y while producing one molecule of Z; the backward reaction does the opposite.

One can verify that this system admits two independent conservation laws

$$x(t) - y(t) = c_1$$
$$x(t) + z(t) = c_2$$

where c_1 and c_2 are constants fixed by the initial conditions.

Using these relations, the system can be reduced to a single equation for $x(t)$. Eliminating y and z, one finds

$$\dot{x} = k^+ x(c_1 - x) + k^-(c_2 - x) \tag{13.9}$$

This is a non-linear first-order differential equation that describes the relaxation of the system toward chemical equilibrium.

13.4.1 Haber process

As a final example, we consider the Haber process

$$\mathrm{N_2} + 3\,\mathrm{H_2} \longrightarrow 2\,\mathrm{NH_3}$$

What matters for our purposes is that the stoichiometric coefficients are not all equal to one. In abstract form, we write

$$\mathrm{X} + 3\,\mathrm{Y} \underset{\mathrm{k^-}}{\overset{\mathrm{k^+}}{\rightleftarrows}} 2\,\mathrm{Z} \tag{13.10}$$

Applying the law of mass action, the deterministic rate equations for the concentrations are

$$\begin{cases} \dot{x} = -k^+ xy^3 + k^- z^2 \\ \dot{y} = -3k^+ xy^3 + 3k^- z^2 \\ \dot{z} = -2k^- z^2 + 2k^+ xy^3 \end{cases} \tag{13.11}$$

Each term follows directly from stoichiometry. The forward reaction consumes one molecule of X and three molecules of Y, hence the factor y^3 in the rate and the coefficients 1 and 3 in $\dot{x}$ and $\dot{y}$. Similarly, the backward reaction consumes two molecules of Z , producing one X and three Y, which explains the z^2 dependence and the coefficients in front of the backward terms.

In general, stoichiometric coefficients affect both:

1. the functional dependence of the reaction rate on concentrations, and
2. the relative speed at which different species are produced or depleted.

For instance, from the equation 13.11 above one sees that

$$3\dot{z} = -2\dot{y}$$

which reflects the fact that a decrease of three molecules of Y corresponds to an increase of two molecules of Z, and vice versa.

Exercise 13.4.1. *At stationarity, the system in Equation 13.11 has a steady state which satisfies*

$$\frac{\bar{x}\bar{y}^3}{\bar{z}^2} = \frac{k^-}{k^+}$$

Moreover, the dynamics possesses the conservation laws

$$\begin{aligned} 3x(t) - y(t) &= c_1 \\ 2x(t) + z(t) &= c_2 \\ 2y(t) + 3z(t) &= c_3 \end{aligned}$$

Explain why these quantities are conserved and why the constants c_1, c_2 and c_3 are not independent

Exercise 13.4.2. *Write the rate equations for the reaction:*

$$2\,\mathrm{X} + \mathrm{Y} \underset{\mathrm{k^-}}{\overset{\mathrm{k^+}}{\rightleftarrows}} 2Z$$

and determine all independent conservation laws.

13.5 Multiple reactions

Molecules or species may interact through several reactions occurring simultaneously. A simple and important example is provided by predator-prey systems. We consider the following set of reactions

$$\begin{aligned} \mathrm{X} &\xrightarrow{\mathrm{k}_1} 2\,\mathrm{X} \quad \text{(reproduction of preys)} \\ \mathrm{Y} + \mathrm{X} &\xrightarrow{\mathrm{k}_2} 2\,\mathrm{Y} \quad \text{(predator eats a prey)} \\ \mathrm{Y} &\xrightarrow{\mathrm{k}_3} \emptyset \quad \text{(predator dies)} \end{aligned} \tag{13.12}$$

In the first reaction, a prey reproduces and generates an offspring. In the second reaction, a predator encounters a prey, consumes it, and reproduces. In the third reaction, a predator dies. All these reactions are assumed to be independent.

Applying the law of mass action, the deterministic rate equations for the concentrations are

$$\begin{cases} \dot{x} = k_1 x - k_2 xy \\ \dot{y} = k_2 xy - k_3 y \end{cases} \tag{13.13}$$

Each term in the equations corresponds to one of the reactions listed above. These equations are known as the **Lotka-Volterra equations** in the mass-action approximation.

Exercise 13.5.1. *Show that the steady state is*

$$\bar{x} = \frac{k_3 54}{k_2}, \quad \bar{y} = \frac{k_1}{k_2}$$

Show that the Jacobian matrix at $(\bar{x}, \bar{y})$ *has eigenvalues* $\lambda = \pm i\sqrt{k_1 k_3}$. *Show also that*

$$V(x, y) = k_2(x + y) - k_3 \ln x - k_1 \ln y$$

is a conserved quantity of the system.

13.6 General chemical reactions

We now address the problem from a general perspective. We have shown through several examples how, starting from simple reversible reactions or from systems involving multiple reactions, the law of mass action leads to a set of deterministic ODEs. We now want to formulate this procedure in full generality, that is, starting from an arbitrary set of reactions involving an arbitrary number of species, we aim to derive both the mass-action equations and their fluctuations.

We consider S different chemical species X_i, with $i = 1, 2, \ldots, S$ involved in R distinct reactions of the form

$$\sum_{i=1}^{S} N_{ir} \mathrm{X}_i \underset{\mathrm{k}_r^-}{\overset{\mathrm{k}_r^+}{\rightleftarrows}} \sum_{i=1}^{S} M_{ir} \mathrm{X}_i, \qquad r = 1, \ldots, R \tag{13.14}$$

where $N_{ir}, M_{ir} \in \mathbb{Z}^+$ are the stoichiometric coefficients. We also introduce the **stoichiometric vectors**

$$\vec{N}_r = (N_{1r}, \ldots, N_{Sr})^T, \qquad \vec{M}_r = (M_{1r}, \ldots, M_{Sr})^T$$

This notation means that, for reaction r, N_{ir} molecules of species X_i are required for the forward reaction to occur with rate constant k_r^+, while M_{ir} molecules of type X_i are required for the backward reaction to occur with rate constant k_r^-.

Applying the law of mass-action, the deterministic evolution equations for the concentrations (ODEs) are

$$\dot{x_i} = \sum_{r=1}^{R} (M_{ir} - N_{ir}) \underbrace{\left(k_r^+ \prod_{j=1}^{S} x_i^{N_{jr}} - k_r^- \prod_{j=1}^{S} x_j^{M_{jr}} \right)}_{\equiv v_r(\vec{x})}, \quad i = 1, \ldots, S \tag{13.15}$$

Inside the parenthesis we have the net contribution of reaction r, given by the difference between the forward and the backward reaction rates. This term is multiplied by $M_{ir} - N_{ir}$, which represents the net stoichiometric change of species i due to a single occurrence of reaction r. Finally, since all reactions are assumed to be independent, the total rate of change x_i is obtained by summing over all reactions.

Interpretation: The time derivative of the concentration of species i is given by the sum, over all reactions, of the number of molecules of species i created or destroyed by a single reaction event, multiplied by the frequency at which that reaction occurs. In this expression, $v_r(\vec{x})$ is the reaction rate of reaction r, namely the number of reaction events per unit time, while $(M_{ir} - N_{ir})$ quantifies the net stoichiometric effect of one reaction event on species i.

13.7 Fluctuations

Molecules, (or, more generally, discrete individuals) collide at random . When they do, reactions occurs at some rate. Because the reacting entities are discrete, reaction events are inherently random, and this randomness generates stochasticity. The mass-action equations 13.15 can approximate the evolution of molecule numbers or concentrations only when the system contains very large numbers of reacting entities. In other words, mass-action is accurate only in specific regimes. A natural question is whether we can justify this approximation by taking an appropriate limit and deriving the mass-action equations from a more fundamental description. To do so, we must write down the master equation of the system.

Let $\vec{n} = (n_1, n_2, \ldots, n_S)^T$ denote the population vector, where n_i is the number of molecules (or individuals) of species X_i. We define the **stoichiometric matrix** through

$$s_{ir} \equiv M_{ir} - N_{ir}$$

and we denote by $\vec{s}_r \equiv (s_{1r}, s_{2r}, \ldots, s_{Sr})^T$ the stoichiometric change vector associated with reaction r.

Starting from state $\vec{n}$, the forward occurrence of reaction r produces the jump

$$\vec{n} \xrightarrow{T_r^+(\vec{n})} \vec{n} + \vec{s}_r$$

meaning that, when reaction r fires in the forward direction, the population of species i changes by $M_{ir} - N_{ir}$. For reaction r to occur, N_{ir} molecules of type X_i must collide from the available n_i. Since the identity of the molecules does not matter (only their species does), the rate at which reaction r occurs is determined by the number of distinct

combinations of reactants that can be formed. This leads to the **transition rates** (also called **propensities**)

$$\begin{aligned} T_r^+(\vec{n}) &= k_r^+ \prod_{i=1}^{S} \binom{n_i}{N_{ir}} \qquad \text{(forward reaction)} \\ T_r^-(\vec{n}) &= k_r^- \prod_{i=1}^{S} \binom{n_i}{M_{ir}} \qquad \text{(backward reaction)} \end{aligned} \tag{13.16}$$

Similarly, the backward firing of reaction r produces

$$\vec{n} \xrightarrow{T_r^-} \vec{n} - \vec{s}_r$$

Using these transition rates, we can write the **chemical master equation** (CME) for a system with R independent reactions:

$$\begin{aligned} \dot{p}(\vec{n},t) = \sum_{r=1}^{R} \big[& T_r^+(\vec{n}-\vec{s}_r)p(\vec{n}-\vec{s}_r,t) - T_r^+(\vec{n})p(\vec{n},t) + \\ & + T_r^-(\vec{n}+\vec{s}_r)p(\vec{n}+\vec{s}_r,t) - T_r^-(\vec{n})p(\vec{n},t) \big] \end{aligned} \tag{13.17}$$

The CME provides the exact stochastic description of the system, and it can be simulated exactly using the Gillespie algorithm.

While exact, the CME is typically difficult to handle analytically. However, Equation 13.17 is in a form suitable for a Kramers-Moyal expansion (non-rigorous but practically useful). Under appropriate conditions, one can approximate the CME by a Fokker-Planck equation, and equivalently by a Langevin description, which is often much easier to simulate.

To proceed, one assumes that each jump $\vec{s}_r$ represents a *small* perturbation and performs a Taylor expansion up to 2nd order. This yields the Fokker-Planck equation:

$$\dot{p}(\vec{x},t) = -\sum_{i=1}^{S} \frac{\partial}{\partial x_i} \left[A_i(\vec{x}) p(\vec{x},t) \right] + \frac{1}{2} \sum_{i,j=1}^{S} \frac{\partial^2}{\partial x_i \partial x_j} \left[B_{ij}(\vec{x}) p(\vec{x},t) \right] \tag{13.18}$$

where $\vec{x}$ is a vector of continuous variables. The drift term A_i encodes the deterministic evolution (the mass-action limit), while B_{ij} collects the contributions of fluctuations:

$$\begin{aligned} A_i(\vec{x}) &\equiv \sum_{r=1}^{R} s_{ir} \left[T_r^+(\vec{x}) - T_r^-(\vec{x}) \right], \qquad (\dot{x}_i = A_i(\vec{x})) \\ B_{ij}(\vec{x}) &\equiv \sum_{r=1}^{R} s_{ir} s_{jr} \left[T_r^+(\vec{x}) + T_r^-(\vec{x}) \right] \qquad \text{(fluctuations)} \end{aligned} \tag{13.19}$$

This is the **chemical Fokker-Planck equation**, i.e. the diffusive approximation of the CME 13.17.

At this point, two questions remain open. First, what precisely does it mean that s_{ir} is *small*? Second, how should we interpret the transition rates in Equation 13.16 when $\vec{n}$ is discrete, while the Fokker-Planck description uses continuous variables $\vec{x}$? In particular, this raises the issue of how the parameters and variables should scale with system size in order for the approximation to be consistent.

13.7.1 Deterministic limit

We now show how the mass-action equations can be obtained as a suitable limit of the CME by deriving what is known as the **deterministic limit**. The goal is to understand under which assumptions the transition rates defined in Equation 13.16 lead to the deterministic mass-action equations when the system size becomes large.

The system size Ω represents an extensive parameter, such as the total volume of the system or the total number of molecules, and we consider the limit $\Omega \to \infty$.

Since reactions are independent, we first focus on a single reaction and later generalize to multiple reactions:

$$\sum_{i=1}^{S} N_i \mathrm{X}_i \underset{\mathrm{k}^-}{\overset{\mathrm{k}^+}{\rightleftarrows}} \sum_{i=1}^{S} M_i \mathrm{X}_i, \qquad \vec{s} = (M_1 - N_1, \ldots, M_S - N_S)^T$$

In contrast with previous sections, we now allow the microscopic kinetic constants $k^{\pm}$ in Equation 13.16 to depend, and thus scaling, on the system size Ω. This dependence is necessary in order to recover a meaningful deterministic limit.

We now show that the rates must scale in an appropriate way if we want to recover the mass-action. The transition rates are

$$T^+(\vec{n}) = k^+(\Omega) \prod_{i=1}^{S} \binom{n_i}{N_i}, \qquad T^-(\vec{n}) = k^-(\Omega) \prod_{i=1}^{S} \binom{n_i}{M_i} \tag{13.20}$$

and the corresponding CME reads

$$\dot{p}(\vec{n}) = T^+(\vec{n} - \vec{s})p(\vec{n} - \vec{s}) - T^+(\vec{n})p(\vec{n}) + T^-(\vec{n} + \vec{s})p(\vec{n} + \vec{s}) - T^-(\vec{n})p(\vec{n}) \tag{13.21}$$

The idea is the following: the mass-action equations describe the evolution of average concentrations. We therefore compute the time evolution of the mean population vector and see if this mean goes into the mass-action. Therefore

$$\langle \vec{n} \rangle = \sum_{\{\vec{n}\}} \vec{n}\, p(\vec{n}, t)$$

Multiplying Equation 13.21 by $\vec{n}$ and summing over all states yields

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \vec{n} \rangle = \vec{s} [\langle T^+(\vec{n}) \rangle - \langle T^-(\vec{n}) \rangle] \tag{13.22}$$

This gives the evolution of the average of all populations

$$\langle T(\vec{n}) \rangle \equiv \sum_{\{\vec{n}\}} T(\vec{n}) p(\vec{n}, t)$$

We now introduce concentration variables $x = n/\Omega$. Dividing by Ω, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \vec{x} \rangle = \vec{s} \left[\frac{\langle T^+(\Omega \vec{x}) \rangle}{\Omega} - \frac{\langle T^-(\Omega \vec{x}) \rangle}{\Omega} \right] \tag{13.23}$$

For this equation to have a non-trivial limit as $\Omega \to \infty$, the transition rates must scale linearly with Ω. Otherwise, the RHS would either vanish or diverge. This means that the

transition rates must be of order $O(1)$, that is $T^{\pm}$ scale linearly with Ω in such a way that the ratio cancels out. We therefore examine the large-Ω behavior of the transition rates.

For the forward reaction we have

$$T^+(\Omega\vec{x}) = k^+(\Omega)\prod_i \binom{\Omega x_i}{N_i} = k^+(\Omega)\prod_i \frac{(\Omega x_i)!}{(\Omega x_i - N_i)!N_i!} \simeq$$
$$\simeq k^+(\Omega)\prod_i \frac{(\Omega x_i)^{N_i}}{N_i!} + \text{h.o.t.} = \underbrace{k^+(\Omega)\frac{\Omega^{\sigma^+}}{\prod_i N_i!}}_{\sim\Omega}\prod_i x_i^{N_i} + \text{h.o.t.}$$

where we used the approximation

$$\binom{x}{n} = \frac{x(x-1)\dots(x-n+1)}{n!} \simeq \frac{x^n}{n!} + \text{h.o.t.}$$

for large x. We also defined

$$\sigma^+ \equiv \sum_{i=1}^{S} N_i$$

which is the order of the forward reaction. Therefore, we can write

$$\frac{k^+(\Omega)\Omega^{\sigma^+}}{\prod_i N_i!} = \Omega\bar{k}^+ + O(1)$$

as $\Omega \to \infty$ and $\bar{k}^+ \sim O(1)$ does not depend on Ω. The same can be done for the backward reaction. To ensures linear scaling with Ω, the kinetic constants must therefore scale as $\Omega \to \infty$

$$\begin{aligned} k^+(\Omega) &= \bar{k}^+\Omega^{1-\sigma^+}\prod_i N_i! + O(\Omega^{-\sigma^+}) \\ k^-(\Omega) &= \bar{k}^-\Omega^{1-\sigma^-}\prod_i M_i! + O(\Omega^{-\sigma^-}) \end{aligned} \tag{13.24}$$

With this scalings, the difference of transition rates obtained in 13.23 satisfies

$$T^+(\Omega\vec{x}) - T^-(\Omega\vec{x}) = \Omega v(\vec{x})$$

where $v(\vec{x})$ is the macroscopic reaction rate. Substituting into the evolution equation, we obtain for a single reaction

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle x_i\rangle = s_i\left[\bar{k}^+\left\langle\prod_i x_i^{N_i}\right\rangle - \bar{k}^-\left\langle\prod_i x_i^{M_i}\right\rangle\right] = s_i\langle v(\vec{x})\rangle \tag{13.25}$$

For multiple independent reactions, summing over r yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle x_i\rangle = \sum_{r=1}^{R} s_{ir}\left[\bar{k}_r^+\left\langle\prod_i x_i^{N_{ir}}\right\rangle - \bar{k}_r^-\left\langle\prod_i x_i^{M_{ir}}\right\rangle\right] = \sum_r s_{ir}\langle v_r(\vec{x})\rangle \tag{13.26}$$

This equation still depends on higher moments and is therefore not closed. To recover deterministic mass-action equations, we introduce a mean-field approximation, valid for

large Ω, which neglects correlations and replaces averages of products with products of averages. Under this approximation,

$$\left\langle \prod_i x^{N_{ir}} \right\rangle \simeq \prod_i \langle x_i \rangle^{N_{ir}}$$
$$\left\langle \prod_i x^{M_{ir}} \right\rangle \simeq \prod_i \langle x_i \rangle^{M_{ir}}$$

This finally leads to the deterministic limit of the mass-action ODEs:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \langle x_i \rangle &= \sum_r s_{ir} \left[\bar{k}_r^+ \prod_i \langle x_i \rangle^{N_{ir}} - \bar{k}_r^- \prod_i \langle x_i \rangle^{M_{ir}} \right] = \\ &= \sum_r s_{ir} v(\langle x_i \rangle) = \\ &= \sum_r s_{ir} \left[v_r^+(\langle x_i \rangle) - v_r^-(\langle x_i \rangle) \right] \end{aligned} \tag{13.27}$$

where $v_r^\pm$ is defined as

$$v_r^+ \equiv \bar{k}_r^+ \prod_i x_i^{N_{ir}}$$
$$v_r^- \equiv \bar{k}^- \prod_i x_i^{M_{ir}}$$

Thus, Equation 13.27 is equivalent to Equation 13.15 if we can also show that the constants $\bar{k}^\pm$ are the same.

In summary, the mass-action equations emerge from the chemical master equation only under specific conditions: the existence of an extensive limit, an appropriate scaling of the microscopic rates with system size, and the neglect of correlations. Outside these conditions, fluctuations and correlations play a crucial role, and the mass-action approximation may fail.

13.7.2 Derivation of the Fokker-Planck equation from the deterministic limit

We now use the deterministic limit derived previously, together with the large-Ω limit, to obtain the **chemical Fokker-Planck equation**, which provides a diffusive approximation of the CME.

The ideas is that, although the underlying variables are discrete and evolve through finite jumps, when the system size Ω is large these jumps become small compared to the typical scale of the variables, and the dynamics can be approximated by continuous stochastic processes.

In this approximation, the PDF $p(\vec{x}, t)$ for the concentration vector $\vec{x}$ satisfies a Fokker-

Planck equation of the form

$$\begin{aligned}
\dot{p}(\vec{x},t) &= -\sum_{i=1}^{S} \frac{\partial}{\partial x_i}\left[A_i(\vec{x})p(\vec{x})\right] + \frac{1}{2}\sum_{ij}\frac{\partial^2}{\partial x_i \partial x_j}\left[B_{ij}(\vec{x})p(\vec{x})\right] \\
A_i(\vec{x}) &= \sum_{r=1}^{R} s_{ir}\left[v_r^+(\vec{x}) - v_r^-(\vec{x})\right], \qquad \dot{x}_i = A_i(\vec{x}) \\
B_{ij}(\vec{x}) &\equiv \frac{1}{\Omega}\sum_{r=1}^{R} s_{ir}s_{jr}\left[v_r^+(\vec{x}) + v_r^-(\vec{x})\right]
\end{aligned} \tag{13.28}$$

This derivation shows how to interpret Equation 13.18 and 13.19 in the diffusive approximation: the drift term $A_i(\vec{x})$ reproduces the deterministic mass-action dynamics, while the diffusion matrix $B_{ij}(\vec{x})$ encodes the fluctuations.

This Fokker-Planck equation corresponds (non-uniquely) to a **chemical Langevin equation** in which each reaction channel is associated with an independent Wiener process. In differential form, the Langevin dynamics is

$$dx_i = \sum_{r=1}^{R} s_{ir}\left[v_r^+(\vec{x}) - v_r^-(\vec{x})\right]\, dt + \frac{1}{\sqrt{\Omega}}\sum_{r=1}^{R} s_{ir}\left[\sqrt{v_r^+(\vec{x})}\, dB_r^+ - \sqrt{v_r^-(\vec{x})}\, dB_r^-\right] \tag{13.29}$$

where $dB^{\pm}$ are R independent Wiener processes and correspond to the forward and backward occurrences of reaction r.

With this choice, the noise correlations satisfy

$$\langle d\eta_i\, d\eta_j | \vec{x}\rangle = B_{ij}(\vec{x})\, dt$$

Finally, notice that the matrix B in Equation 13.28 can be written as a tensor product of the vector $\vec{s}_r \equiv (s_{1r}, \ldots, s_{Sr})^T$

$$B(\vec{x}) = \frac{1}{\Omega}\sum_r \vec{s}_r \odot \vec{s}_r^T (v^+ + v^-) = \frac{1}{\Omega}\sum_r \begin{pmatrix} s_{1r} \\ \vdots \\ s_{SR} \end{pmatrix} \odot \begin{pmatrix} s_{1r} & \ldots & s_{SR} \end{pmatrix} (v^+ + v^-)$$

Exercise 13.7.1. *Define*

$$d\eta_i \coloneqq \frac{1}{\sqrt{\Omega}}\sum_{r=1}^{R} s_{ir}\left[\sqrt{v_r^+}\, dB_r^+ - \sqrt{v_r^-}\, dB_r^-\right]$$

Calculate $\mathbb{E}\left[d\eta_i\, d\eta_j | \vec{x}\right]$ *assuming that* $dB_r^{\pm} dB_s^{\pm} = \delta_{rs}\, dt$ *and* $dB_r^+\, dB_s^- = 0$. *Show that*

$$\mathbb{E}\left[d\eta_i d\eta_j | \vec{x}\right] = \frac{1}{\Omega}\sum_r s_{ir}s_{jr}(v_r^+ + v_r^-)\, dt = B_{ij}(\vec{x})\, dt$$

This means that $d\eta_i$ *is a correct noise that generates the diffusive matrix* $B(\vec{x})$.

Example 13.7.1

Consider the reversible reaction with one species

$$\mathrm{A} \underset{\mathrm{k}^-}{\overset{\mathrm{k}^+}{\rightleftarrows}} 2\,\mathrm{A}$$

For the CME we need the transition rates. The forward rate is

$$T^+(n) = k^+(\Omega)\binom{n}{1} = k^+(\Omega)n = \bar{k}^+\Omega x$$

where n is the number of the molecules and k^+ does not scale because it is $O(1)$, that is $k^+(\Omega) = \bar{k}^+$ (see Equation 13.24). The backward rate is

$$T^-(n) = k^-(\Omega)\binom{n}{2} = k^-(\Omega)\frac{n(n-1)}{2} = \frac{\bar{k}^-}{\Omega}n(n-1)$$

where $k^-(\Omega) = 2\bar{k}^-/\Omega$. Therefore,

$$T^-(\Omega x) = \frac{\bar{k}^-}{\Omega}\Omega x(\Omega x - 1) \quad \xrightarrow{\Omega\to\infty} \quad \Omega\bar{k}^- x^2 + \text{h.o.t.}$$

and $\bar{k}^- x^2 = v^-(x)$ in Equation 13.27.
The mean field equation is therefore

$$\dot{x} = \bar{k}^+ x - \bar{k}^- x^2$$

($\langle x^2 \rangle \simeq \langle x \rangle^2$).
The CME reads

$$\begin{aligned}\dot{p}(n) &= T^+(n-1)p(n-1) - T^+(n)p(n) + T^-(n+1)p(n+1) - T^-(n)p(n) = \\ &= k^+(n-1)p(n-1) - k^+(n)p(n) + \frac{\bar{k}^-}{\Omega}n(n+1)p(n+1) - \frac{\bar{k}^-}{\Omega}n(n-1)p(n)\end{aligned}$$

The diffusive approximation yields

$$B(x) = \frac{1}{\Omega}(\bar{k}^+ x + \bar{k}^- x^2)$$

The corresponding Langevin equation is

$$dx = (\bar{k}^+ x - \bar{k}^- x^2)\,dt + \frac{1}{\sqrt{\Omega}}\left(\sqrt{\bar{k}^+ x}\,dB^+ - \sqrt{\bar{k}^- x^2}\,dB^-\right)$$

which is valid for large populations and away from the boundaries.

Example 13.7.2

Consider the reversible reaction with three species

$$\mathrm{A} + 3\,\mathrm{B} \underset{\mathrm{k}^-}{\overset{\mathrm{k}^+}{\rightleftarrows}} 2\,\mathrm{C}$$

The stoichiometric vectors are $\vec{N} = (1, 3, 0) \equiv (\mathrm{A}, \mathrm{B}, \mathrm{C})$ for the reactants and $\vec{M} = (0, 0, 2)$ for the products, so that $\vec{s} = (-1, -3, 2)^T$.
The transition rates are

$$T^+(\vec{n}) = k^+(\Omega)\binom{n_A}{1}\binom{n_B}{3} = \frac{k^+(\Omega)}{6} n_A n_B (n_B - 1)(n_B - 2)$$
$$T^-(\vec{n}) = k^-(\Omega)\binom{n_C}{2} = \frac{k^-(\Omega)}{2} n_C (n_C - 1)$$

Notice that if we want the rate to be larger then zero, we need at least 3 molecules of B. Check this always with rates. In the deterministic limit,

$$v^+(\vec{x}) = \bar{k}^+ x_A x_B^3$$
$$v^-(\vec{x}) = \bar{k}^- x_C^2$$

The ODEs (mass-action equation)

$$\begin{pmatrix} \dot{x}_A \\ \dot{x}_B \\ \dot{x}_C \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} (v^+ - v^-) = \begin{pmatrix} -(\bar{k}^+ x_A x_B^3 - \bar{k}^- x_C^2) \\ -3(\dots) \\ 2(\dots) \end{pmatrix}$$

which coincide with Equation 13.11.
The diffusive matrix is

$$B(\vec{x}) = \frac{v^+ + v^-}{\Omega} \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} \odot \begin{pmatrix} -1 & -3 & 2 \end{pmatrix} = \frac{v^+ + v^-}{\Omega} \begin{pmatrix} 1 & 3 & -2 \\ 3 & 9 & -6 \\ -2 & -6 & 4 \end{pmatrix}$$

Example 13.7.3

Consider the three irreversible reactions (Equation 13.12)

$$\mathrm{X} \xrightarrow{\mathrm{k_1}} 2\,\mathrm{X}$$
$$\mathrm{X} + \mathrm{Y} \xrightarrow{\mathrm{k_2}} 2\,\mathrm{Y}$$
$$\mathrm{Y} \xrightarrow{\mathrm{k_3}} \emptyset$$

For the first reaction, the stoichiometric vectors and macroscopic rates are

$$\vec{N}_1 = (1, 0), \quad \vec{M}_1 = (2, 0), \quad \vec{s}_1 = (1, 0)^T$$
$$T_1 = k_1 \binom{n_X}{1} \quad \xrightarrow{\Omega\to\infty} \quad v_1 = \bar{k}_1 x$$

For the second, we have:

$$\vec{N}_2 = (1, 1), \quad \vec{M}_2 = (0, 2), \quad \vec{s}_2 = (-1, 1)^T$$
$$T_2 = k_2 \binom{n_X}{1}\binom{n_Y}{1} \quad \xrightarrow{\Omega\to\infty} \quad v_2 = \bar{k}_2 xy$$

For the third, we have:

$$\vec{N}_3 = (0,1), \quad \vec{M}_3 = (0,0), \quad \vec{s}_3 = (0,-1)^T$$
$$T_3 = k_3 \begin{pmatrix} n_Y \\ 1 \end{pmatrix} \quad \xrightarrow{\Omega\to\infty} \quad v_3 = \bar{k}_3 y$$

The ODEs in the mean-field limit are

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} v_1 + \begin{pmatrix} -1 \\ 1 \end{pmatrix} v_2 + \begin{pmatrix} 0 \\ -1 \end{pmatrix} v_3 = \begin{pmatrix} \bar{k}_1 x - \bar{k}_2 xy \\ \bar{k}_2 xy - \bar{k}_3 y \end{pmatrix}$$

This is Equation 13.13. The diffusive matrix

$$\begin{aligned} B(x) &= \frac{1}{\Omega}\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 \end{pmatrix} v_1 + \begin{pmatrix} -1 \\ 1 \end{pmatrix} \odot \begin{pmatrix} -1 & 1 \end{pmatrix} v_2 + \begin{pmatrix} 0 \\ -1 \end{pmatrix} \odot \begin{pmatrix} 0 & -1 \end{pmatrix} v_3\right] = \\ &= \frac{1}{\Omega}\left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \bar{k}_1 x + \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \bar{k}_2 xy + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \bar{k}_3 y +\right] = \\ &= \frac{1}{\Omega} \begin{pmatrix} \bar{k}_1 x + \bar{k}_2 xy & -\bar{k}_2 xy \\ -\bar{k}_2 xy & \bar{k}_2 xy + \bar{k}_3 y \end{pmatrix} \end{aligned}$$

The Langevin approximation is

$$\begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} \bar{k}_1 x - \bar{k}_2 xy \\ \bar{k}_2 xy - \bar{k}_3 y \end{pmatrix} dt + \frac{1}{\sqrt{\Omega}} \begin{pmatrix} \sqrt{\bar{k}_1 x} dB_1 - \sqrt{\bar{k}_2 xy}\, dB_2 \\ \sqrt{\bar{k}_2 xy} dB_2 - \sqrt{\bar{k}_3 y} dB_3 \end{pmatrix}$$